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(54) **TRANSFORM CODING BASED ON
MATRIX-BASED INTRA PREDICTION**

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(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,303,325 B2 12/2007 Takata
8,335,484 B1 12/2012 Arad

(Continued)

FOREIGN PATENT DOCUMENTS

CN 1738434 A 2/2006
CN 1784015 A 6/2006

(Continued)

OTHER PUBLICATIONS

US 11,496,767 B2, 11/2022, Deng et al. (withdrawn)

(Continued)

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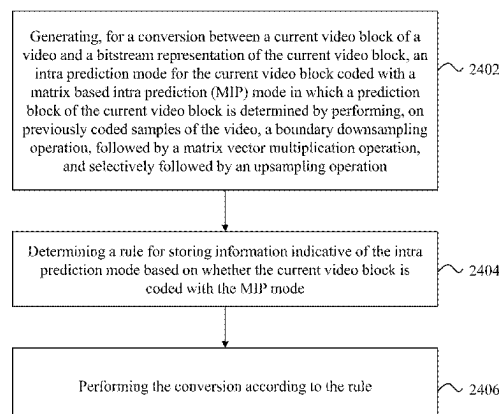
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(57) **ABSTRACT**

Devices, systems and methods for digital video coding, which includes matrix-based intra prediction methods for video coding, are described. In a representative aspect, a method for video processing includes performing a conversion between a current video block of a video and a bitstream representation of the current video block according to a rule, where the rule specifies a relationship between applicability of a matrix based intra prediction (MIP) mode or a transform mode during the conversion, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, and where the trans-

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form mode specifies use of a transform operation for the determining the prediction block for the current video block.

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(56) References Cited

U.S. PATENT DOCUMENTS

9,723,313 B2	8/2017	Yang et al.	2015/0358631 A1	12/2015	Zhang et al.
10,142,627 B2	11/2018	Zhao et al.	2016/0073107 A1	3/2016	Moon et al.
10,284,860 B1	5/2019	Zhao et al.	2016/0100189 A1	4/2016	Pang et al.
10,368,107 B2	7/2019	Zhang	2016/0165258 A1	6/2016	Lei
10,390,050 B2	8/2019	An et al.	2016/0182913 A1	6/2016	Joshi et al.
10,721,489 B2	7/2020	Chen	2016/0227251 A1	8/2016	Lee et al.
10,841,593 B2	11/2020	Zhao	2016/0261885 A1	9/2016	Li et al.
11,134,257 B2	9/2021	Zhao et al.	2016/0286232 A1	9/2016	Li et al.
11,197,025 B2	12/2021	Ramasubramonian et al.	2016/0330457 A1	11/2016	Ye et al.
11,425,389 B2	8/2022	Deng et al.	2016/0360210 A1	12/2016	Xiu et al.
11,451,782 B2	9/2022	Deng et al.	2016/0373742 A1	12/2016	Zhao
11,451,784 B2	9/2022	Deng et al.	2016/0373743 A1	12/2016	Zhao et al.
11,457,207 B2	9/2022	Deng et al.	2016/0373770 A1	12/2016	Zhao et al.
11,457,220 B2	9/2022	Deng et al.	2016/0373782 A1	12/2016	Zhao et al.
11,463,702 B2	10/2022	Deng et al.	2016/0373785 A1	12/2016	Said et al.
11,463,729 B2	10/2022	Deng et al.	2017/0006293 A1	1/2017	Lee et al.
11,546,633 B2	1/2023	Deng	2017/0094313 A1	3/2017	Zhao et al.
11,606,570 B2	3/2023	Deng	2017/0094314 A1	3/2017	Zhao
11,659,185 B2	5/2023	Deng et al.	2017/0142418 A1	5/2017	Li et al.
11,805,275 B2	10/2023	Deng et al.	2017/0150183 A1	5/2017	Zhang et al.
11,831,877 B2	11/2023	Deng	2017/0272745 A1	9/2017	Liu et al.
11,943,444 B2	3/2024	Deng et al.	2017/0272748 A1	9/2017	Seregin et al.
2004/0202250 A1	10/2004	Kong et al.	2017/0302934 A1	10/2017	Ikai et al.
2008/0170615 A1	7/2008	Sekiguchi et al.	2017/0332098 A1	11/2017	Rusanovskyy et al.
2008/0187044 A1	8/2008	Kim et al.	2017/0359595 A1	12/2017	Zhang et al.
2010/0246676 A1	9/2010	Cao et al.	2017/0374369 A1	12/2017	Chuang et al.
2012/0057630 A1	3/2012	Saxena et al.	2018/0020223 A1	1/2018	King
2012/0069906 A1	3/2012	Sato	2018/0048889 A1	2/2018	Zhang et al.
2012/0140815 A1	6/2012	Zhou et al.	2018/0098064 A1	4/2018	Seregin et al.
2012/0177112 A1	7/2012	Guo et al.	2018/0103252 A1	4/2018	Hsieh et al.
2012/0201303 A1	8/2012	Yang et al.	2018/0103254 A1	4/2018	Tanaka et al.
2012/0236929 A1	9/2012	Liu	2018/0115787 A1	4/2018	Koo et al.
2013/0272391 A1	10/2013	Zheng et al.	2018/0160113 A1	6/2018	Jeong et al.
2013/0279569 A1	10/2013	Gao et al.	2018/0176601 A1	6/2018	Jeong et al.
2014/0078250 A1	3/2014	Zhang et al.	2018/0184082 A1	6/2018	Yoo et al.
2014/0098883 A1	4/2014	Hannuksela et al.	2018/0199062 A1	7/2018	Zhang et al.
2014/0105283 A1	4/2014	Li et al.	2018/0220138 A1	8/2018	He
2014/0286413 A1	9/2014	Joshi et al.	2018/0309984 A1	10/2018	Son et al.
2014/0286423 A1	9/2014	Chen et al.	2018/0324417 A1	11/2018	Karczewicz et al.
2015/0043637 A1	2/2015	Morigami et al.	2018/0324463 A1	11/2018	Zhang et al.
2015/0103892 A1	4/2015	Zhou et al.	2019/0052886 A1	2/2019	Chiang et al.
2015/0110172 A1	4/2015	Ye et al.	2019/0082194 A1	3/2019	Chang
2015/0256828 A1	9/2015	Dong et al.	2019/0104303 A1	4/2019	Xiu et al.
			2019/0116381 A1	4/2019	Lee et al.
			2019/0313130 A1	10/2019	Zhang
			2020/0045322 A1	2/2020	Ye et al.
			2020/0162749 A1	5/2020	Zhang et al.
			2020/0204814 A1	6/2020	Yoo et al.
			2020/0221099 A1	7/2020	Pham Van et al.
			2020/0221138 A1	7/2020	Lee
			2020/0260096 A1	8/2020	Ikai et al.
			2020/0322620 A1 *	10/2020	Zhao H04N 19/593
			2020/0344468 A1	10/2020	Lin et al.
			2020/0359033 A1	11/2020	Ramasubramonian et al.
			2020/0359037 A1	11/2020	Ramasubramonian et al.
			2020/0359050 A1	11/2020	Van Der Auwera et al.
			2021/0092405 A1	3/2021	Biatek et al.
			2021/0105499 A1 *	4/2021	Lee H04N 19/159
			2021/0195176 A1	6/2021	Kim
			2021/0211738 A1	7/2021	Yin et al.
			2021/0218960 A1	7/2021	Pfaff et al.
			2021/0227213 A1	7/2021	Kim et al.
			2021/0266568 A1	8/2021	Huo et al.
			2021/0297661 A1	9/2021	Deng et al.
			2021/0297671 A1	9/2021	Deng
			2021/0306654 A1	9/2021	Lee
			2021/0306666 A1 *	9/2021	Lee H04N 19/105
			2021/0314562 A1 *	10/2021	Kang H04N 19/139
			2021/0314599 A1	10/2021	Lee
			2021/0314618 A1 *	10/2021	Pfaff H04N 19/176
			2021/0321090 A1	10/2021	Deng et al.
			2021/0329248 A1	10/2021	Deng et al.
			2021/0344915 A1	11/2021	Wang et al.
			2021/0385438 A1	12/2021	Deng et al.
			2021/0392331 A1 *	12/2021	Pfaff H04N 19/13
			2022/0007029 A1	1/2022	Deng et al.
			2022/0014752 A1	1/2022	Deng et al.
			2022/0014783 A1	1/2022	Deng et al.
			2022/0014785 A1	1/2022	Deng

(56)

References Cited

U.S. PATENT DOCUMENTS

2022/0038698 A1 2/2022 Koo et al.
 2022/0058239 A1 2/2022 Pfaff et al.
 2022/0060745 A1 2/2022 Deng et al.
 2022/0124342 A1 4/2022 Deng et al.
 2022/0224898 A1 7/2022 Chiang et al.
 2022/0239919 A1 7/2022 Zhang et al.
 2022/0264124 A1 8/2022 Deng
 2023/0057982 A1 2/2023 Deng et al.

FOREIGN PATENT DOCUMENTS

CN 101646081 A 2/2010
 CN 101977316 A 2/2011
 CN 102377993 A 3/2012
 CN 102427530 A 4/2012
 CN 102598663 A 7/2012
 CN 102823254 A 12/2012
 CN 103098473 A 5/2013
 CN 103140877 A 6/2013
 CN 103391443 A 11/2013
 CN 103503452 A 1/2014
 CN 103636208 A 3/2014
 CN 104050688 A 9/2014
 CN 104126304 A 10/2014
 CN 104247425 A 12/2014
 CN 104488270 A 4/2015
 CN 105264888 A 1/2016
 CN 105338348 A 2/2016
 CN 105359532 A 2/2016
 CN 105723707 A 6/2016
 CN 105765974 A 7/2016
 CN 105765979 A 7/2016
 CN 105830440 A 8/2016
 CN 106464866 A 2/2017
 CN 106664416 A 5/2017
 CN 103959782 A 7/2017
 CN 103988437 A 8/2017
 CN 107046642 A 8/2017
 CN 107633538 A 1/2018
 CN 107646195 A 1/2018
 CN 107736023 A 2/2018
 CN 107736027 A 2/2018
 CN 107743705 A 2/2018
 CN 107749987 A 3/2018
 CN 107770534 A 3/2018
 CN 108028919 A 5/2018
 CN 108141596 A 6/2018
 CN 108605130 A 9/2018
 CN 108712650 A 10/2018
 CN 108712651 A 10/2018
 CN 109076237 A 12/2018
 CN 109076243 A 12/2018
 CN 109151468 A 1/2019
 CN 109196867 A 1/2019
 CN 109196869 A 1/2019
 CN 109479129 A 3/2019
 CN 109565602 A 4/2019
 CN 109691099 A 4/2019
 CN 109792521 A 5/2019
 CN 110708559 A 1/2020
 CN 113678453 B 11/2021
 CN 113785588 B 11/2023
 CN 113748676 B 5/2024
 CN ZL202080028042 X 5/2024
 CN 114051735 B 7/2024
 CN 114641997 B 7/2024
 EP 2237217 A2 10/2010
 EP 2557793 A1 2/2013
 EP 3244614 A1 11/2017
 EP 3301916 A1 4/2018
 EP 3935854 A1 1/2022
 EP 3935833 A4 6/2022
 ID P000094526 B 7/2024
 IN 201727039800 A 8/2014
 IN 552774 10/2024

IN 554495 11/2024
 IN 260616 2/2025
 IN 559576 2/2025
 JP 2004531154 A 10/2004
 JP 2018537898 A 12/2018
 JP 2020017970 A 1/2020
 JP 2022514870 A 2/2022
 JP 2022516846 A 3/2022
 JP 7403555 B2 6/2022
 JP 2022531902 A 7/2022
 JP 2022535726 A 8/2022
 JP 2022501896 A 9/2022
 JP 2022553789 A 12/2022
 JP 7303325 B2 6/2023
 JP 7524433 B2 7/2024
 JP 7612795 B2 1/2025
 KR 100323676 B1 2/2002
 KR 20090074886 A 7/2009
 KR 20100088034 A 8/2010
 KR 20160146591 A 12/2016
 KR 20180136555 A 12/2018
 KR 20190028575 A 3/2019
 KR 2019006732 A 6/2019
 KR 102698314 B1 8/2024
 KR 102736164 B1 11/2024
 KR 102744182 B1 12/2024
 KR 102744572 B1 12/2024
 MX 420196 1/2025
 RU 2595515 C2 8/2016
 RU 2669005 C2 12/2017
 SG 11202110936 P 11/2021
 SG 11202110939 T 11/2021
 SG 11202110939T A 11/2021
 TW 201729292 A 8/2017
 WO 2005086981 A2 9/2005
 WO 2012014461 A1 2/2012
 WO 2012172779 A1 12/2012
 WO 2015131388 A1 9/2015
 WO 2015165030 A1 11/2015
 WO 2015187978 A1 12/2015
 WO 2016205684 A1 12/2016
 WO 2016205718 A1 12/2016
 WO 2017019818 A1 2/2017
 WO 2017052174 A1 3/2017
 WO 2017058615 A1 4/2017
 WO 2017192898 A1 11/2017
 WO 2017204427 A1 11/2017
 WO 2017214420 A1 12/2017
 WO 2018052224 A1 3/2018
 WO 2018064948 A1 4/2018
 WO 2018080135 A1 5/2018
 WO 2018128417 A1 7/2018
 WO 2018132710 A1 7/2018
 WO 2018199675 A1 11/2018
 WO 2018208349 A1 11/2018
 WO 2018224004 A1 12/2018
 WO 2018236051 A1 12/2018
 WO 2019009540 A1 1/2019
 WO 2019022099 A1 1/2019
 WO 2019172676 A1 9/2019
 WO 2019194496 A1 10/2019
 WO 2020009556 A1 1/2020
 WO 2020127811 A3 6/2020
 WO 2020127881 A2 6/2020
 WO 2020205705 A1 10/2020
 WO 2020207498 A1 10/2020
 WO 2020207502 A1 10/2020
 WO 2020221373 A1 11/2020
 WO 2020222484 A1 11/2020
 WO 2020229394 A1 11/2020
 WO 2020233663 A1 11/2020
 WO 202046806 A1 12/2020

OTHER PUBLICATIONS

EP19160705—Priority Document for US 2021/0392331 (Year: 2019).
 Bross et al. “Versatile Video Coding (Draft 5),” Joint Video Experts
 Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG

(56)

References Cited**OTHER PUBLICATIONS**

11, 14th Meeting, Geneva, CH, Mar. 19-27, 2019, document JVET-N1001, 2019.

Bross et al. "Versatile Video Coding (Draft 7)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 16th Meeting, Geneva, CH, Oct. 1-11, 2019, document JVET-P2001, 2019.

Chiang et al. "CE6-Related: Constraint and Simplification for LFNST Signalling," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting, Gothenburg, SE, Jul. 3-12, 2019, document JVET-O0291, 2019.

Choi et al. "CE6-Related: Unified Matrix for Transform," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 12th Meeting, Macao, CN, Oct. 3-12, 2018, document JVET-L0060, 2018.

Choi et al. "CE6: Unified Matrix for Transform (Test 6-1.2a)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 13th Meeting, Marrakech, MA, Jan. 9-18, 2019, document JVET-M0200, 2019.

Filippov et al. "CE3-Related: Simplification of Matrix-based Intra Prediction (MIP)," Joint Video Experts Team (JVET) of ITU-T SG WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting, Gothenburg, SE, Jul. 3-12, 2019, document JVET-O0203, 2019.

He et al. "Cross-Check of JVET-K0172: CE3-Related: Gradient-Based Boundary Filtering in Intra Prediction," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 11th Meeting, Ljubljana, SI, Jul. 10-18, 2018, document JVET-K0458, 2018.

Koo et al. "CE6: Reduced Secondary Transform (RST) (CE6-3.1)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 14th Meeting, Geneva, CH, Mar. 19-27, 2019, document JVET-N0193, 2019.

Lainema, Jani. "CE6: Shape Adaptive Transform Selection (Test 3.1)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 13th Meeting, Marrakech, MA, Jan. 9-18, 2019, document JVET-M0303, 2019.

Lu et al. "AHG16: Simplification of Reshaper Information," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 14th Meeting, Geneva, CH, Mar. 19-27, 2019, document JVET-N0220, 2019.

Pfaff et al. "CE3: Affine Linear Weighted Intra Prediction (test 1.2.1, test 1.2.2)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 13th Meeting, Marrakech, MA, Jan. 9-18, 2019, document JVET-M0043, 2019.

Pfaff et al. "CE3: Affine Linear Weighted Intra Prediction (CE3-4.1, CE3-4.2)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 14th Meeting, Geneva, CH, Mar. 19-27, 2019, document JVET-N0217, 2019.

Pfaff et al. "Non-CE3: Harmonization of 8-Bit MIP with Unified-MPM and LFNST," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting, Gothenburg, SWE, Jul. 3-12, 2019, document JVET-O0485, 2019.

Ramasubramanian et al. "CE3-Related: Unification of MPM Derivation for Luma Intra Modes," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 14th Meeting, Geneva, CH, Mar. 19-27, 2019, document JVET-N0433, 2019.

Wang et al. "CE3-Related: A Unified MPM List for Intra Coding," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 14th Meeting, Geneva, CH, Mar. 19-27, 2019, document JVET-N0185, 2019.

Yoon et al. "Non-CE3: MIP Modifications," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting, Gothenburg, SE, Jul. 3-12, 2019, document JVET-O0202, 2019.

Zhang et al. "Non-CE3: Fixed MPMs for MIP," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting, Gothenburg, SE, Jul. 3-12, 2019, document JVET-O0255, 2019.

Zhou et al. "Study on the Development of Video Coding Standard VVC," Radio & TV Broadcast Engineering, ISSN 1002-4522, Dec. 31, 2018, 45(9):26-31.

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/084455 dated Jul. 10, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/084462 dated Jul. 6, 2020 (10 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/084472 dated Jul. 9, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/084486 dated Jul. 9, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/084488 dated Jul. 9, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/084499 dated Jun. 28, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/084505 dated Jul. 17, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/085050 dated Jul. 15, 2020 (13 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/088583 dated Jul. 1, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/088584 dated Aug. 3, 2020 (11 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/091542 dated Aug. 20, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/092905 dated Aug. 27, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/092906 dated Aug. 17, 2020 (11 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/094513 dated Sep. 7, 2020 (9 pages).

International Search Report and Written Opinion from International Patent Application No. PCT/CN2020/124317 dated Jan. 18, 2021 (13 pages).

Non Final Office Action from U.S. Appl. No. 17/343,980 dated Oct. 15, 2021.

Non Final Office Action from U.S. Appl. No. 17/342,951 dated Oct. 25, 2021.

Non Final Office Action from U.S. Appl. No. 17/343,041 dated Oct. 29, 2021.

Notice of Allowance from U.S. Appl. No. 17/342,996 dated Nov. 23, 2021.

Notice of Allowance from U.S. Appl. No. 17/399,879 dated Dec. 21, 2021.

Notice of Allowance from U.S. Appl. No. 17/478,511 dated Dec. 24, 2021.

Non Final Office Action from U.S. Appl. No. 17/453,135 dated Feb. 3, 2022.

Final Office Action from U.S. Appl. No. 17/343,041 dated Feb. 24, 2022.

Notice of Allowance from U.S. Appl. No. 17/342,951 dated Mar. 4, 2022.

Bross et al. "Versatile Video Coding (Draft 4)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 13th Meeting, Marrakech, MA, Jan. 9-18, 2019, document JVET-M1001, 2019.

Chen et al. "Algorithm Description for Versatile Video Coding and Test Model 5 (VTM 5)," Joint Video Experts Team (JVET) of

(56)

References Cited**OTHER PUBLICATIONS**

ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 14th Meeting: Geneva, CH, Mar. 19-27, 2019, document JVET-N1002, 2019.

Helle et al. "CE3-related: Non-Linear Weighted Intra Prediction (Cross-Check Report in JVET-K0262)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 11th Meeting: Ljubljana, SI, Jul. 10-18, 2018, document JVET-K0196, 2018.

Extended European Search Report from European Patent Application 20791054.8 dated Apr. 11, 2022.

Extended European Search Report from European Patent Application 20788667.2 dated May 9, 2022.

Extended European Search Report from European Patent Application 20813478.3 dated May 30, 2022.

Examination Report from Indian Patent Application 2021470558869 dated May 18, 2022.

Final Office Action from U.S. Appl. No. 17/399,879 dated Apr. 7, 2022.

Non Final Office Action from U.S. Appl. No. 17/479,360 dated Apr. 18, 2022.

Examination Report from Indian Patent Application 202147052371 dated May 25, 2022.

Examination Report from Indian Patent Application 202147052279 dated Jul. 13, 2022.

Extended European Search Report from European Patent Application 20786940.5 dated May 6, 2022.

Final Office Action from U.S. Appl. No. 17/453,135 dated Jun. 15, 2022.

Non Final Office Action from U.S. Appl. No. 17/731,472 dated Aug. 16, 2022.

Bross et al. "Versatile Video Coding (Draft 8)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 17th Meeting: Brussels, BE, Jan. 7-17, 2020, document JVET-Q2001, 2020.

Bross et al. "Versatile Video Coding (Draft 9)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference Apr. 15-24, 2020, document JVET-R2001, 2020.

Chen et al. "Algorithm Description for Joint Exploration Test Model 1," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 1st Meeting: Geneva, CH, Oct. 19-21, 2015, document JVET-A1001, 2015.

Fernandes, Felix C. "Low Complexity Rotational Transform," Joint Collaborative Team on Video Coding (JCT-VC) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 3rd Meeting, Guangzhou CN, Oct. 7-15, 2010, document JCTVC-C096, 2010.

Lin et al. "Non-CE3: Simplifications of MIP," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 15th Meeting: Gothenburg, SE, Jul. 3-12, 2019, document JVET-O0312, 2019.

Murakami et al. "High Efficiency Video Coding," HVEC / H.265, 2013.

Pfaff et al. "MIP for All Channels in the Case of 4:4:4-13 Chroma Format and of Single Tree," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: Alpbach, AT, Apr. 15-24, 2020, document JVET-R0350, 2020.

Extended European Search Report from European Patent Application 2081929.2 dated Nov. 10, 2022.

Notice of Allowance from U.S. Appl. No. 17/453,135 dated Jan. 6, 2023.

Examination Report from Indian Patent Application 202147049480 dated Aug. 18, 2022.

Non Final Office Action from U.S. Appl. No. 17/646,412 dated Oct. 28, 2022.

Chen et al. "Algorithm Description for Versatile Video Coding and Test Model 6 (VTM 6)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 15th Meeting: Gothenburg, SE Jul. 3-12, 2019, document JVET-O2002, 2019.

Liu et al. "Non-Ce3: Up- Sampling with a Fixed Order in MIP," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 15th Meeting, Gothenburg, SE, Jul. 3-12, 2019, document JVET-O0571, 2019.

Pfaff et al. "Intra Prediction Modes Based on Neural Networks," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 10th Meeting, San Diego, US Apr. 10-20, 2018, document JVET-J0037, 2018.

Partial Supplementary European Search Report from European Patent Application 20880736.2 dated Feb. 13, 2023.

Extended European Search Report from European Patent Application 20810710.2 dated Mar. 16, 2023.

Extended European Search Report from European Patent Application 20880736.2 dated May 8, 2023.

Bross et al. "Versatile Video Coding (Draft3)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 12th Meeting, Macao, CN, Oct. 3-12, 2018, document JVET-L1001, 2018.

Choi et al. "CE3: Results on Multi-Combined Intra Prediction in JVET-J0024," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 11th Meeting: Ljubljana, SI, Jul. 10-18, 2018, document JVET-K0180, 2018.

Lainema, Jani. "CE6-Related: Latency Reduction for LFNST Signaling," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 16th Meeting, Geneva, CH, Oct. 1-11, 2019, document JVET-P0196, 2019.

"Supervised by Sakae Okubo, "H.265/HVEC Textbook"", First Edition, Inc., Oct. 21, 2013, pp. 38-48, 120-121, ISBN: 978-4-8443-3468-2H.65/HEVC text."

Notice of Allowance from U.S. Appl. No. 17/895,511 dated Jul. 6, 2023.

Huang et al. "CE2: Alignment of Affine Control-Point Motion Vector and Subblock Motion Vector (Test 2.-5.2)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 14th Meeting: Geneva, CH, Mar. 19-27, 2019, document JVET-N0257, 2019.

Van Der Auwera et al. "CE3: Summary Report on Intra Prediction and Mode Coding," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 14th Meeting: Geneva, CH, Mar. 19-27, 2019, document VET-N0023, 2019.

Notice of Allowance from U.S. Appl. No. 17/895,511 dated Aug. 7, 2023.

Fujisawa, Yukio, Special Collection RX Introductory Programming with a Microcontroller Chapter 3 Universal IO A-D Converter Tata Touch Panel Control Method, Interface, vol. 37, No. 5 (May 2011), CQ Publishing Co., Ltd., May 1, 2011, pp. 52—Yukiho Fujisawa, "Special feature: Embedded programming starting with RX microcontroller Chapter 3: Touch panel control method using general-purpose I/O and A-D converter", Interface, vol. 37, No. 5 (May 2011 issue), CQ Publishing Co., Ltd., May 1, 2011, pp. 52-57, ISSN: 0387-9569.

Yasumi, Yoshikawa, The Sum of How Computers Work, First Edition, Feb. 27, 2017, Shizuki Corporation, pp. 146-148, 27th, Coronasha Co., Ltd., ISBN: 978-4-339-02867-6.

Notice of Reasons for Refusal from Japanese Patent Application No. 2023-102125 dated May 21, 2024.

Substantive Examination Report from Phillipine Patent Application No. 1-2021-552534 mailed Apr. 15, 2024.

Notification to Grant Patent Right for Invention from Chinese Patent Application No. 202080028042.X dated Mar. 1, 2024.

Notification to Grant Patent Right for Invention from Chinese Patent Application No. 20208076251.1 dated Apr. 15, 2024.

Document: JVET-R2001-vA, Bross, B., "Versatile Video Coding (Draft 9)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference, Apr. 15-24, 2020, 524 pages.

Document: JVET-R0350-v1, PFAFF, J., et al., "MIP for all channels in the case of 4:4:4-chroma format and of single tree," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: Alpbach, AT, Apr. 15-24, 2020, 4 pages. Document: JVET-O0261, Deng, Z., et al., "Non-CE3: MIP with one-stage boundary sample downsampling," Joint Video Experts

(56)

References Cited**OTHER PUBLICATIONS**

Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 15th Meeting: Gothenburg, SE, Jul. 3-12, 2019, 4 pages.

Document: JVET-O0843-va, Chen, J., et al., "Cross-check of JVET-O0571(Non-CE3: Up-Sampling with a fixed order in MIP)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 15th Meeting: Gothenburg, SE, Jul. 3-12, 2019, 4 pages.

Chen J., et al., "Algorithm description for Versatile Video Coding and Test Model 5 (VTM 5)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 14th Meeting: Geneva, CH, Mar. 19-27, 2019, Document: JVET-N1002-v1, 71 Pages.

Bross B., et al., "Versatile Video Coding (Draft 3)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 12th Meeting: Macao, CN, Oct. 3-12, 2018, Document: JVET-L1001-v12, 19 Pages.

Bross B., et al., "Versatile Video Coding (Draft 7)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 16th Meeting: Geneva, CH, Oct. 1-11, 2019, Document: JVET-P2001-v9, 495 Pages.

Bross B., et al., "Versatile Video Coding (Draft 9)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 18th Meeting: by teleconference, Apr. 15-24, 2020, Document: JVET-R2001-vA, JVET-S0152-v5_SpecText, 531 Pages.

Chen J., et al., "Test Model 6 of Versatile Video Coding (VTM 6)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 15th Meeting: Gothenburg, SE, Jul. 3-12, 2019, Document: JVET-O2002-v1, 82 pages.

Chiang M-S., et al., "CE6-Related: Constraint and Simplification for LFNST Signaling," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 15th Meeting: Gothenburg, SE, Jul. 3-12, 2019, Document: JVET-O0291-v2, 424 Pages.

Document: JVET-N0023-v1, Van Der Auwera, G., et al., "CE3: Summary Report on Intra Prediction and Mode Coding," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 14th Meeting: Geneva, CH, Mar. 19-27, 2019, 18 pages.

Document: JVET-N0223-v2, Li, L., et al., "CE3-related; reference sample filtering simplification," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11 14th Meeting: Geneva, CH, Mar. 19-27, 2019, 5 pages.

Pfaff J., et al., "Non-CE3: Harmonization of 8-Bit MIP with Unified-MPM and LFNST," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting: Gothenburg, SWE, Jul. 3-12, 2019, Document: JVET-O0485-r2, 408 Pages.

Ramasubramanian A.K., et al., "CE3-Related: Unification of MPM Derivation for Luma Intra Modes," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 14th Meeting: Geneva, CH, Mar. 19-27, 2019, Document: JVET-N0433-v3, 4 Pages.

Zhang K., et al., "Non-CE3: Fixed MPMs for MIP," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting: Gothenburg, SE, Jul. 3-12, 2019, Document: JVET-O0255-v1, 10 Pages.

Bross B., et al., "Versatile Video Coding (Draft 7)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 16th Meeting: Geneva, CH, Oct. 1-11, 2019, Document: JVET-P2001-vA, 491 Pages.

Bross B., et al., "Versatile Video Coding (Draft 7)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 16th Meeting: Geneva, CH, Oct. 1-11, 2019, Document: JVET-P2001-vE, 494 Pages.

Canadian Office Action from Canadian Application No. 3,135,966 dated Oct. 12, 2023, 6 pages.

Canadian Office Action from Canadian Patent Application No. 3,135,944 dated Aug. 21, 2024, 3 pages.

Communication Pursuant to Article 94(3) EPC for European Application No. 20788567.4, mailed Nov. 3, 2023, 4 Pages.

Extended European Search Report for European Application No. 20787123.7, mailed May 6, 2022, 7 pages.

Extended European Search Report for European Application No. 20788326.5, mailed May 4, 2022, 7 pages.

Extended European Search Report for European Application No. 20788567.4, mailed May 9, 2022, 6 Pages.

Extended European Search Report for European Application No. 20798302.4, mailed Apr. 26, 2022, 11 Pages.

Flippov A., "CE3-Related: Simplification of Matrix-Based Intra Prediction (MIP)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29/WG 11, 15th Meeting: Gothenburg, SE Jul. 3-12, 2019, Document: JVET-O0203-v1, 775 Pages.

Final Office Action for U.S. Appl. No. 17/399,879, mailed Apr. 7, 2022, 20 Pages.

Final Office Action for U.S. Appl. No. 17/646,412, mailed Apr. 28, 2023, 34 Pages.

Final Office Action for U.S. Appl. No. 17/897,878, mailed Dec. 14, 2023, 62 Pages.

Final Office Action from U.S. Appl. No. 18/319,658 dated Sep. 3, 2024, 29 Pages.

Foreign Communication From A Related Counterpart Application, Extended European Search Report for European Application No. 20810710.2, mailed Mar. 16, 2023, 20 Pages.

Japanese Office Action from Japanede Patent Application No. 2023-190715 dated Sep. 10, 2024, 12 pages.

Japanese Office Action from Japanede Patent Application No. 2023-198738 dated Aug. 27, 2024, 8 pages.

Mexican Office Action from Mexican Patent Application No. MX/A/2021/012229 dated Aug. 23, 2024, 10 pages.

Mexican Office Action from Mexican Patent Application No. MX/A/2021/012230 dated Aug. 16, 2024, 10 pages.

Non-Final Office Action for U.S. Appl. No. 17/479,338, mailed Jan. 21, 2022, 19 Pages.

Non-Final Office Action for U.S. Appl. No. 17/479,360, mailed Apr. 18, 2022, 42 Pages.

Non-Final Office Action for U.S. Appl. No. 17/942,552, mailed Mar. 16, 2023, 43 Pages.

Notice of Allowance for U.S. Appl. No. 17/646,412, mailed Jul. 14, 2023, 15 Pages.

Notice of Allowance for U.S. Appl. No. 17/731,472, mailed Dec. 20, 2022, 17 Pages.

Notice of Allowance for U.S. Appl. No. 17/895,511, mailed Oct. 26, 2023, 10 Pages.

Notice of Allowance for U.S. Appl. No. 17/942,552, mailed Jul. 13, 2023, 13 Pages.

Notice of Allowance for U.S. Appl. No. 17/942,552, mailed Oct. 19, 2023, 16 Pages.

Notice of Allowance for U.S. Appl. No. 17/942,552, mailed Aug. 8, 2023, 25 Pages.

Notice of Allowance for U.S. Appl. No. 18/073,191, mailed on Oct. 2, 2024, 7 pages.

Notice of Allowance for U.S. Appl. No. 18/295,553, mailed Feb. 14, 2024, 16 Pages.

Notice of Hearing for Indian Application No. 202147044962, mailed Oct. 25, 2024, 3 pages.

Notice of Reasons for Refusal for Japanese Application No. 2021-559287, mailed Oct. 29, 2024, 8 pages.

Notice of Reasons for Refusal for Japanese Application No. 2023-102125, mailed Oct. 22, 2024, 15 pages.

Notice of Reasons for Refusal for Japanese Application No. 2023-197883, mailed Oct. 29, 2024, 33 pages.

Notification of Initial Substantive Examination for Indonesian Application No. P00202108306, mailed Sep. 8, 2023, 4 Pages.

Office Action for Canadian Application No. 3,135,944, mailed Oct. 26, 2023, 7 Pages.

Priority Document EP1916705 Mar. 5, 2019, 16 pages.

Ramasubramanian A.K., et al., "CE3-Related: Unification of MPM Derivation for Luma Intra Modes," Joint Video Experts Team

(56)

References Cited**OTHER PUBLICATIONS**

(JVET) of ITU-T SG 16 WP 3 and ISO/IEC 29/WG 11, 14th Meeting: Geneva, CH, Mar. 19-27, 2019, Document: JVET-N0433-v1, 30 Pages.

Singapore Office Action Singapore Patent Application No. 11202112517Q dated Jun. 25, 2024, 11 pages.

Wang B., et al., "CE3-Related: A Unified MPM List for Intra Mode Coding," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC 29/WG 11, 14th Meeting: Geneva, CH, Mar. 19-27, 2019, Document: JVET-N0185-r2, 306 Pages.

Bross B., et al. "Versatile Video Coding (Draft 3)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC 29/WG 11, 12th Meeting: Macao, CN, Oct. 3-12, 2018, Document: JVET-L1001-v8, 229 Pages.

Chen et al. "Algorithm Description for Joint Exploration Test Model 1," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC 29/WG 11, 14th Meeting: Geneva, CH, Oct. 19-21, 2015, Document No. JVET-A1001, 2015.

Document: JVET-O0255-v1, Zhang K., et al., "Non-CE3: Fixed MPMs for MIP," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC 29/WG 11, 14th Meeting: Gothenburg, SE, Jul. 3-12, 2019, 8 Pages.

Document: JVET-O0485-r2, Pfaff, J., et al., "Harmonization of 8-Bit MIP with Unifies-MPM and LFNST," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC 29/WG 11, 15th Meeting: Gothenburg, SE, Jul. 3-12, 2019, 4 Pages.

Document: JVET-R2001-v1, Bross, B., et al., "Versatile Video Coding (Draft)," Joint Video Experts Team (JVET) of ITU-T SG 16 WP 3 and ISO/IEC 29/WG 11, 18th Meeting: by teleconference, Apr. 15-24, 2020, 519 pages.

Edited by Murakami, Tokumichi (and 2 others), "Efficient Image Coding Technology HEVC/H.265 and Its Application", 1st Edition, ohm-Sha, Ltd., Feb. 25, 2013, * 69-73,95-98 Page, ISBN: 978- [4, Jan. 2006, 274] to [21329], 29 pages. With English Translation. Japanese Office Action from Japanese Patent Application No. 2023-150924 dated Jun. 18, 2024, 32 pages.

Mexican Office Action from Mexican Application No. MX/A/2021/012230 dated Aug. 30, 2024, 7 pages.

Notice of Allowance from U.S. Appl. No. 17/343,086 dated Aug. 30, 2024, 19 pages.

Office Action for European Patent Application No. 20787123.7, mailed on Oct. 23, 2023, 5 pages.

Priority Document EP19160705 (Year: 2019), 111 pages.

Tokumichi M., "High Efficiency Video Coding/H.265 and Its Application," Feb. 25, 2013, pp. 1794-1795, 2 Pages.

Notice of Allowance for U.S. Appl. No. 18/174,168, mailed Nov. 14, 2024, 14 pages.

Notice of Allowance for U.S. Appl. No. 18/295,553, mailed Dec. 4, 2024, 29 pages.

Zhang, L., et al., "Multiple Direct Mode for Intra Coding," VCIP 2017, Dec. 10-13, 2017, St Petersburg, U.S.A., 4 pages.

Li, S., et al., "A Linear Model For YUV 4:2:0 Chroma Intra Prediction" IEEE, 2019, 5 pages.

Notice of Allowance from Japanese Patent Application No. 2023-150924 dated Nov. 26, 2024, 6 pages.

Notice of Allowance from U.S. Appl. No. 18/295,553 dated Jan. 24, 2025, 14 pages.

Examination Report No. 1 for Australian Application No. 2020257009, mailed Dec. 11, 2024, 3 pages.

Notice of Allowance for U.S. Appl. No. 18/319,658, mailed Dec. 11, 2024, 25 pages.

Office Action from Japanese Patent Application No. 2021-570148 dated Feb. 18, 2025, 10 pages.

* cited by examiner

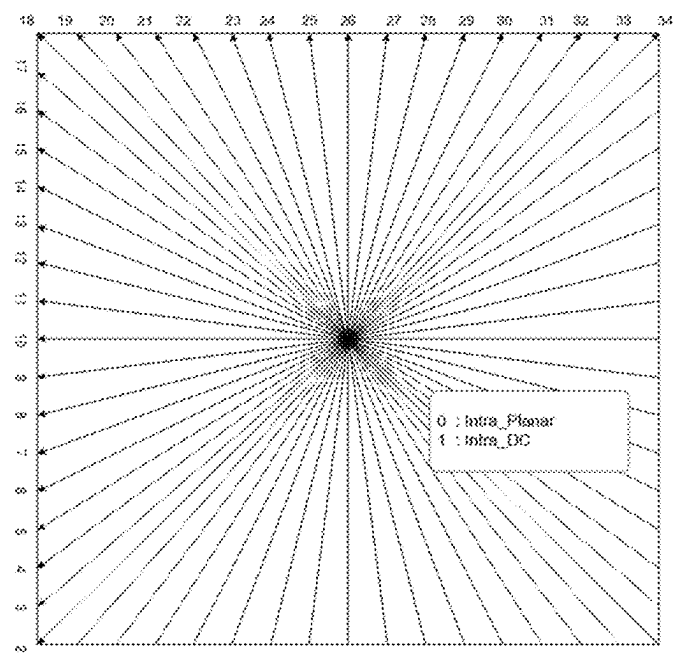


FIG. 1

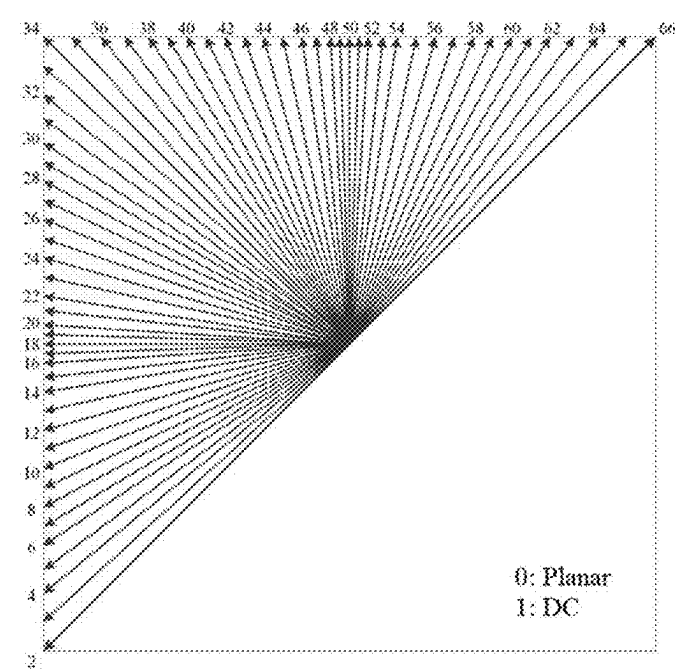


FIG. 2

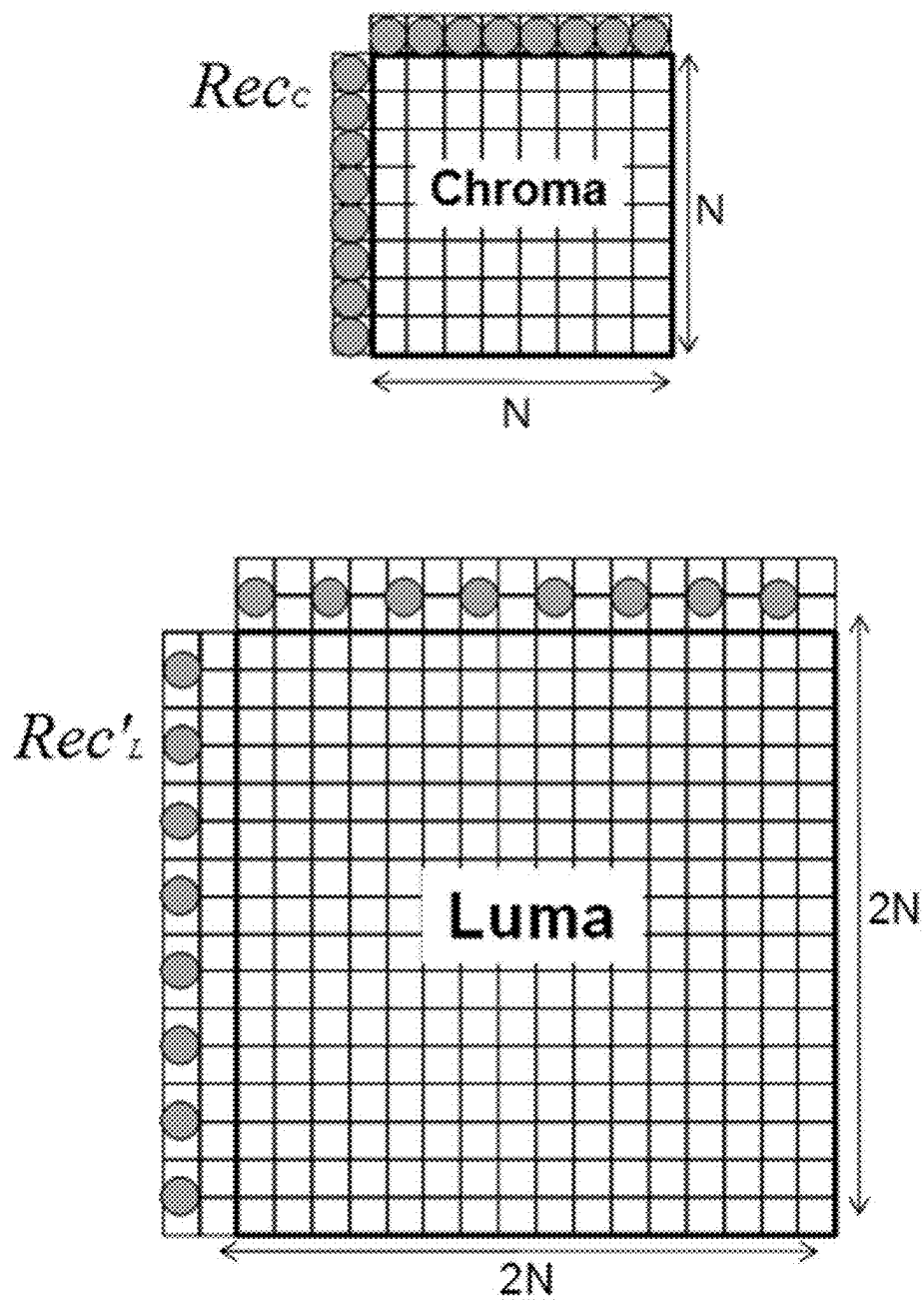


FIG. 3

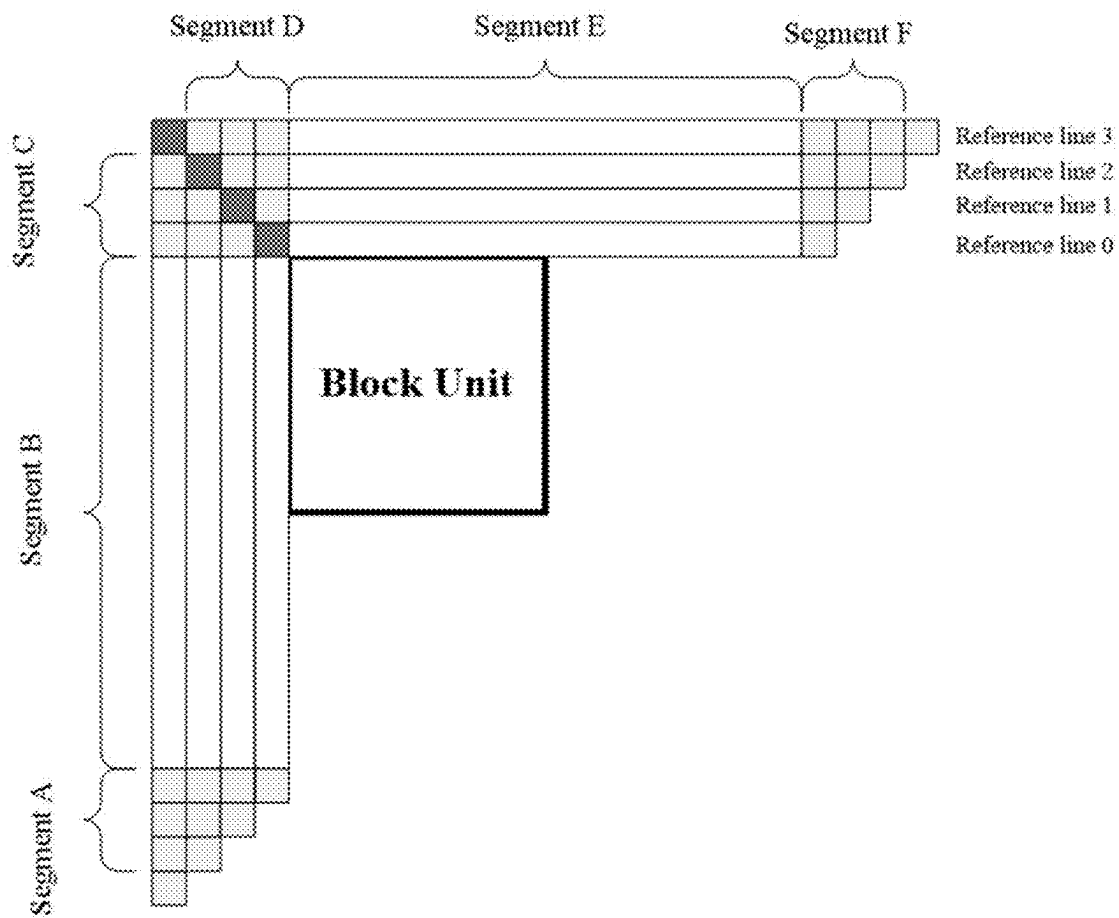


FIG. 4

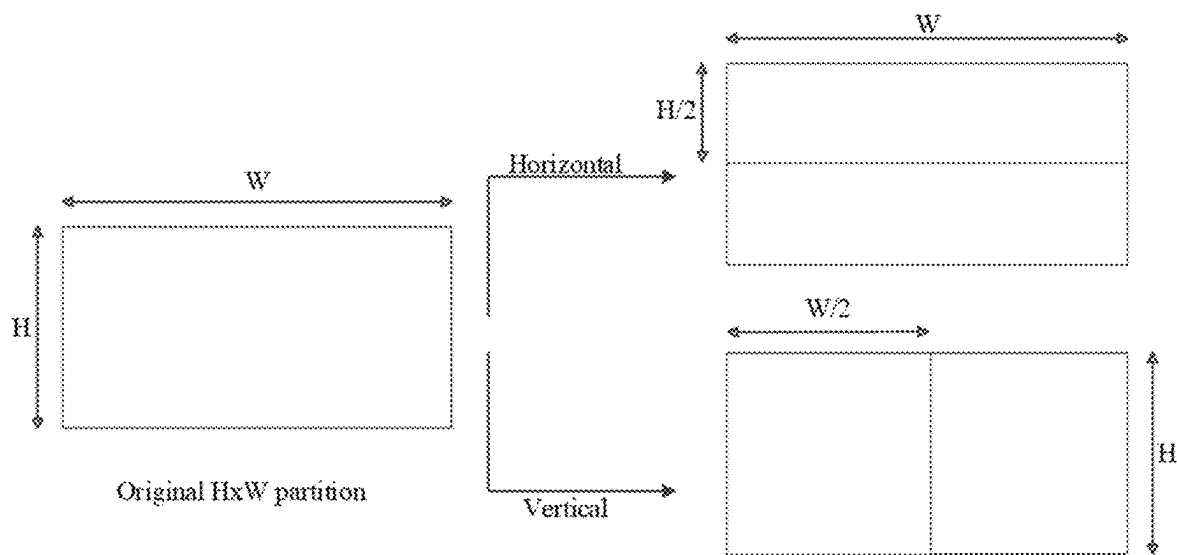


FIG. 5A

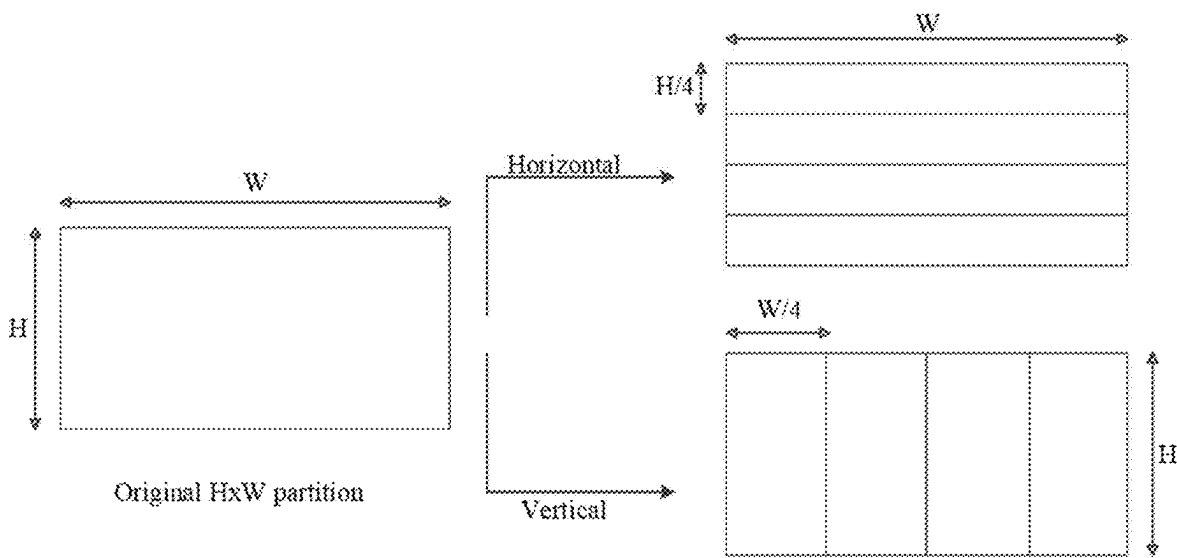


FIG. 5B

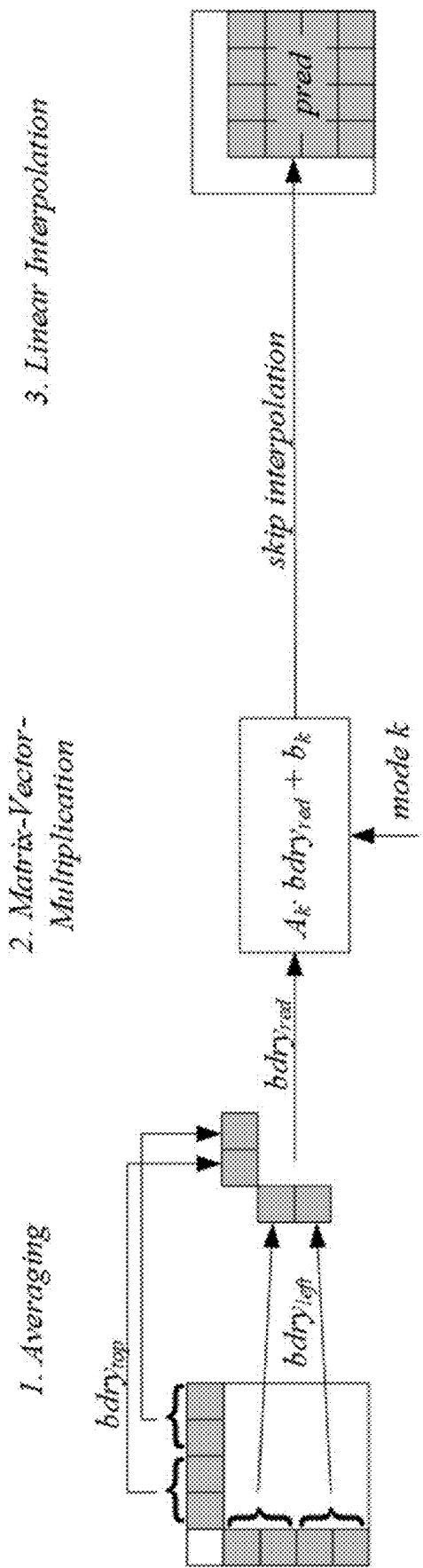


FIG. 6

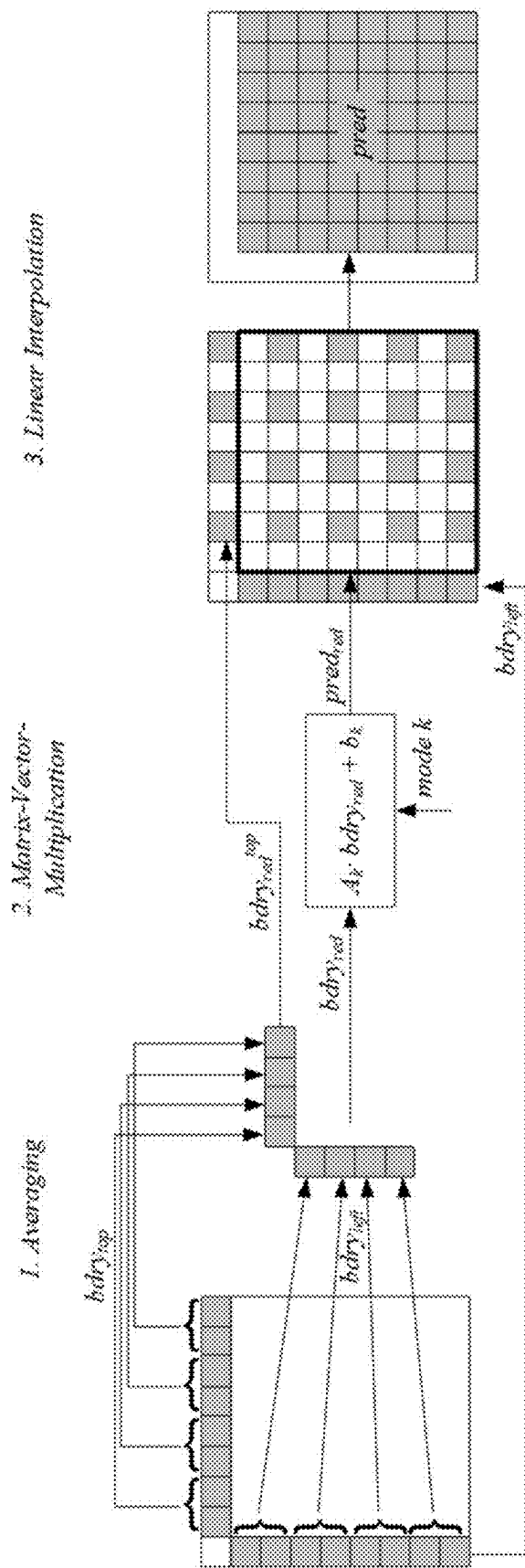


FIG. 7

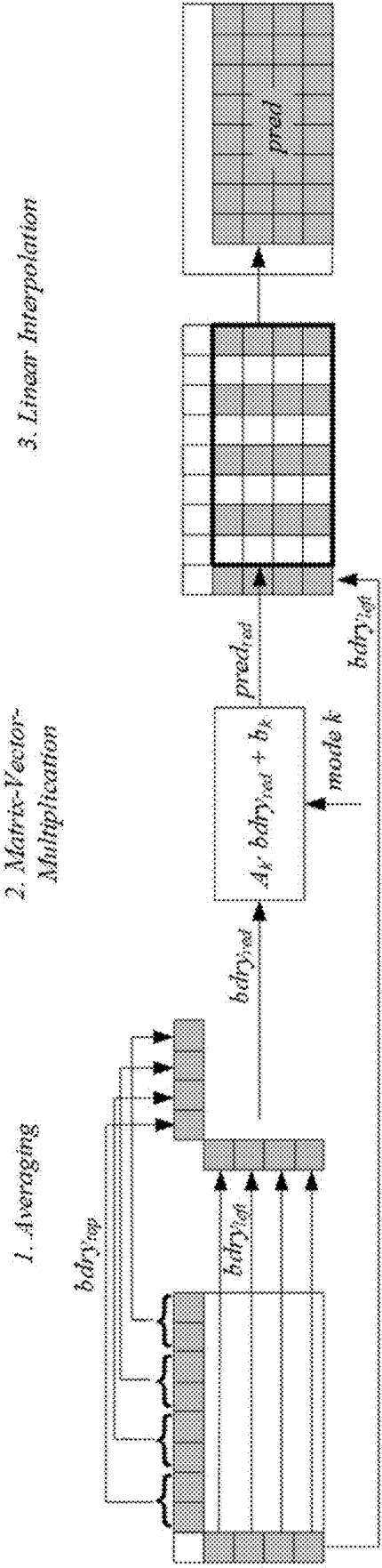


FIG. 8

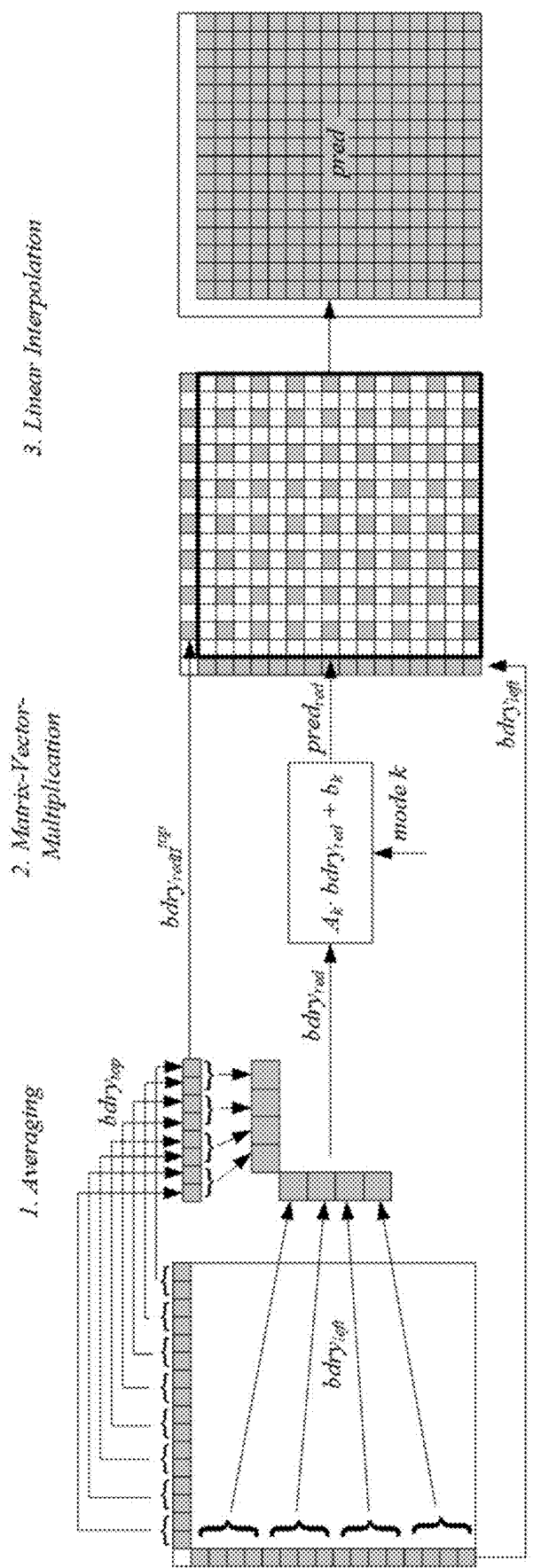


FIG. 9

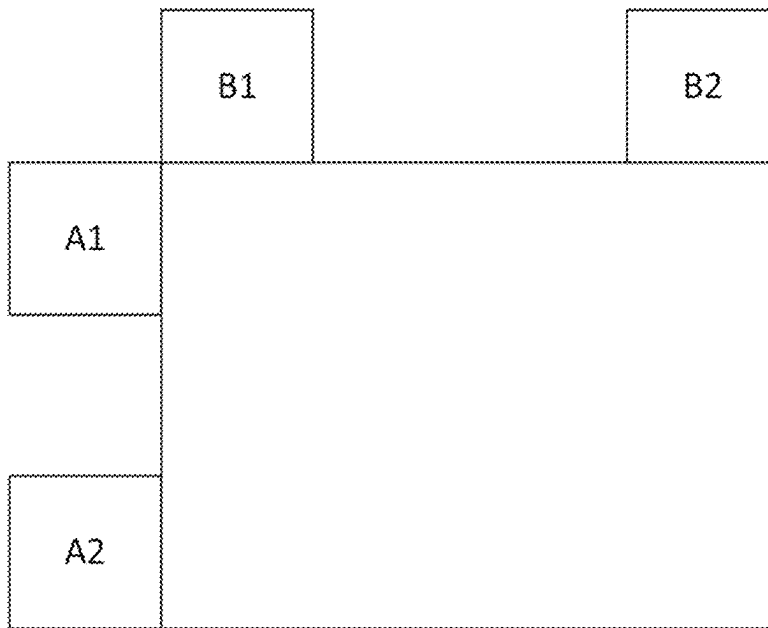


FIG. 10

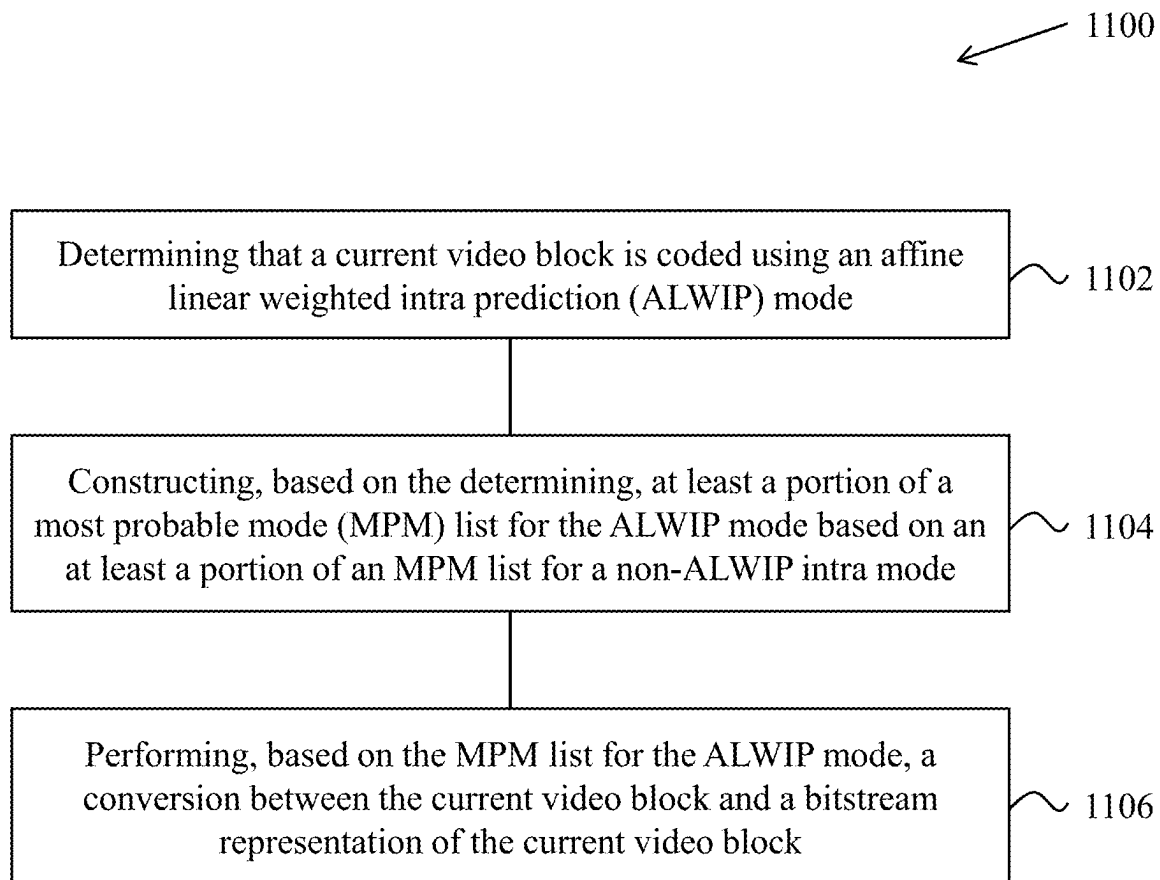


FIG. 11

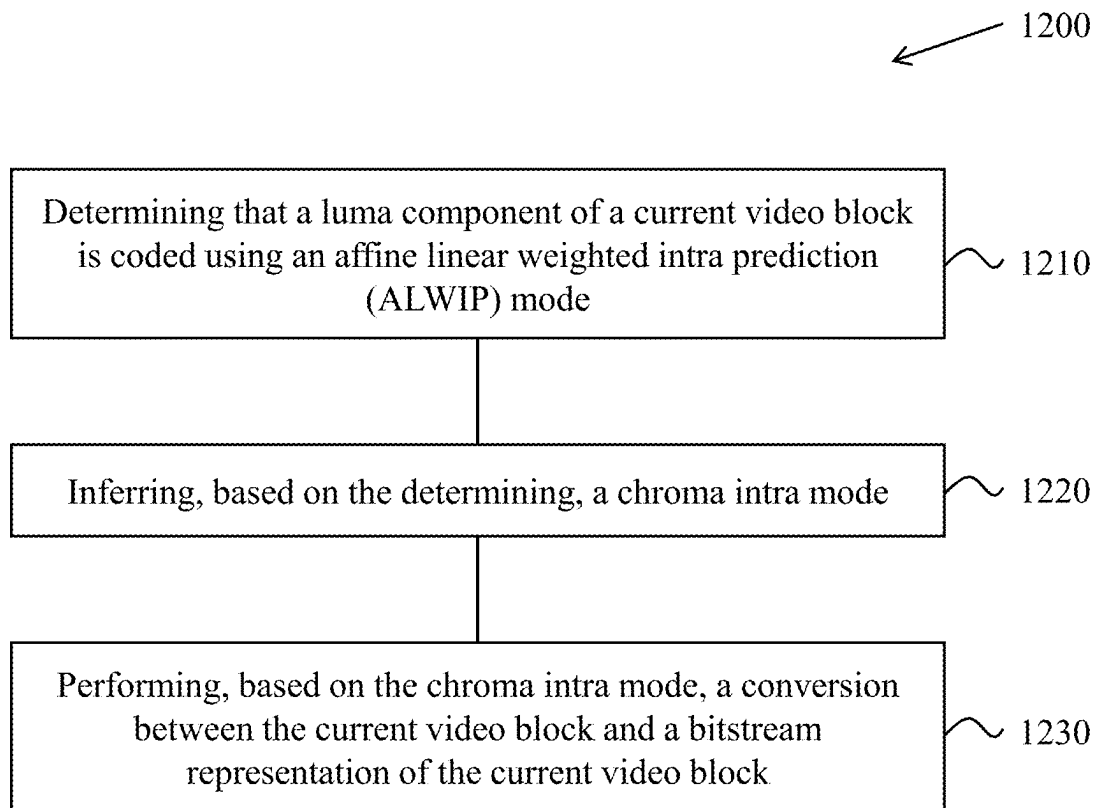


FIG. 12

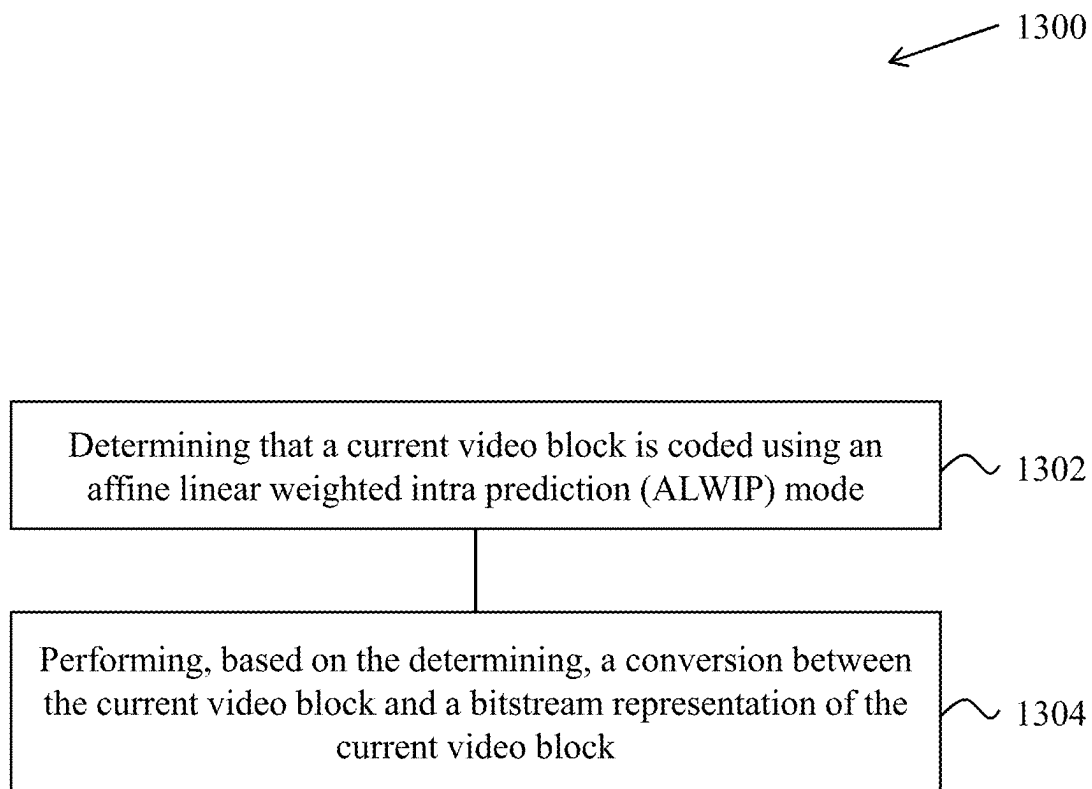


FIG. 13

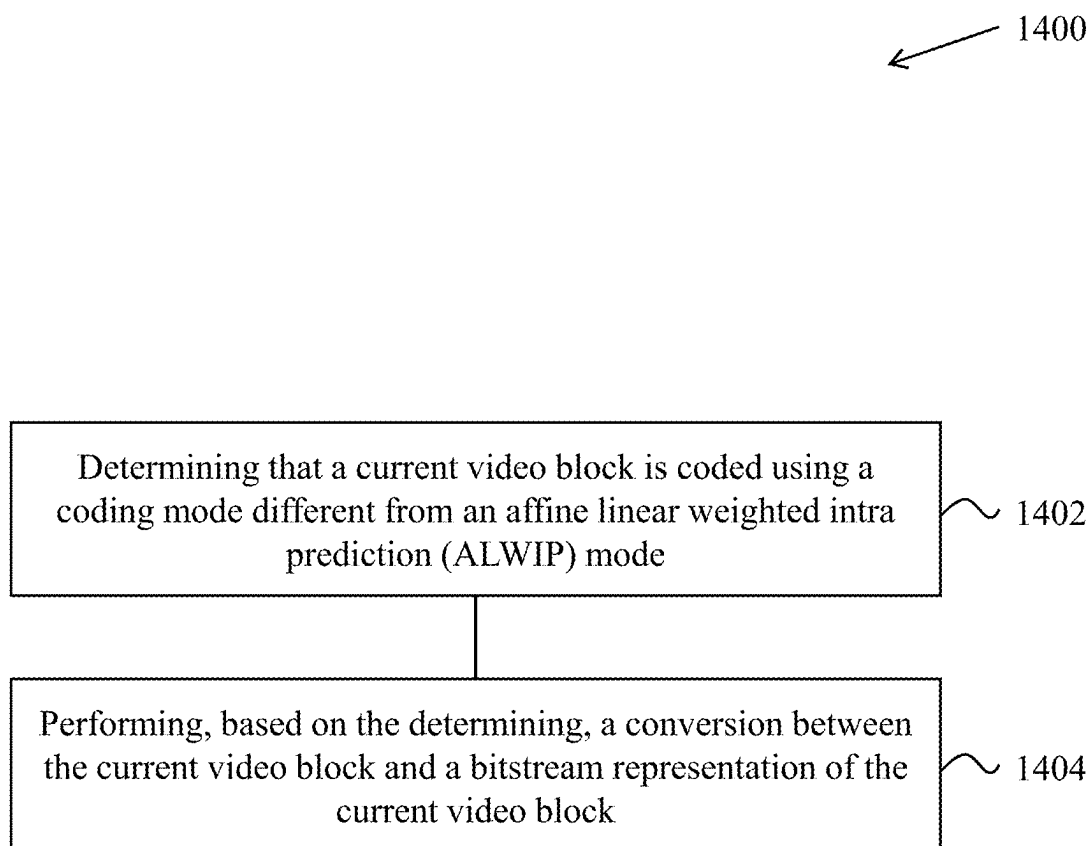


FIG. 14

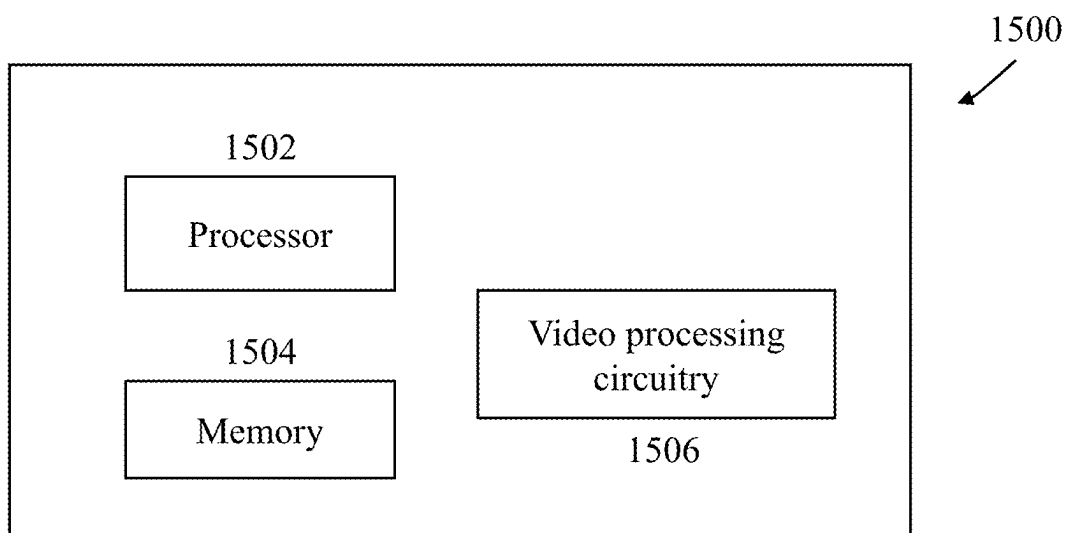


FIG. 15

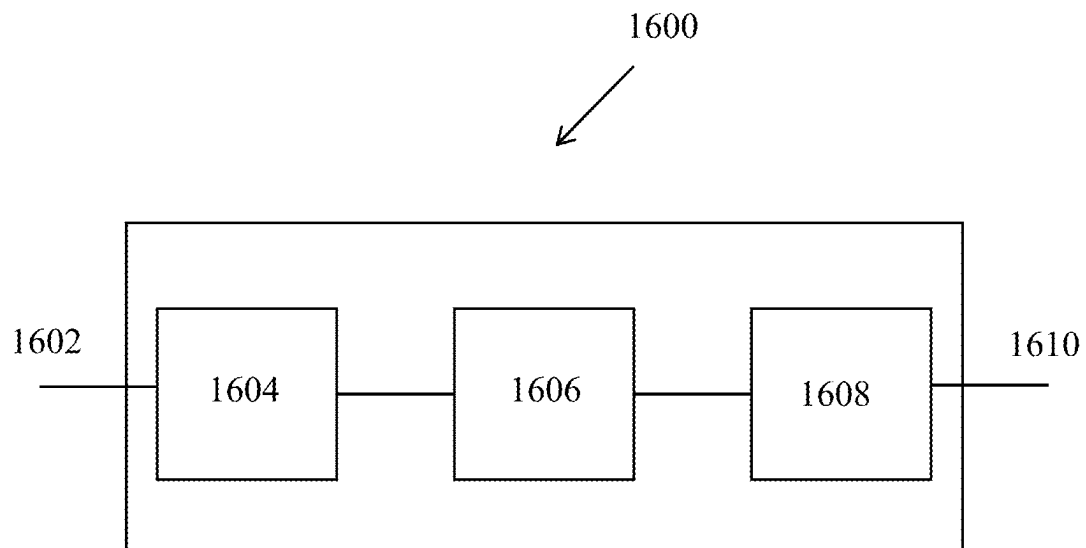


FIG. 16

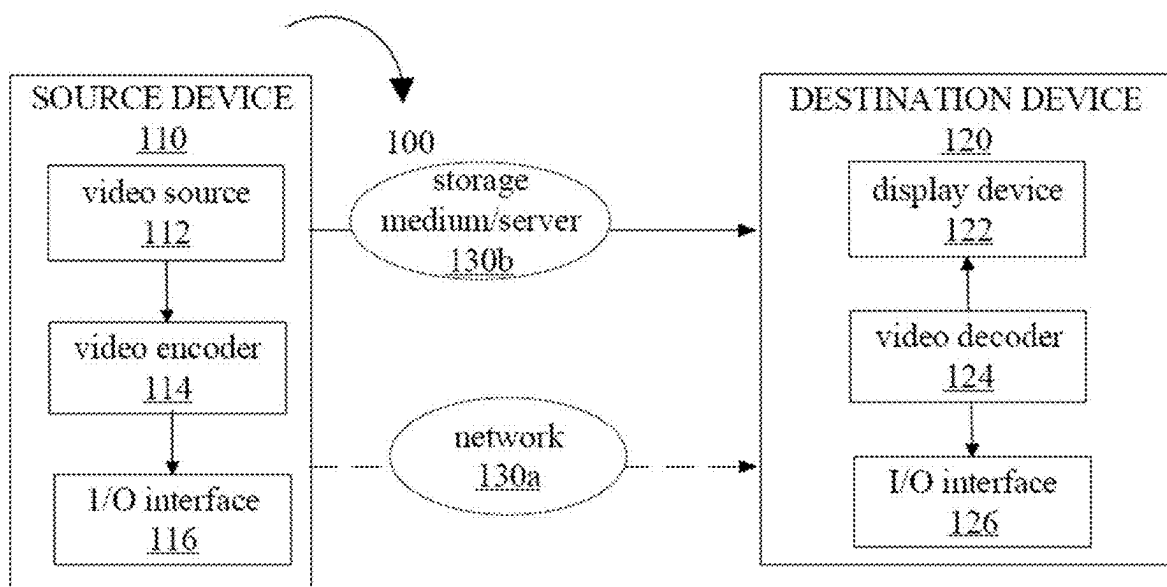


FIG. 17

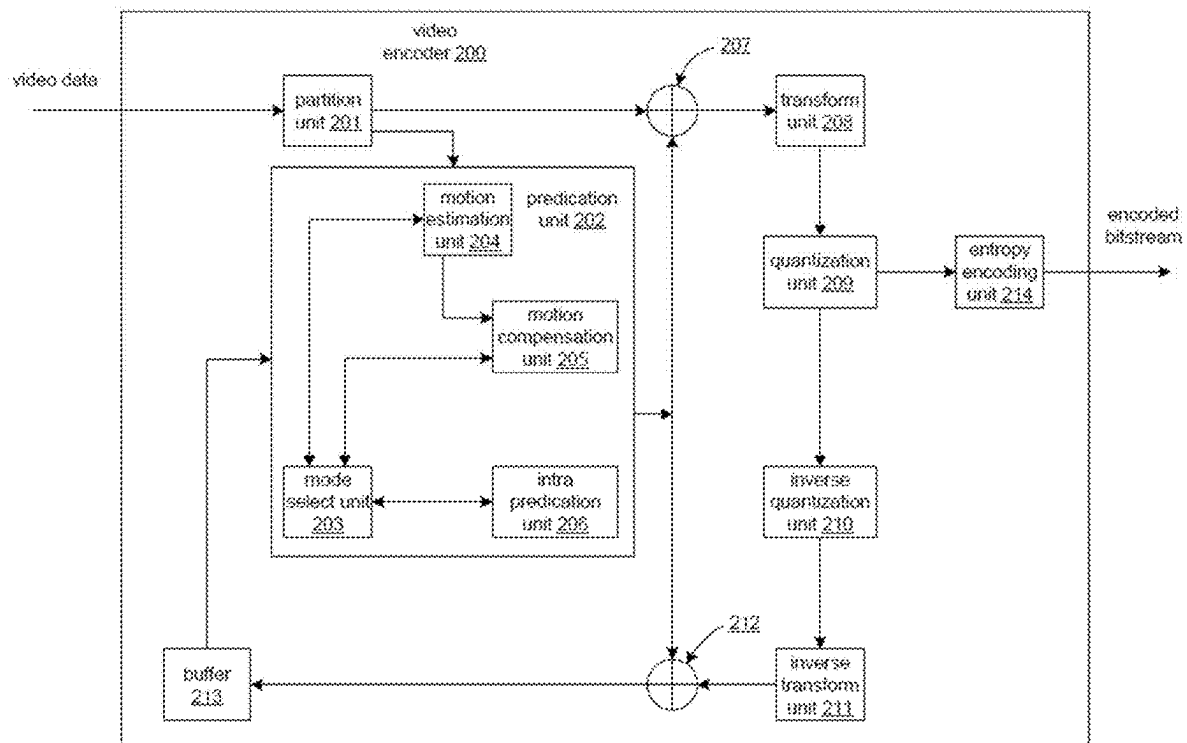


FIG. 18

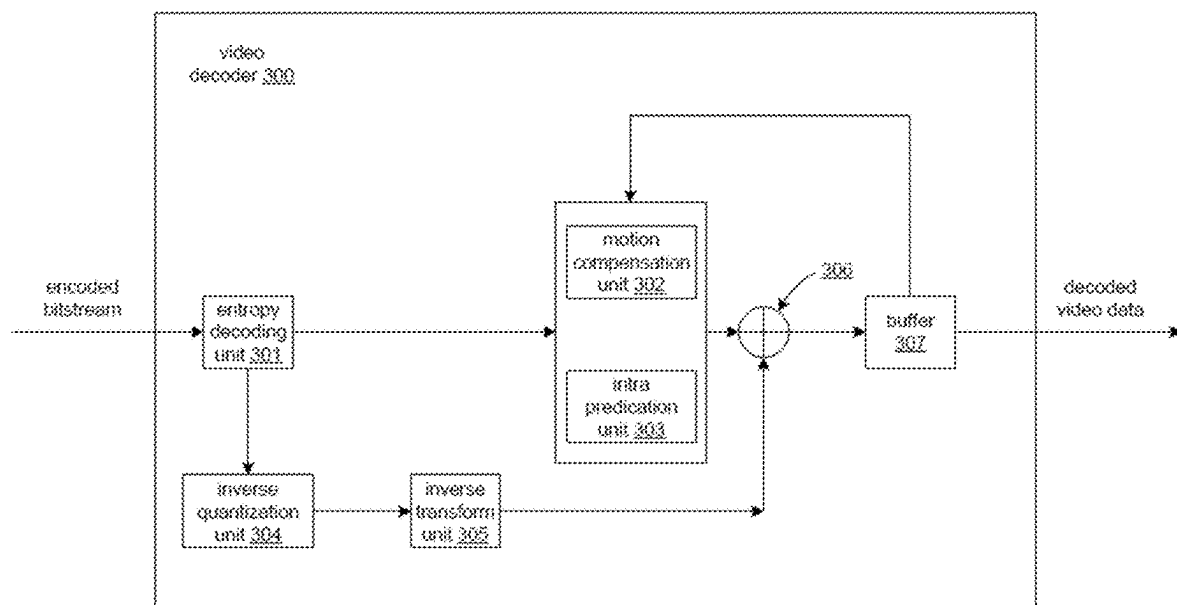


FIG. 19

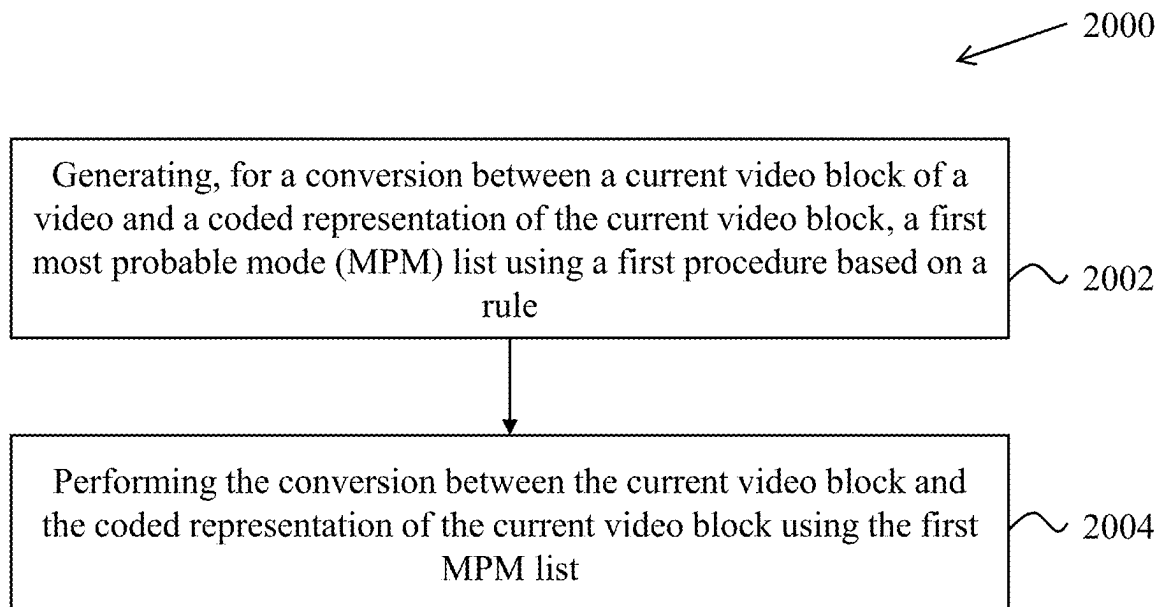


FIG. 20

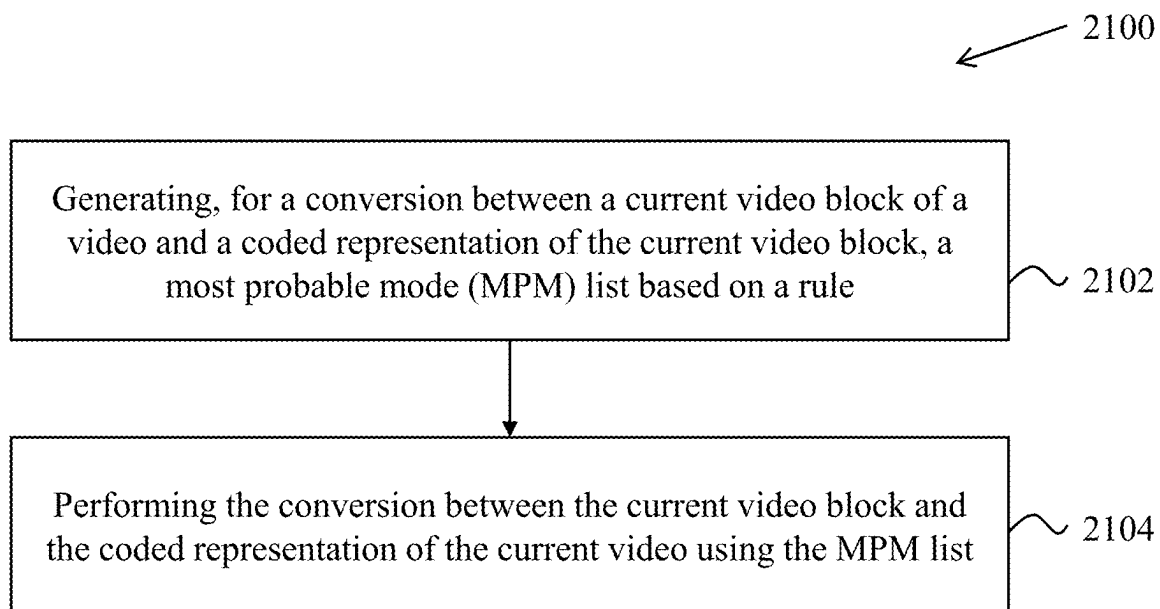


FIG. 21

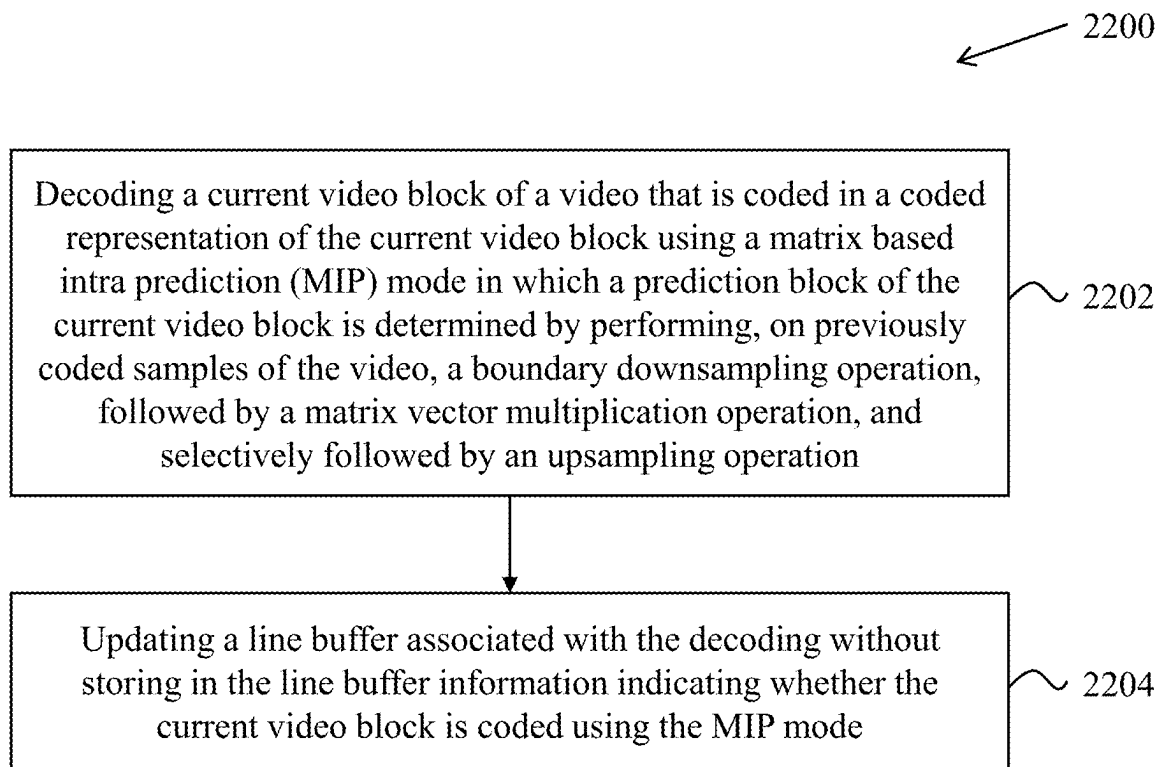


FIG. 22

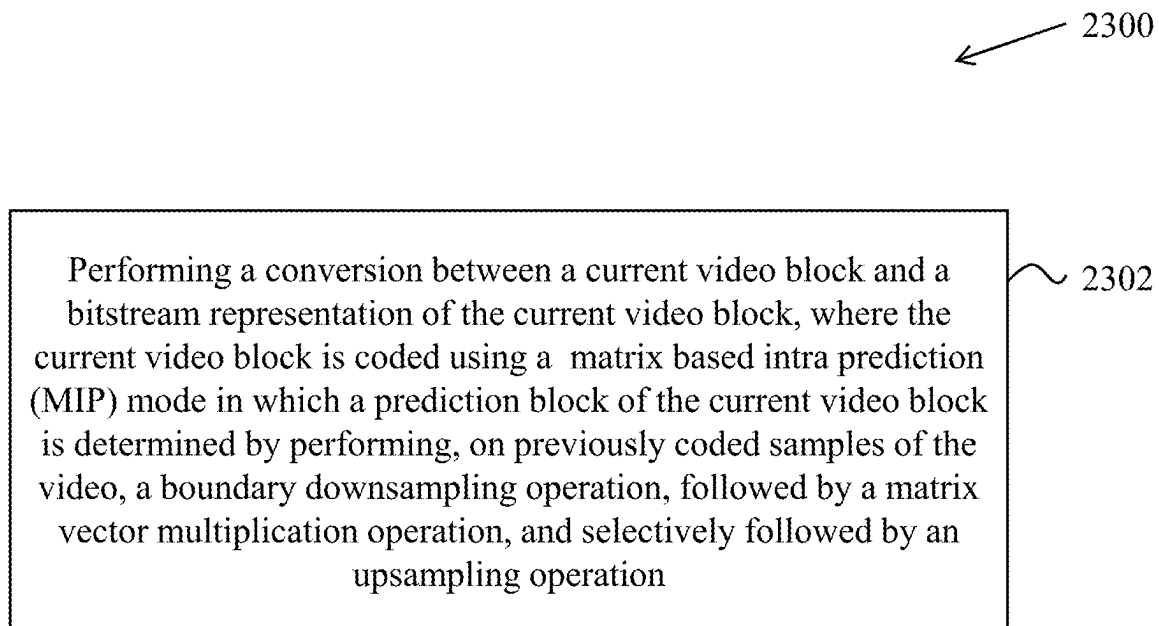


FIG. 23

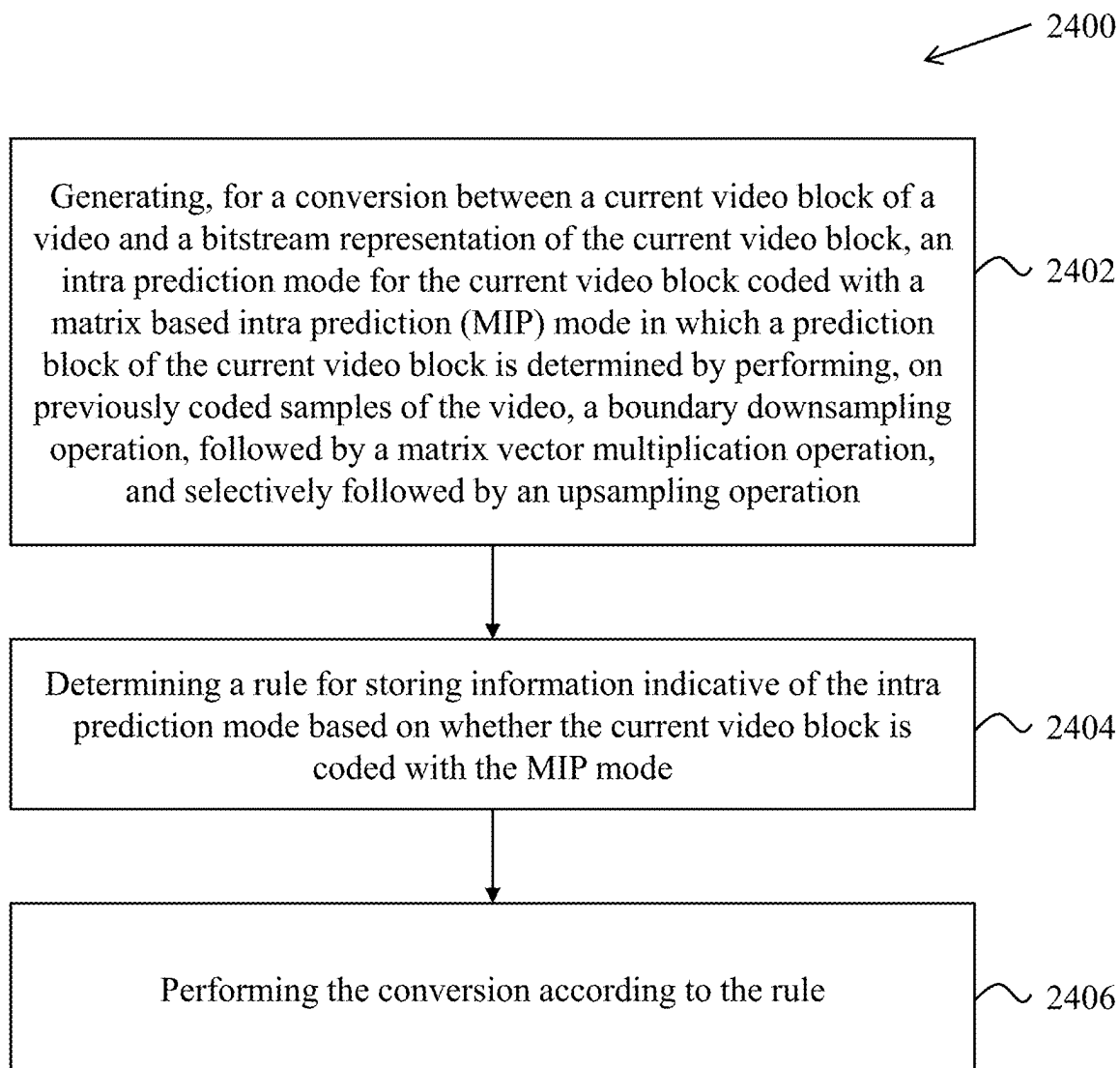


FIG. 24

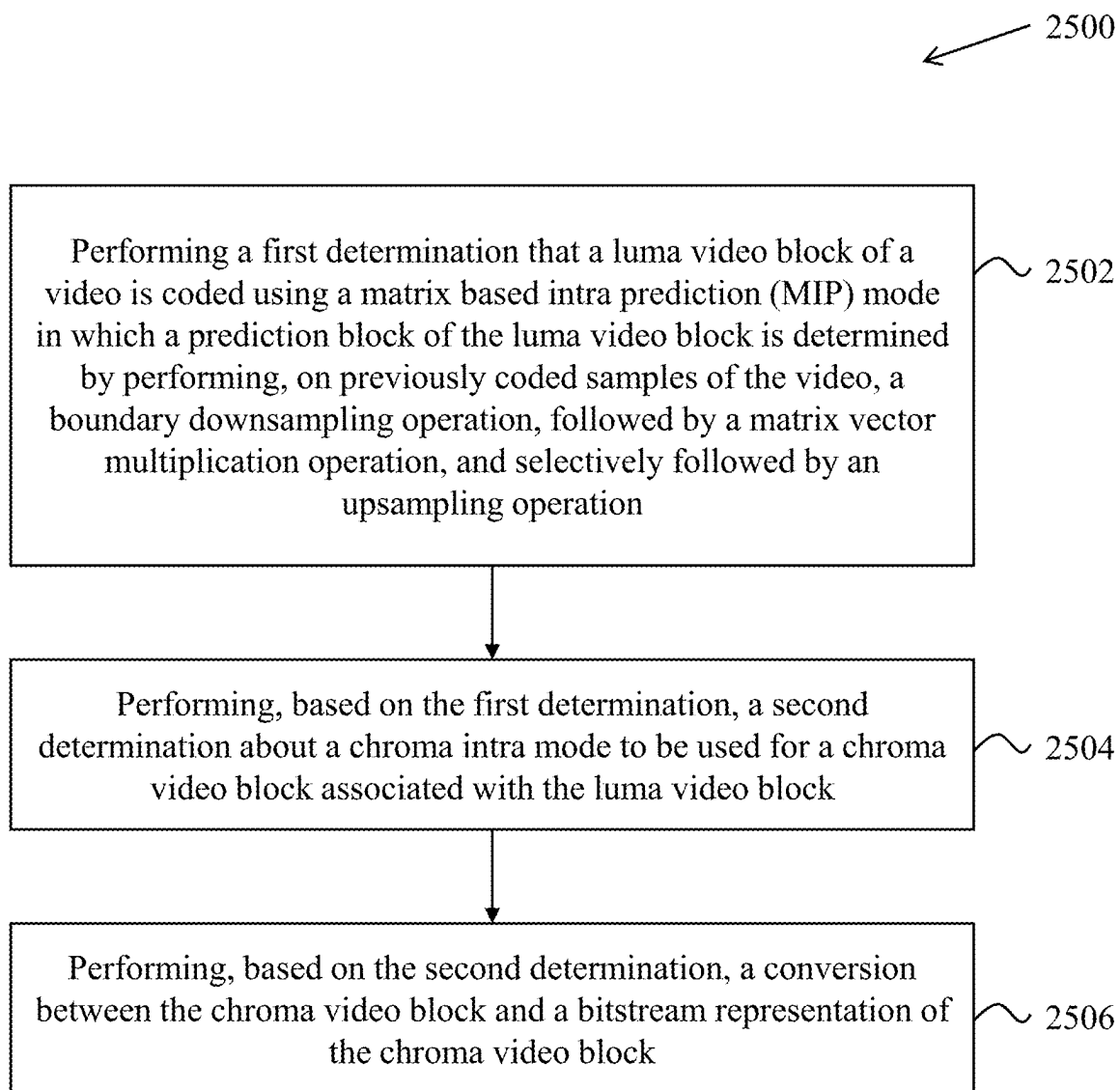


FIG. 25

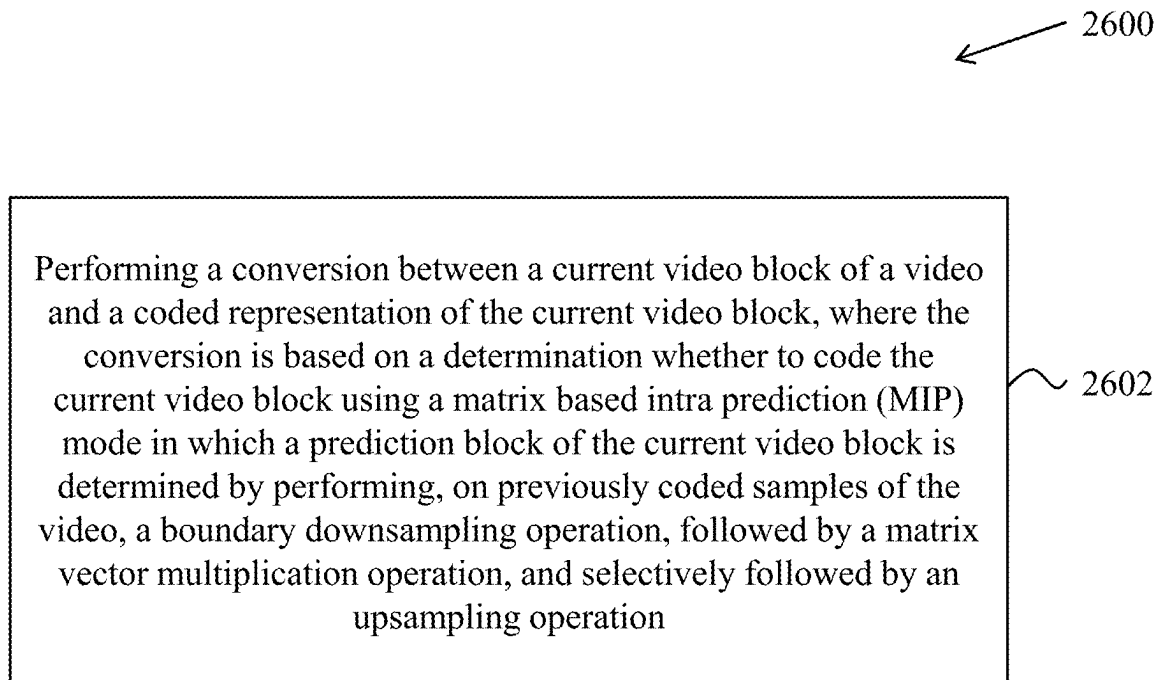


FIG. 26

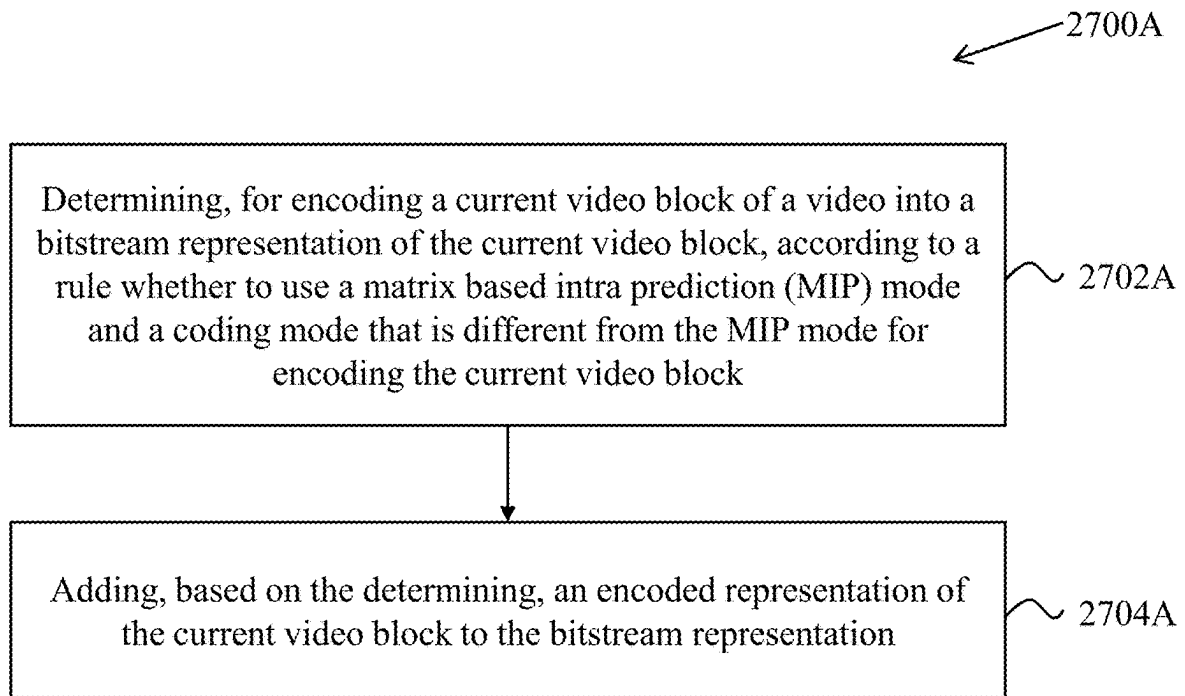


FIG. 27A

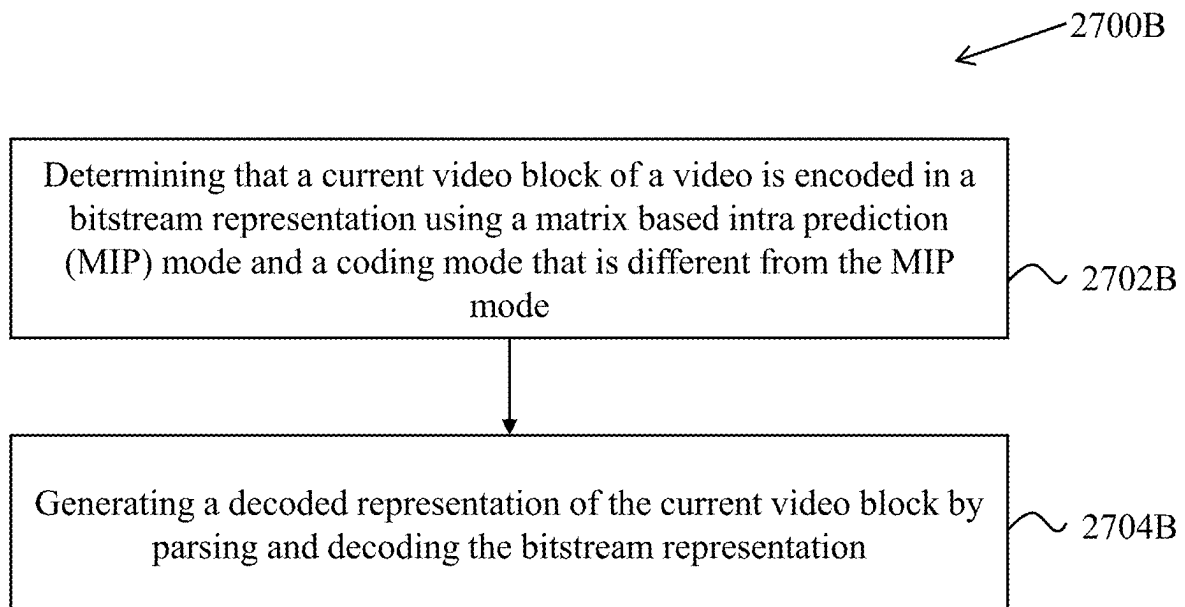


FIG. 27B

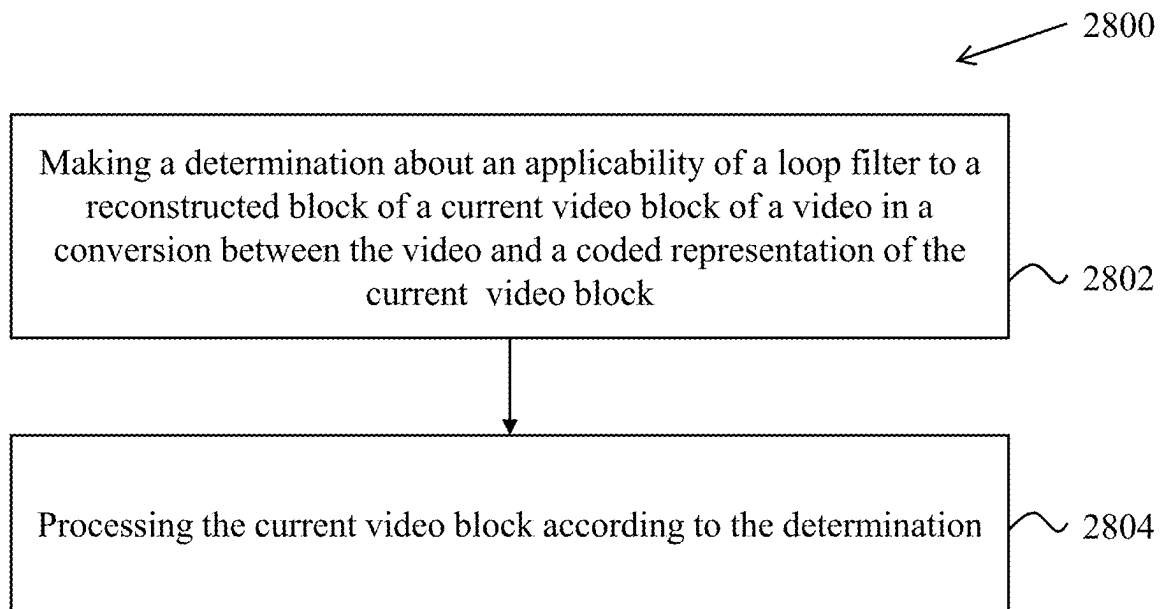


FIG. 28

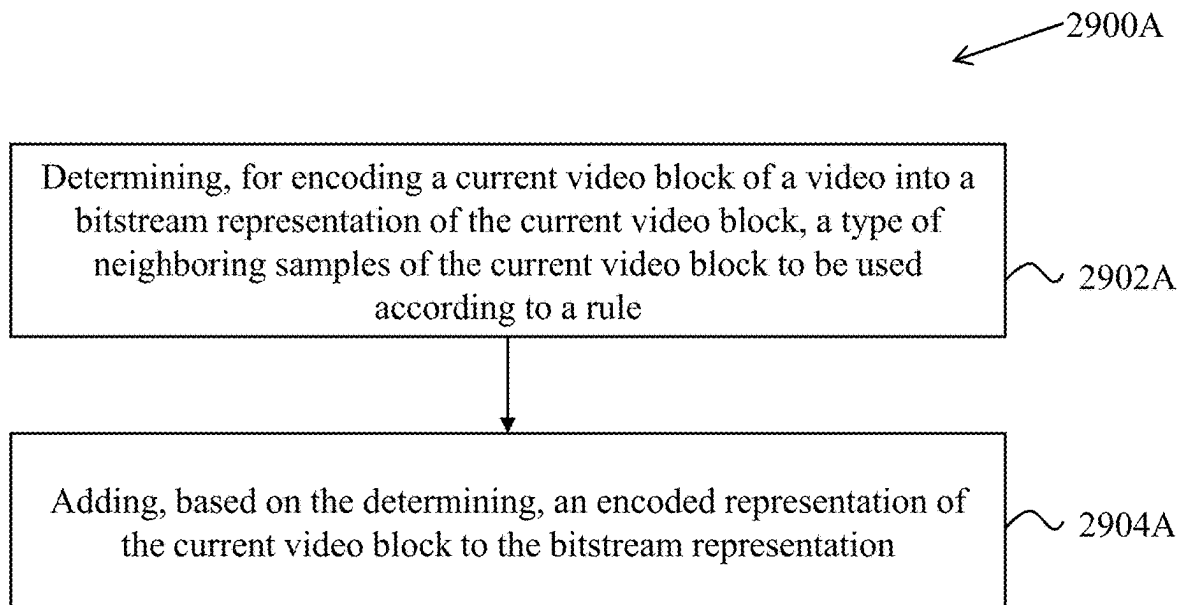


FIG. 29A

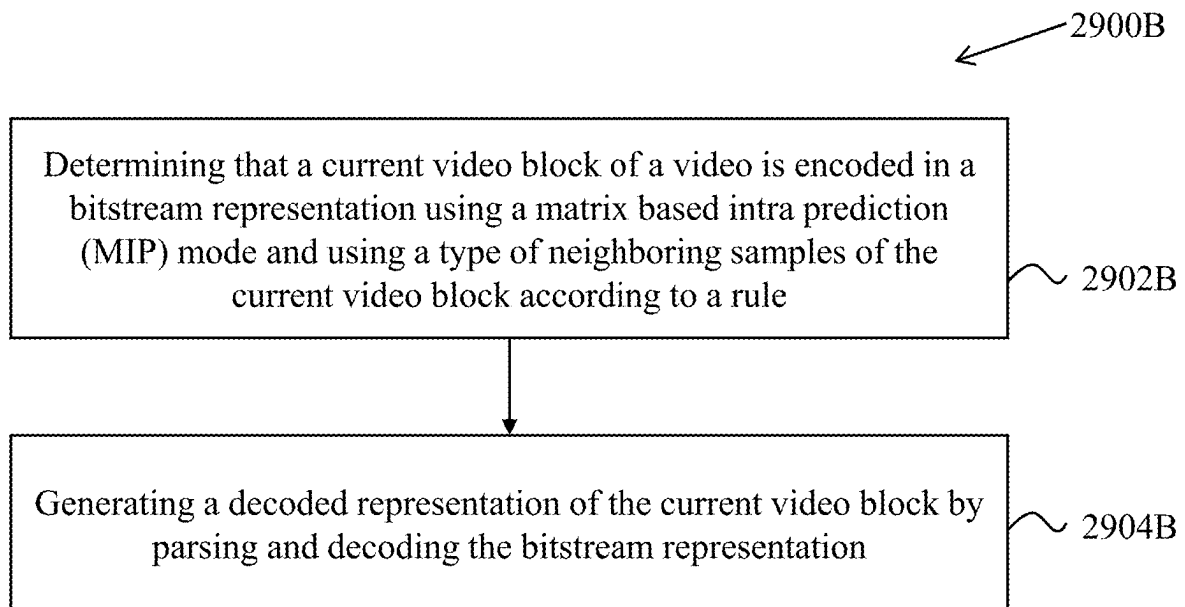


FIG. 29B

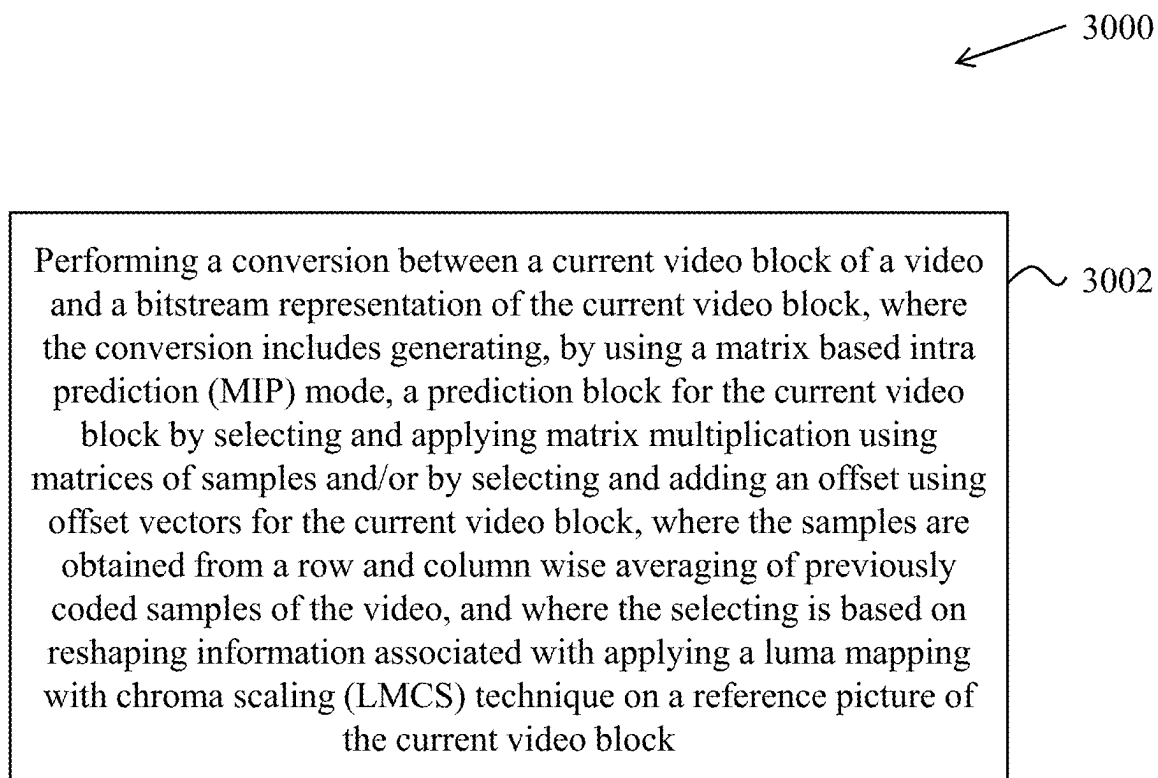


FIG. 30

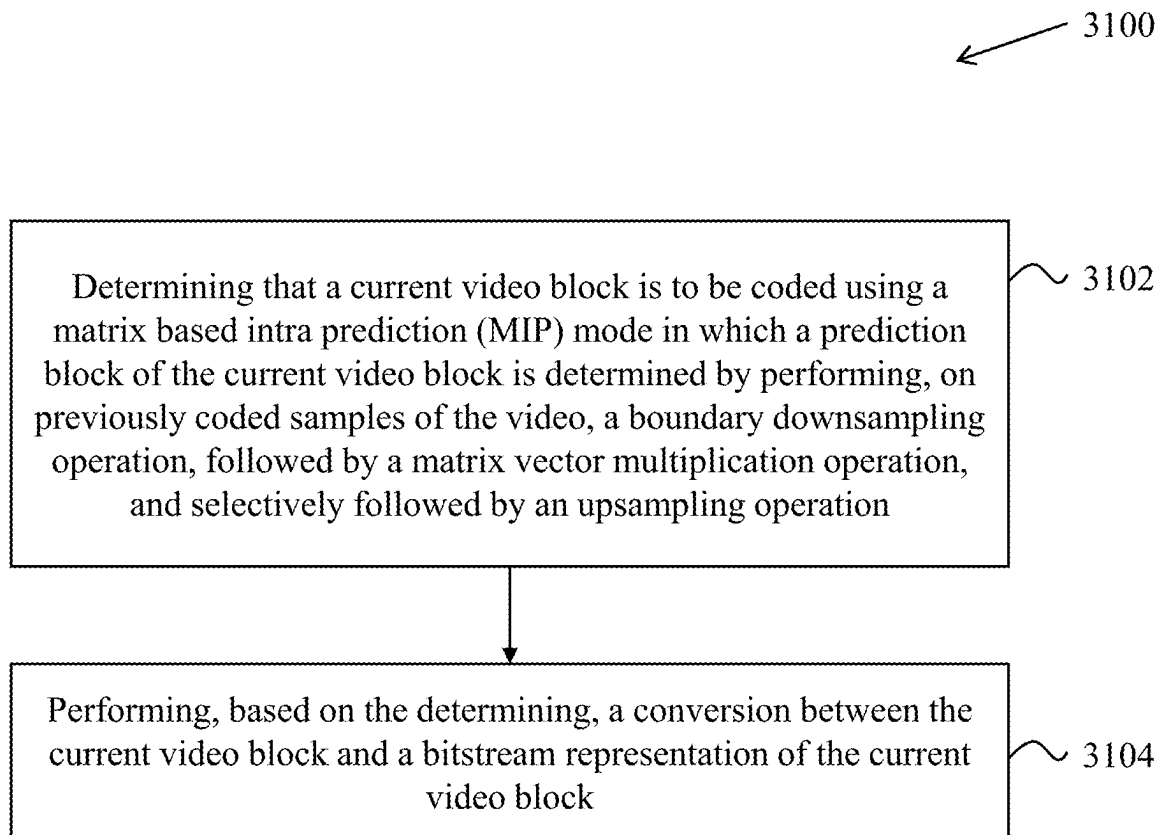


FIG. 31

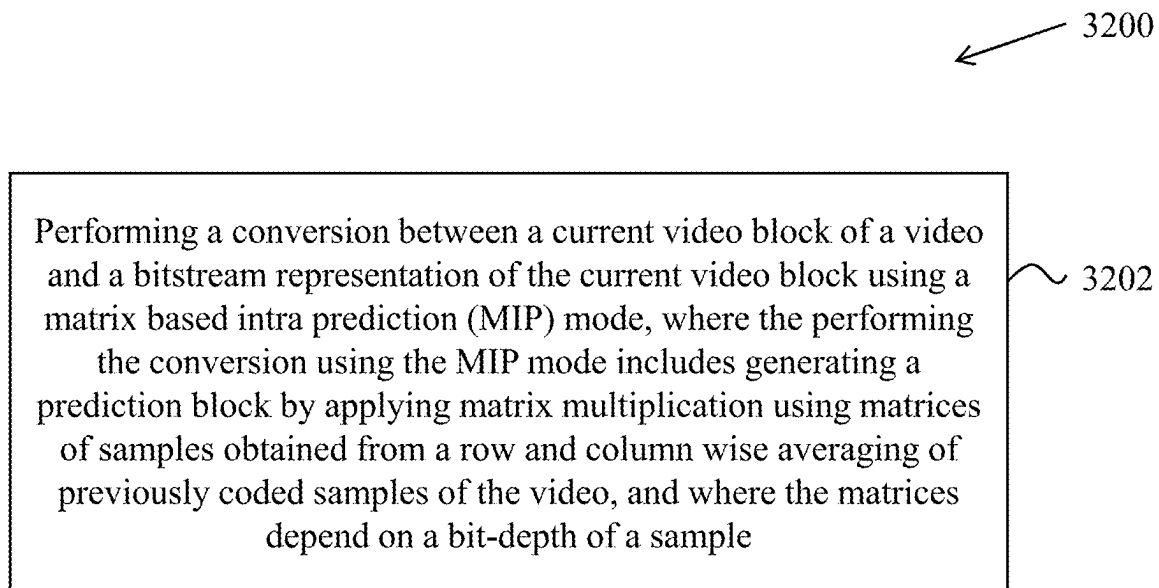


FIG. 32

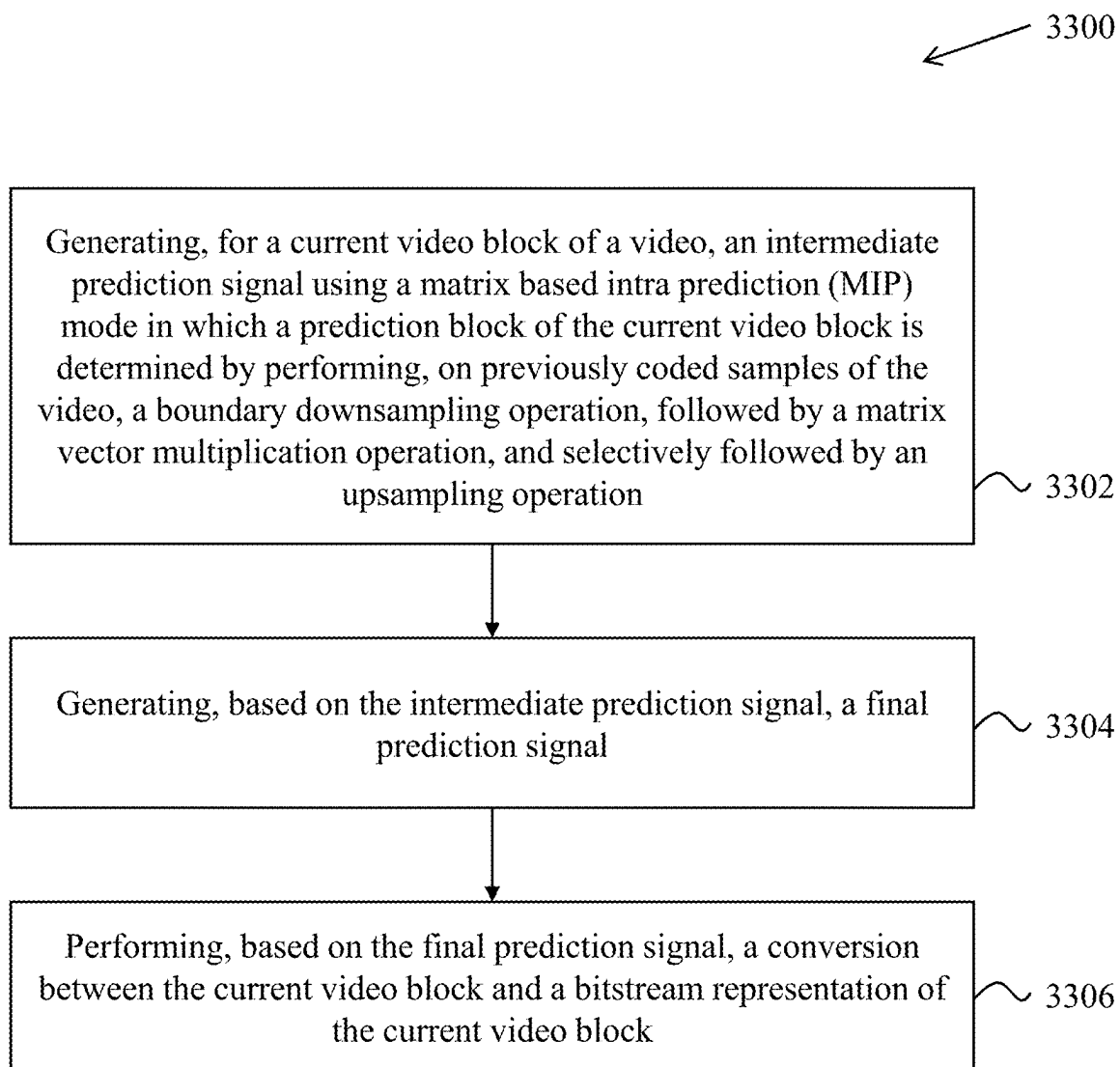


FIG. 33

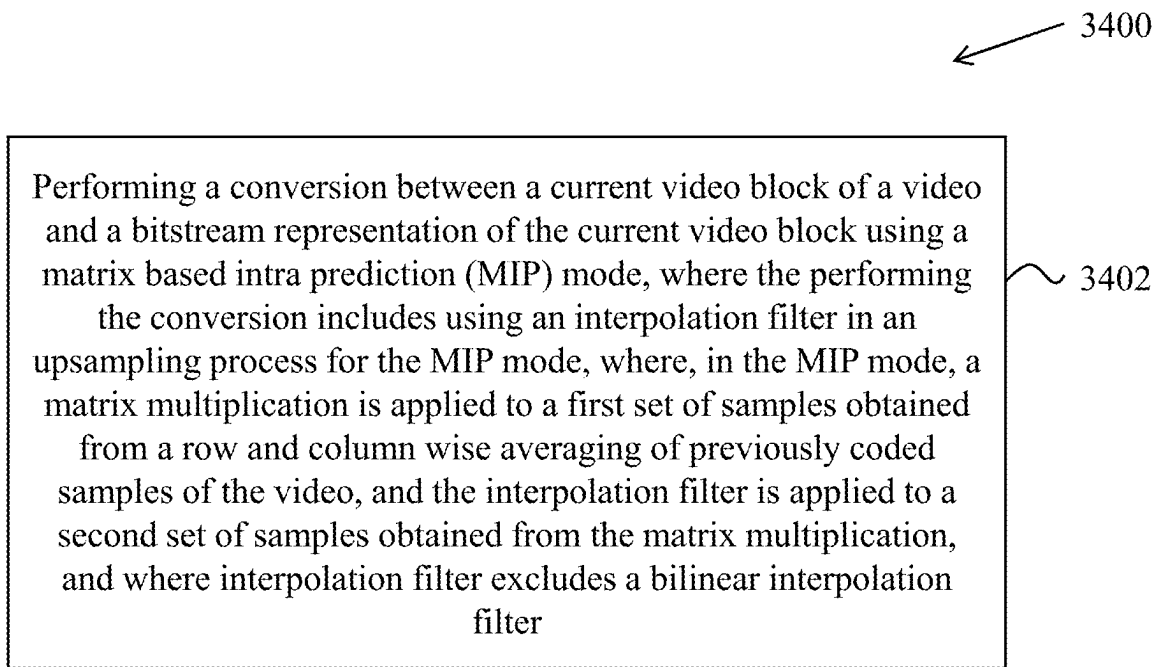


FIG. 34

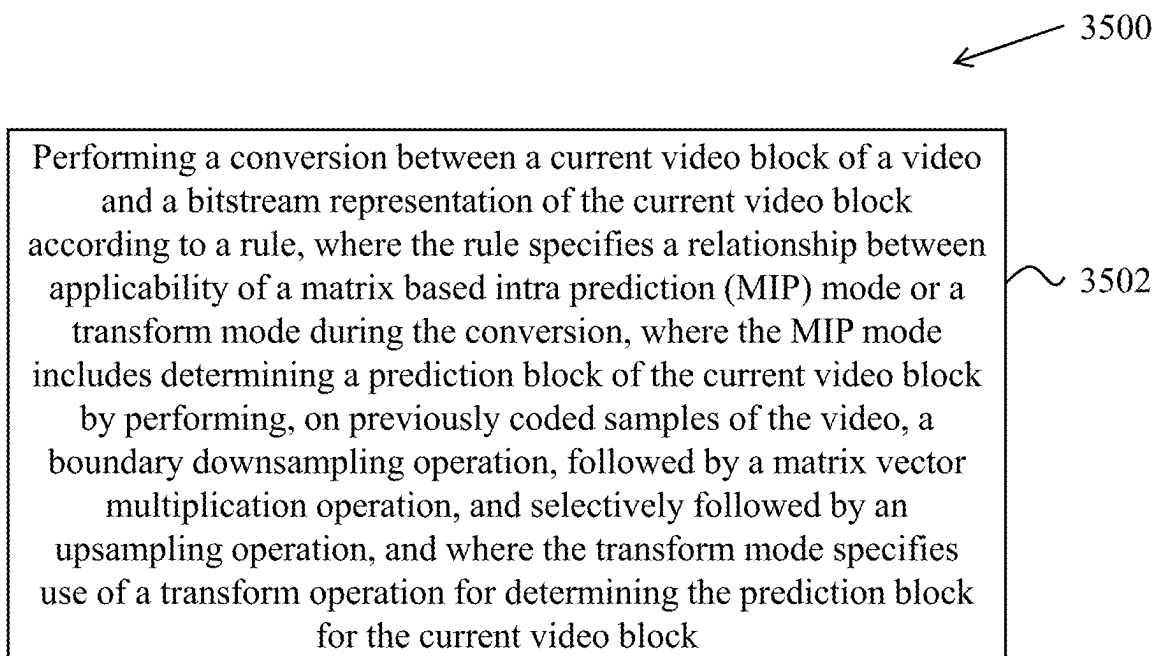


FIG. 35

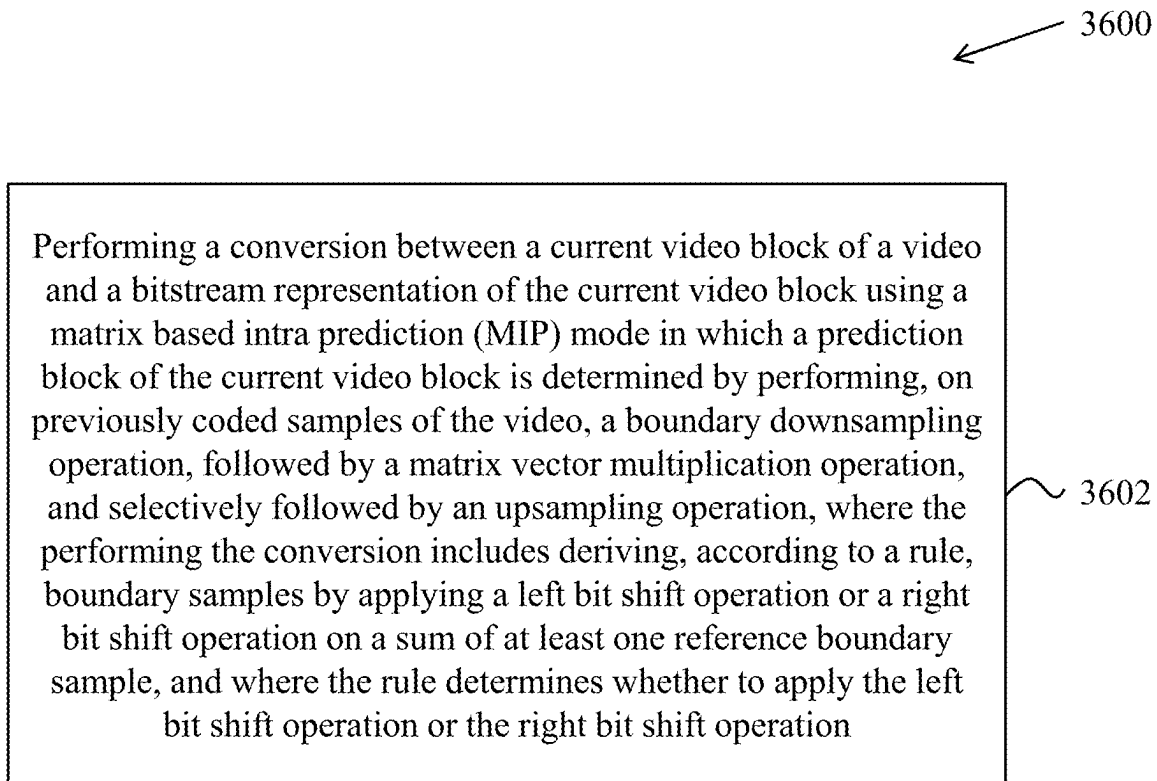


FIG. 36

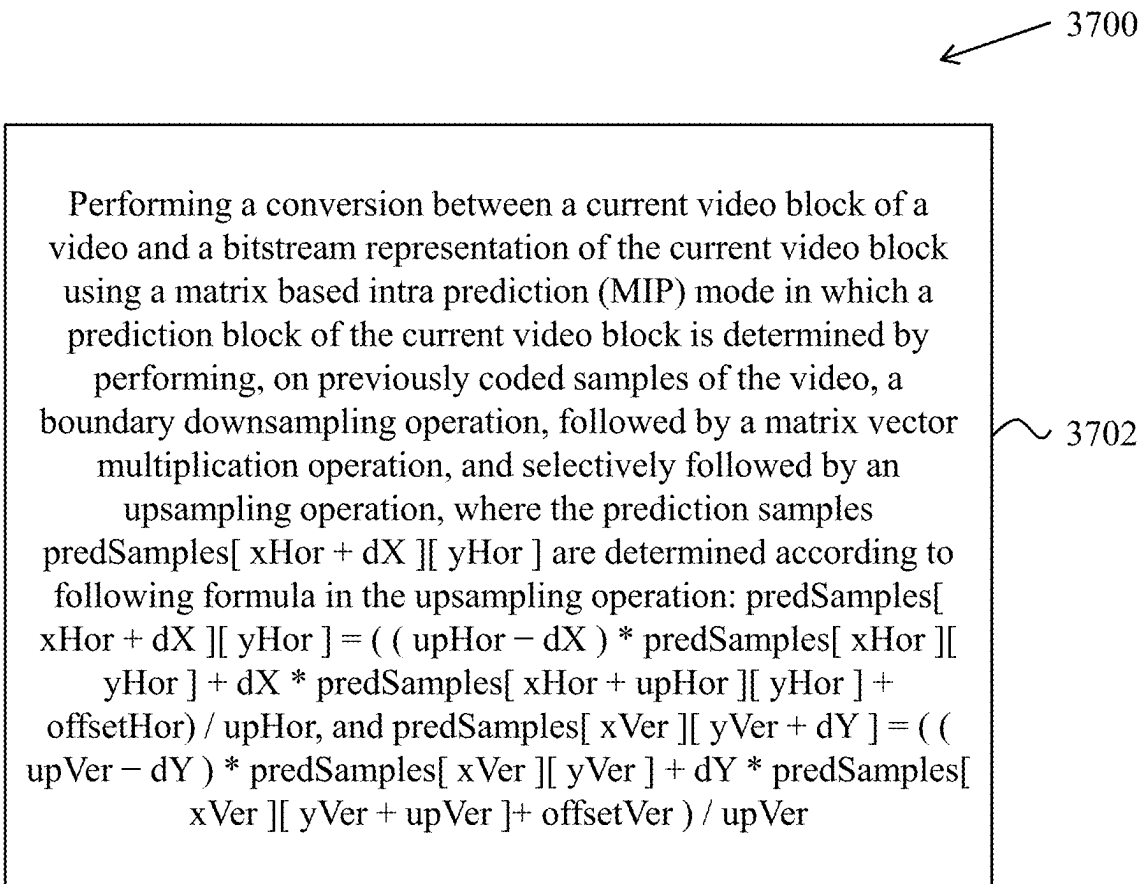


FIG. 37

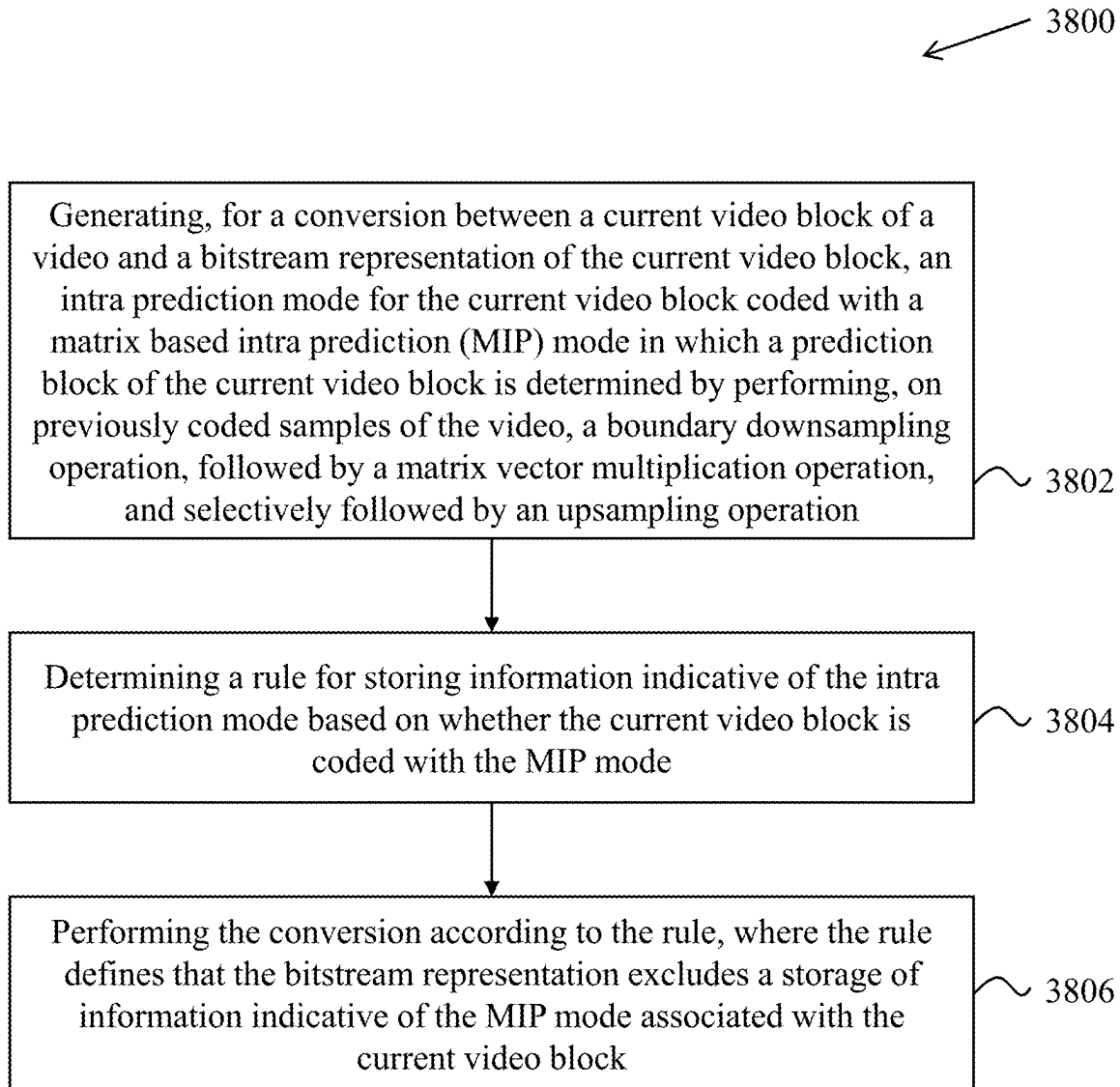


FIG. 38

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TRANSFORM CODING BASED ON MATRIX-BASED INTRA PREDICTION

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Patent Application No. PCT/CN2020/084472, filed on Apr. 13, 2020, which claims the priority to and benefits of International Patent Application No. PCT/CN2019/082424, filed on Apr. 12, 2019. All the aforementioned patent applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

This patent document relates to video coding techniques, devices, and systems.

BACKGROUND

In spite of the advances in video compression, digital video still accounts for the largest bandwidth use on the internet and other digital communication networks. As the number of connected user devices capable of receiving and displaying video increases, it is expected that the bandwidth demand for digital video usage will continue to grow.

SUMMARY

Devices, systems, and methods related to digital video coding, and specifically, matrix-based intra prediction methods for video coding are described. The described methods may be applied to both the existing video coding standards (e.g., High Efficiency Video Coding (HEVC)) and future video coding standards (e.g., Versatile Video Coding (VVC)) or codecs.

A first example video processing method includes generating, for a conversion between a current video block of a video and a coded representation of the current video block, a first most probable mode (MPM) list using a first procedure based on a rule; and performing the conversion between the current video block and the coded representation of the current video block using the first MPM list, where the conversion of the current video block uses a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where the rule specifies that the first procedure used for generation of the first MPM list is same as a second procedure used for generation of a second MPM list for conversion of other video blocks of the video that are coded using a non-MIP intra mode that is different from the MIP mode, and where at least a portion of the first MPM list is generated based on at least a portion of the second MPM list.

A second example video processing method includes generating, for a conversion between a current video block of a video and a coded representation of the current video block, a most probable mode (MPM) list based on a rule, where the rule is based on whether a neighboring video block of the current video block is coded with a matrix based intra prediction (MIP) mode in which a prediction block of the neighboring video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector mul-

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tiplication operation, and selectively followed by an upsampling operation; and performing the conversion between the current video block and the coded representation of the current video block using the MPM list, where the conversion applies a non-MIP mode to the current video block, and where the non-MIP mode is different from the MIP mode.

A third example video processing method includes decoding a current video block of a video that is coded in a coded representation of the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; and updating a line buffer associated with the decoding without storing in the line buffer information indicating whether the current video block is coded using the MIP mode.

A fourth example video processing method includes performing a conversion between a current video block and a bitstream representation of the current video block, where the current video block is coded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where a flag is coded in the bitstream presentation using at most K contexts in an arithmetic coding or a decoding process, where the flag indicates whether the current video block is coded using the MIP mode, and where K is greater than or equal to zero.

A fifth example video processing method includes generating, for a conversion between a current video block of a video and a bitstream representation of the current video block, an intra prediction mode for the current video block coded with a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; determining a rule for storing information indicative of the intra prediction mode based on whether the current video block is coded with the MIP mode; and performing the conversion according to the rule, where the rule defines that a syntax element for the intra prediction mode is stored in the bitstream representation for the current video block, and where the rule defines that a mode index for the MIP mode for the current video block is not stored in the bitstream representation.

A sixth example video processing method includes performing a first determination that a luma video block of a video is coded using a matrix based intra prediction (MIP) mode in which a prediction block of the luma video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; performing, based on the first determination, a second determination about a chroma intra mode to be used for a chroma video block associated with the luma video block; and performing, based on the second determination, a conversion between the chroma video block and a bitstream representation of the chroma video block.

A seventh example video processing method includes performing a conversion between a current video block of a video and a coded representation of the current video block,

where the conversion is based on a determination whether to code the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation.

An eight example video processing method includes determining, for encoding a current video block of a video into a bitstream representation of the current video block, according to a rule whether to use a matrix based intra prediction (MIP) mode and a coding mode that is different from the MIP mode for encoding the current video block, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; and adding, based on the determining, an encoded representation of the current video block to the bitstream representation.

A ninth example video processing method includes determining that a current video block of a video is encoded in a bitstream representation using a matrix based intra prediction (MIP) mode and a coding mode that is different from the MIP mode, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; and generating a decoded representation of the current video block by parsing and decoding the bitstream representation.

A tenth example video processing method includes making a determination about an applicability of a loop filter to a reconstructed block of a current video block of a video in a conversion between the video and a coded representation of the current video block, where the current video block is coded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; and processing the current video block according to the determination.

An eleventh example video processing method includes determining, for encoding a current video block of a video into a bitstream representation of the current video block, a type of neighboring samples of the current video block to be used according to a rule; and adding, based on the determining, an encoded representation of the current video block to the bitstream representation, where the current video block is encoded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation.

A twelfth example video processing method includes determining that a current video block of a video is encoded in a bitstream representation using a matrix based intra prediction (MIP) mode and using a type of neighboring samples of the current video block according to a rule, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation,

followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; and generating a decoded representation of the current video block by parsing and decoding the bitstream representation.

A thirteenth example video processing method includes performing a conversion between a current video block of a video and a bitstream representation of the current video block, where the conversion includes generating, by using a matrix based intra prediction (MIP) mode, a prediction block for the current video block by selecting and applying matrix multiplication using matrices of samples and/or by selecting and adding an offset using offset vectors for the current video block, where the samples are obtained from a row and column wise averaging of previously coded samples of the video, and where the selecting is based on reshaping information associated with applying a luma mapping with chroma scaling (LMCS) technique on a reference picture of the current video block.

A fourteenth example video processing method includes determining that a current video block is to be coded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; and performing, based on the determining, a conversion between the current video block and a bitstream representation of the current video block, where the performing the conversion is based on a rule of co-application of the MIP mode and another coding technique.

A fifteenth example video processing method includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode, where the performing the conversion using the MIP mode includes generating a prediction block by applying matrix multiplication using matrices of samples obtained from a row and column wise averaging of previously coded samples of the video, and where the matrices depend on a bit-depth of a sample.

A sixteenth example video processing method includes generating, for a current video block of a video, an intermediate prediction signal using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; generating, based on the intermediate prediction signal, a final prediction signal; and performing, based on the final prediction signal, a conversion between the current video block and a bitstream representation of the current video block.

A seventeenth example video processing method includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode, where the performing the conversion includes using an interpolation filter in an upsampling process for the MIP mode, where, in the MIP mode, a matrix multiplication is applied to a first set of samples obtained from a row and column wise averaging of previously coded samples of the video, and the interpolation filter is applied to a second set of samples obtained from the matrix multiplication, and where interpolation filter excludes a bilinear interpolation filter.

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An eighteenth example video processing method includes performing a conversion between a current video block of a video and a bitstream representation of the current video block according to a rule, where the rule specifies a relationship between applicability of a matrix based intra prediction (MIP) mode or a transform mode during the conversion, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, and where the transform mode specifies use of a transform operation for the determining the prediction block for the current video block.

A nineteenth example video processing method includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where the performing the conversion includes deriving, according to a rule, boundary samples by applying a left bit shift operation or a right bit shift operation on a sum of at least one reference boundary sample, and where the rule determines whether to apply the left bit shift operation or the right bit shift operation.

A twentieth example video processing method includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where the prediction samples $\text{predSamples}[\text{xHor}+\text{dX}][\text{yHor}]$ are determined according to following formula in the upsampling operation: $\text{predSamples}[\text{xHor}+\text{dX}][\text{yHor}] = ((\text{upHor}-\text{dX}) * \text{predSamples}[\text{xHor}][\text{yHor}] + \text{dX} * \text{predSamples}[\text{xHor}+\text{upHor}][\text{yHor}] + \text{offsetHor}) / \text{upHor}$, and $\text{predSamples}[\text{xVer}][\text{yVer}+\text{dY}] = ((\text{upVer}-\text{dY}) * \text{predSamples}[\text{xVer}][\text{yVer}] + \text{dY} * \text{predSamples}[\text{xVer}][\text{yVer}+\text{upVer}] + \text{offsetVer}) / \text{upVer}$, where offsetHor and offsetVer are integers, where upHor is a function of the current video block width and a pre-defined value based on the current video block size, where upVer is a function of the current video block height and the pre-defined value based on the current video block size; where dX is $1 \dots \text{upHor}-1$, where dY is $1 \dots \text{upVer}-1$, and where xHor are positions based on the upHor and yHor are positions based on the upVer .

A twenty-first example video processing method includes generating, for a conversion between a current video block of a video and a bitstream representation of the current video block, an intra prediction mode for the current video block coded with a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation; determining a rule for storing information indicative of the intra prediction mode based on whether the current video block is coded with the MIP mode; and performing the conversion according to the rule, where the rule defines that the bitstream representation

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excludes a storage of information indicative of the MIP mode associated with the current video block.

In yet another representative aspect, the disclosed technology may be used to provide a method for video processing. This exemplary method includes determining that a current video block is coded using an affine linear weighted intra prediction (ALWIP) mode, constructing, based on the determining, at least a portion of a most probable mode (MPM) list for the ALWIP mode based on an at least a portion of an MPM list for a non-ALWIP intra mode, and performing, based on the MPM list for the ALWIP mode, a conversion between the current video block and a bitstream representation of the current video block.

In yet another representative aspect, the disclosed technology may be used to provide a method for video processing. This exemplary method includes determining that a luma component of a current video block is coded using an affine linear weighted intra prediction (ALWIP) mode, inferring, based on the determining, a chroma intra mode, and performing, based on the chroma intra mode, a conversion between the current video block and a bitstream representation of the current video block.

In yet another representative aspect, the disclosed technology may be used to provide a method for video processing. This exemplary method includes determining that a current video block is coded using an affine linear weighted intra prediction (ALWIP) mode, and performing, based on the determining, a conversion between the current video block and a bitstream representation of the current video block.

In yet another representative aspect, the disclosed technology may be used to provide a method for video processing. This exemplary method includes determining that a current video block is coded using a coding mode different from an affine linear weighted intra prediction (ALWIP) mode, and performing, based on the determining, a conversion between the current video block and a bitstream representation of the current video block.

In yet another representative aspect, the above-described method is embodied in the form of processor-executable code and stored in a computer-readable program medium.

In yet another representative aspect, a device that is configured or operable to perform the above-described method is disclosed. The device may include a processor that is programmed to implement this method.

In yet another representative aspect, a video decoder apparatus may implement a method as described herein.

The above and other aspects and features of the disclosed technology are described in greater detail in the drawings, the description and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an example of 33 intra prediction directions.

FIG. 2 shows an example of 67 intra prediction modes.

FIG. 3 shows an example of locations of samples used for the derivation of the weights of the linear model.

FIG. 4 shows an example of four reference lines neighboring a prediction block.

FIG. 5A and FIG. 5B show examples of sub-partitions depending on block size.

FIG. 6 shows an example of ALWIP for 4x4 blocks.

FIG. 7 shows an example of ALWIP for 8x8 blocks.

FIG. 8 shows an example of ALWIP for 8x4 blocks.

FIG. 9 shows an example of ALWIP for 16x16 blocks.

FIG. 10 shows an example of neighboring blocks using in MPM list construction.

FIG. 11 shows a flowchart of an example method for matrix-based intra prediction, in accordance with the disclosed technology.

FIG. 12 shows a flowchart of another example method for matrix-based intra prediction, in accordance with the disclosed technology.

FIG. 13 shows a flowchart of yet another example method for matrix-based intra prediction, in accordance with the disclosed technology.

FIG. 14 shows a flowchart of yet another example method for matrix-based intra prediction, in accordance with the disclosed technology.

FIG. 15 is a block diagram of an example of a hardware platform for implementing a visual media decoding or a visual media encoding technique described in the present document.

FIG. 16 is a block diagram showing an example video processing system in which various techniques disclosed herein may be implemented.

FIG. 17 is a block diagram that illustrates an example video coding system that may utilize the techniques of this disclosure.

FIG. 18 is a block diagram illustrating an example of video encoder.

FIG. 19 is a block diagram illustrating an example of video decoder.

FIGS. 20-38 show example flowcharts of additional example methods for matrix-based intra prediction, in accordance with the disclosed technology.

DETAILED DESCRIPTION

Due to the increasing demand of higher resolution video, video coding methods and techniques are ubiquitous in modern technology. Video codecs typically include an electronic circuit or software that compresses or decompresses digital video, and are continually being improved to provide higher coding efficiency. A video codec converts uncompressed video to a compressed format or vice versa. There are complex relationships between the video quality, the amount of data used to represent the video (determined by the bit rate), the complexity of the encoding and decoding algorithms, sensitivity to data losses and errors, ease of editing, random access, and end-to-end delay (latency). The compressed format usually conforms to a standard video compression specification, e.g., the High Efficiency Video Coding (HEVC) standard (also known as H.265 or MPEG-H Part 2), the Versatile Video Coding (VVC) standard to be finalized, or other current and/or future video coding standards.

Embodiments of the disclosed technology may be applied to existing video coding standards (e.g., HEVC, H.265) and future standards to improve runtime performance. Section headings are used in the present document to improve readability of the description and do not in any way limit the discussion or the embodiments (and/or implementations) to the respective sections only.

1 A Brief Review on HEVC

1.1 Intra Prediction in HEVC/H.265

Intra prediction involves producing samples for a given TB (transform block) using samples previously reconstructed in the considered color channel. The intra prediction mode is separately signaled for the luma and chroma channels, with the chroma channel intra prediction mode optionally dependent on the luma channel intra prediction mode via the 'DM_CHROMA' mode. Although the intra prediction mode is signaled at the PB (prediction block) level, the

intra prediction process is applied at the TB level, in accordance with the residual quad-tree hierarchy for the CU, thereby allowing the coding of one TB to have an effect on the coding of the next TB within the CU, and therefore reducing the distance to the samples used as reference values.

HEVC includes 35 intra prediction modes—a DC mode, a planar mode and 33 directional, or 'angular' intra prediction modes. The 33 angular intra prediction modes are illustrated in FIG. 1.

For PBs associated with chroma color channels, the intra prediction mode is specified as either planar, DC, horizontal, vertical, 'DM_CHROMA' mode or sometimes diagonal mode '34'.

Note for chroma formats 4:2:2 and 4:2:0, the chroma PB may overlap two or four (respectively) luma PBs; in this case the luma direction for DM_CHROMA is taken from the top left of these luma PBs.

The DM_CHROMA mode indicates that the intra prediction mode of the luma color channel PB is applied to the chroma color channel PBs. Since this is relatively common, the most-probable-mode coding scheme of the intra_chroma_pred_mode is biased in favor of this mode being selected.

2 Examples of Intra Prediction in VVC

2.1 Intra Mode Coding with 67 Intra Prediction Modes

To capture the arbitrary edge directions presented in natural video, the number of directional intra modes is extended from 33, as used in HEVC, to 65. The additional directional modes are depicted as red dotted arrows in FIG. 2, and the planar and DC modes remain the same. These denser directional intra prediction modes apply for all block sizes and for both luma and chroma intra predictions.

2.2 Examples of the Cross-Component Linear Model (CCLM)

In some embodiments, and to reduce the cross-component redundancy, a cross-component linear model (CCLM) prediction mode (also referred to as LM), is used in the JEM, for which the chroma samples are predicted based on the reconstructed luma samples of the same CU by using a linear model as follows:

$$\text{pred}_c(i,j) = \alpha \cdot \text{rec}_L'(i,j) + \beta \quad (1)$$

Here, $\text{pred}_c(i,j)$ represents the predicted chroma samples in a CU and $\text{rec}_L'(i,j)$ represents the downsampled reconstructed luma samples of the same CU. Linear model parameter α and β are derived from the relation between luma values and chroma values from two samples, which are luma sample with minimum sample value and with maximum sample inside the set of downsampled neighboring luma samples, and their corresponding chroma samples. FIG. 3 shows an example of the location of the left and above samples and the sample of the current block involved in the CCLM mode.

This parameter computation is performed as part of the decoding process and is not just as an encoder search operation. As a result, no syntax is used to convey the α and β values to the decoder.

For chroma intra mode coding, a total of 8 intra modes are allowed for chroma intra mode coding. Those modes include five traditional intra modes and three cross-component linear model modes (CCLM, LM_A, and LM_L). Chroma mode coding directly depends on the intra prediction mode of the corresponding luma block. Since separate block partitioning structure for luma and chroma components is enabled in I slices, one chroma block may correspond to multiple luma blocks. Therefore, for Chroma DM mode, the intra predic-

tion mode of the corresponding luma block covering the center position of the current chroma block is directly inherited.

2.3 Multiple Reference Line (MRL) Intra Prediction

Multiple reference line (MRL) intra prediction uses more reference lines for intra prediction. In FIG. 4, an example of 4 reference lines is depicted, where the samples of segments A and F are not fetched from reconstructed neighboring samples but padded with the closest samples from Segment B and E, respectively. HEVC intra-picture prediction uses the nearest reference line (i.e., reference line 0). In MRL, 2 additional lines (reference line 1 and reference line 3) are used. The index of selected reference line (*mrl_idx*) is signalled and used to generate intra predictor. For reference line *idx*, which is greater than 0, only include additional reference line modes in MPM list and only signal *mpm* index without remaining mode.

2.4 Intra Sub-Partitions (ISP)

The Intra Sub-Partitions (ISP) tool divides luma intra-predicted blocks vertically or horizontally into 2 or 4 sub-partitions depending on the block size. For example, minimum block size for ISP is 4x8 (or 8x4). If block size is greater than 4x8 (or 8x4) then the corresponding block is divided by 4 sub-partitions. FIG. 5 shows examples of the two possibilities. All sub-partitions fulfill the condition of having at least 16 samples.

For each sub-partition, reconstructed samples are obtained by adding the residual signal to the prediction signal. Here, a residual signal is generated by the processes such as entropy decoding, inverse quantization and inverse transform. Therefore, the reconstructed sample values of each sub-partition are available to generate the prediction of the next sub-partition, and each sub-partition is processed repeatedly. In addition, the first sub-partition to be processed is the one containing the top-left sample of the CU and then continuing downwards (horizontal split) or rightwards (vertical split). As a result, reference samples used to generate the sub-partitions prediction signals are only located at the left and above sides of the lines. All sub-partitions share the same intra mode.

2.5 Affine Linear Weighted Intra Prediction (ALWIP or Matrix-Based Intra Prediction)

Affine linear weighted intra prediction (ALWIP, a.k.a. Matrix based intra prediction (MIP)) is proposed in JVET-N0217.

In JVET-N0217, two tests are conducted. In test 1, ALWIP is designed with a memory restriction of 8K bytes and at most 4 multiplications per sample. Test 2 is similar to test 1, but further simplifies the design in terms of memory requirement and model architecture.

Single set of matrices and offset vectors for all block shapes.

Reduction of number of modes to 19 for all block shapes.
Reduction of memory requirement to 5760 10-bit values, that is 7.20 Kilobyte.

Linear interpolation of predicted samples is carried out in a single step per direction replacing iterative interpolation as in the first test.

2.5.1 Test 1 of JVET-N0217

For predicting the samples of a rectangular block of width *W* and height *H*, affine linear weighted intra prediction (ALWIP) takes one line of *H* reconstructed neighboring boundary samples left of the block and one line of *W* reconstructed neighboring boundary samples above the block as input. If the reconstructed samples are unavailable, they are generated as it is done in the conventional intra prediction.

The generation of the prediction signal is based on the following three steps:

Out of the boundary samples, four samples in the case of *W*=*H*=4 and eight samples in all other cases are extracted by averaging.

A matrix vector multiplication, followed by addition of an offset, is carried out with the averaged samples as an input. The result is a reduced prediction signal on a subsampled set of samples in the original block.

The prediction signal at the remaining positions is generated from the prediction signal on the subsampled set by linear interpolation which is a single step linear interpolation in each direction.

The matrices and offset vectors needed to generate the prediction signal are taken from three sets *S*₀, *S*₁, *S*₂ of matrices. The set *S*₀ consists of 18 matrices *A*₀^{*i*}, *i* ∈ {0, . . . , 17} each of which has 16 rows and 4 columns and 18 offset vectors *b*₀^{*i*}, *i* ∈ {0, . . . , 17} each of size 16. Matrices and offset vectors of that set are used for blocks of size 4x4. The set *S*₁ consists of 10 matrices *A*₁^{*i*}, *i* ∈ {0, . . . , 9}, each of which has 16 rows and 8 columns and 10 offset vectors *b*₁^{*i*}, *i* ∈ {0, . . . , 9} each of size 16. Matrices and offset vectors of that set are used for blocks of sizes 4x8, 8x4 and 8x8. Finally, the set *S*₂ consists of 6 matrices *A*₂^{*i*}, *i* ∈ {0, . . . , 5}, each of which has 64 rows and 8 columns and of 6 offset vectors *b*₂^{*i*}, *i* ∈ {0, . . . , 5} of size 64. Matrices and offset vectors of that set or parts of these matrices and offset vectors are used for all other block-shapes.

The total number of multiplications needed in the computation of the matrix vector product is always smaller than or equal to 4x*W*x*H*. In other words, at most four multiplications per sample are required for the ALWIP modes.

2.5.2 Averaging of the Boundary

In a first step, the input boundaries *bdry*^{*top*} and *bdry*^{*left*} are reduced to smaller boundaries *bdry*_{*red*}^{*top*} and *bdry*_{*red*}^{*left*}. Here, *bdry*_{*red*}^{*top*} and *bdry*_{*red*}^{*left*} both consists of 2 samples in the case of a 4x4-block and both consist of 4 samples in all other cases.

In the case of a 4x4-block, for 0 ≤ *i* < 2, one defines

$$bdry_{red}^{top}[i] = \left(\left(\sum_{j=0}^1 bdry^{top}[i \cdot 2 + j] \right) + 1 \right) \gg 1$$

and defines *bdry*_{*red*}^{*left*} analogously.

Otherwise, if the block-width *W* is given as *W*=4·2^{*k*}, for 0 ≤ *i* < 4, one defines

$$bdry_{red}^{top}[i] = \left(\left(\sum_{j=0}^{2^k-1} bdry^{top}[i \cdot 2^k + j] \right) + (1 \ll (k-1)) \right) \gg k$$

and defines *bdry*_{*red*}^{*left*} analogously.

The two reduced boundaries *bdry*_{*red*}^{*top*} and *bdry*_{*red*}^{*left*} are concatenated to a reduced boundary vector *bdry*_{*red*} which is thus of size four for blocks of shape 4x4 and of size eight for blocks of all other shapes. If mode refers to the ALWIP-mode, this concatenation is defined as follows:

$$bdry_{red} = \begin{cases} [bdry_{red}^{top}, bdry_{red}^{left}] & \text{for } W = H = 4 \text{ and mode} < 18 \\ [bdry_{red}^{left}, bdry_{red}^{top}] & \text{for } W = H = 4 \text{ and mode} \geq 18 \\ [bdry_{red}^{top}, bdry_{red}^{left}] & \text{for } \max(W, H) = 8 \text{ and mode} < 10 \\ [bdry_{red}^{left}, bdry_{red}^{top}] & \text{for } \max(W, H) = 8 \text{ and mode} \geq 10 \\ [bdry_{red}^{top}, bdry_{red}^{left}] & \text{for } \max(W, H) > 8 \text{ and mode} < 6 \\ [bdry_{red}^{left}, bdry_{red}^{top}] & \text{for } \max(W, H) > 8 \text{ and mode} \geq 6. \end{cases}$$

Finally, for the interpolation of the subsampled prediction signal, on large blocks a second version of the averaged boundary is needed. Namely, if $\min(W, H) > 8$ and $W \geq H$, one writes $W = 8 \cdot 2^l$, and, for $0 \leq l < 8$, defines

$$bdry_{redll}^{top}[i] = \left(\sum_{j=0}^{2^l-1} bdry_{red}^{top}[i \cdot 2^l + j] \right) + (1 \ll (l-1)) \gg l.$$

If $\min(W, H) > 8$ and $H > W$, one defines $bdry_{redll}^{left}$ analogously.

2.5.3 Generation of the Reduced Prediction Signal by Matrix Vector Multiplication

Out of the reduced input vector $bdry_{red}$ one generates a reduced prediction signal $pred_{red}$. The latter signal is a signal on the downsampled block of width W_{red} and height H_{red} . Here, W_{red} and H_{red} are defined as:

$$W_{red} = \begin{cases} 4 & \text{for } \max(W, H) \leq 8 \\ \min(W, 8) & \text{for } \max(W, H) > 8 \end{cases}$$

$$H_{red} = \begin{cases} 4 & \text{for } \max(W, H) \leq 8 \\ \min(W, 8) & \text{for } \max(W, H) > 8 \end{cases}$$

The reduced prediction signal $pred_{red}$ is computed by calculating a matrix vector product and adding an offset:

$$pred_{red} = A \cdot bdry_{red} + b$$

Here, A is a matrix that has $W_{red} \cdot H_{red}$ rows and 4 columns if $W = H = 4$ and 8 columns in all other cases. b is a vector of size $W_{red} \cdot H_{red}$.

The matrix A and the vector b are taken from one of the sets S_0, S_1, S_2 as follows. One defines an index $idx = idx(W, H)$ as follows:

$$idx(W, H) = \begin{cases} 0 & \text{for } W = H = 4 \\ 1 & \text{for } \max(W, H) = 8 \\ 2 & \text{for } \max(W, H) > 8. \end{cases}$$

Moreover, one puts m as follows:

$$m = \begin{cases} \text{mode} & \text{for } W = H = 4 \text{ and mode} < 18 \\ \text{mode} - 17 & \text{for } W = H = 4 \text{ and mode} \geq 18 \\ \text{mode} & \text{for } \max(W, H) = 8 \text{ and mode} < 10 \\ \text{mode} - 9 & \text{for } \max(W, H) = 8 \text{ and mode} \geq 10 \\ \text{mode} & \text{for } \max(W, H) > 8 \text{ and mode} < 6 \\ \text{mode} - 5 & \text{for } \max(W, H) > 8 \text{ and mode} \geq 6. \end{cases}$$

Then, if $idx \leq 1$ or $idx = 2$ and $\min(W, H) > 4$, one puts $A = A_{idx}^m$ and $b = b_{idx}^m$. In the case that $idx = 2$ and

$\min(W, H) = 4$, one lets A be the matrix that arises by leaving out every row of A_{idx}^m that, in the case $W = 4$, corresponds to an odd x-coordinate in the downsampled block, or, in the case $H = 4$, corresponds to an odd y-coordinate in the downsampled block.

Finally, the reduced prediction signal is replaced by its transpose in the following cases:

$$\begin{aligned} & W = H = 4 \text{ and mode} \geq 18 \\ & \max(W, H) = 8 \text{ and mode} \geq 10 \\ & \max(W, H) > 8 \text{ and mode} \geq 6 \end{aligned}$$

The number of multiplications required for calculation of $pred_{red}$ is 4 in the case of $W = H = 4$ since in this case A has 4 columns and 16 rows. In all other cases, A has 8 columns and $W_{red} \cdot H_{red}$ rows and one immediately verifies that in these cases $8 \cdot W_{red} \cdot H_{red} \leq 4 \cdot W \cdot H$ multiplications are required, i.e. also in these cases, at most 4 multiplications per sample are needed to compute $pred_{red}$.

2.5.4 Illustration of the Entire ALWIP Process

The entire process of averaging, matrix vector multiplication and linear interpolation is illustrated for different shapes in FIGS. 6-9. Note, that the remaining shapes are treated as in one of the depicted cases.

1. Given a 4×4 block, ALWIP takes two averages along each axis of the boundary. The resulting four input samples enter the matrix vector multiplication. The matrices are taken from the set S_0 . After adding an offset, this yields the 16 final prediction samples. Linear interpolation is not necessary for generating the prediction signal. Thus, a total of $(4 \cdot 16) / (4 \cdot 4) = 4$ multiplications per sample are performed.

2. Given an 8×8 block, ALWIP takes four averages along each axis of the boundary. The resulting eight input samples enter the matrix vector multiplication. The matrices are taken from the set S_1 . This yields 16 samples on the odd positions of the prediction block. Thus, a total of $(8 \cdot 16) / (8 \cdot 8) = 2$ multiplications per sample are performed. After adding an offset, these samples are interpolated vertically by using the reduced top boundary. Horizontal interpolation follows by using the original left boundary.

3. Given an 8×4 block, ALWIP takes four averages along the horizontal axis of the boundary and the four original boundary values on the left boundary. The resulting eight input samples enter the matrix vector multiplication. The matrices are taken from the set S_1 . This yields 16 samples on the odd horizontal and each vertical positions of the prediction block. Thus, a total of $(8 \cdot 16) / (8 \cdot 4) = 4$ multiplications per sample are performed. After adding an offset, these samples are interpolated horizontally by using the original left boundary.

4. Given a 16×16 block, ALWIP takes four averages along each axis of the boundary. The resulting eight input samples enter the matrix vector multiplication. The matrices are taken from the set S_2 . This yields 64 samples on the odd positions of the prediction block. Thus, a total of $(8 \cdot 64) / (16 \cdot 16) = 2$ multiplications per sample are performed. After adding an offset, these samples are interpolated vertically by using eight averages of the top boundary. Horizontal interpolation follows by using the original left boundary. The interpolation process, in this case, does not add any multiplications. Therefore, totally, two multiplications per sample are required to calculate ALWIP prediction.

For larger shapes, the procedure is essentially the same and it is easy to check that the number of multiplications per sample is less than four.

For $W \times 8$ blocks with $W > 8$, only horizontal interpolation is necessary as the samples are given at the odd horizontal and each vertical positions.

Finally for $W \times 4$ blocks with $W > 8$, let A_k be the matrix that arises by leaving out every row that corresponds to an odd entry along the horizontal axis of the downsampled block. Thus, the output size is 32 and again, only horizontal interpolation remains to be performed.

The transposed cases are treated accordingly.

2.5.5 Single Step Linear Interpolation

For a $W \times H$ block with $\max(W, H) \geq 8$, the prediction signal arises from the reduced prediction signal pred_{red} on $W_{red} \times H_{red}$ by linear interpolation. Depending on the block shape, linear interpolation is done in vertical, horizontal or both directions. If linear interpolation is to be applied in both directions, it is first applied in horizontal direction if $W < H$ and it is first applied in vertical direction, else.

Consider without loss of generality a $W \times H$ block with $\max(W, H) \geq 8$ and $W \geq H$. Then, the one-dimensional linear interpolation is performed as follows. Without loss of generality, it suffices to describe linear interpolation in vertical direction. First, the reduced prediction signal is extended to the top by the boundary signal. Define the vertical upsampling factor $U_{ver} = H/H_{red}$ and write $U_{ver} = 2^{u_{ver}} > 1$. Then, define the extended reduced prediction signal by

$$\text{pred}_{red}[x][y] = \begin{cases} \text{bdr}_{red}^{top}[x] & \text{for } W = 8 \\ \text{bdr}_{red}^{top}[x] & \text{for } W > 8. \end{cases}$$

Then, from this extended reduced prediction signal, the vertically linear interpolated prediction signal is generated by

$$\text{pred}_{red}^{ups,ver}[x][U_{ver} \cdot y + k] = \left((U_{ver} - k - 1) \cdot \text{pred}_{red}[x][y - 1] + (k + 1) \cdot \text{pred}_{red}[x][y] + \frac{U_{ver}}{2} \right) \gg u_{ver}$$

for $0 \leq x < W_{red}$, $0 \leq y < H_{red}$ and $0 \leq k < U_{ver}$.

2.5.6 Signalization of the Proposed Intra Prediction Modes

For each Coding Unit (CU) in intra mode, a flag indicating if an ALWIP mode is to be applied on the corresponding Prediction Unit (PU) or not is sent in the bitstream. The signalization of the latter index is harmonized with MRL in the same way as in JVET-M0043. If an ALWIP mode is to be applied, the index predmode of the ALWIP mode is signaled using a MPM-list with 3 MPMS.

Here, the derivation of the MPMS is performed using the intra-modes of the above and the left PU as follows. There are three fixed tables $\text{map_angular_to_alwip}_{idx}$, $idx \in \{0, 1, 2\}$ that assign to each conventional intra prediction mode $\text{predmode}_{Angular}$ an ALWIP mode

$$\text{predmode}_{ALWIP} = \text{map_angular_to_alwip}_{idx}[\text{predmode}_{Angular}].$$

For each PU of width W and height H one defines an index

$$\text{idx}(\text{PU}) = \text{idx}(W, H) \in \{0, 1, 2\}$$

that indicates from which of the three sets the ALWIP-parameters are to be taken as in Section 2.5.3.

If the above Prediction Unit PU_{above} is available, belongs to the same CTU as the current PU and is in intra mode, if

$\text{idx}(\text{PU}) = \text{idx}(\text{PU}_{above})$ and if ALWIP is applied on PU_{above} with ALWIP-mode $\text{predmode}_{ALWIP}^{above}$, one puts

$$\text{mode}_{ALWIP}^{above} = \text{predmode}_{ALWIP}^{above}.$$

If the above PU is available, belongs to the same CTU as the current PU and is in intra mode and if a conventional intra prediction mode $\text{predmode}_{Angular}^{above}$ is applied on the above PU, one puts

$$\text{mode}_{ALWIP}^{above} = \text{map_angular_to_alwip}_{idx(\text{PU}_{above})}[\text{predmode}_{Angular}^{above}].$$

In all other cases, one puts

$$\text{mode}_{ALWIP}^{above} = -1,$$

which means that this mode is unavailable. In the same way but without the restriction that the left PU needs to belong to the same CTU as the current PU, one derives a mode $\text{mode}_{ALWIP}^{left}$.

Finally, three fixed default lists list_{idx} , $idx \in \{0, 1, 2\}$ are provided, each of which contains three distinct ALWIP modes. Out of the default list $\text{list}_{idx(\text{PU})}$ and the modes $\text{mode}_{ALWIP}^{above}$ and $\text{mode}_{ALWIP}^{left}$, one constructs three distinct MPMS by substituting -1 by default values as well as eliminating repetitions.

The left neighboring block and above neighboring block used in the ALWIP MPM list construction is A1 and B1 as shown in FIG. 10.

2.5.7 Adapted MPM-List Derivation for Conventional Luma and Chroma Intra-Prediction Modes

The proposed ALWIP-modes are harmonized with the MPM-based coding of the conventional intra-prediction modes as follows. The luma and chroma MPM-list derivation processes for the conventional intra-prediction modes uses fixed tables $\text{map_alwip_to_angular}_{idx}$, $idx \in \{0, 1, 2\}$, mapping an ALWIP-mode predmode_{ALWIP} on a given PU to one of the conventional intra-prediction modes

$$\text{predmode}_{Angular} = \text{map_alwip_to_angular}_{idx(\text{PU})}[\text{predmode}_{ALWIP}].$$

For the luma MPM-list derivation, whenever a neighboring luma block is encountered which uses an ALWIP-mode predmode_{ALWIP} , this block is treated as if it was using the conventional intra-prediction mode $\text{predmode}_{Angular}$. For the chroma MPM-list derivation, whenever the current luma block uses an LWIP-mode, the same mapping is used to translate the ALWIP-mode to a conventional intra prediction mode.

2.5.8 Corresponding Modified Working Draft

In some embodiments, as described in this section, portions related to `intra_lwip_flag`, `intra_lwip_mpm_flag`, `intra_lwip_mpm_idx` and `intra_lwip_mpm_remainder` have been added to the working draft based on embodiments of the disclosed technology.

In some embodiments, as described in this section, the `<begin>` and `<end>` tags are used to denote additions and modifications to the working draft based on embodiments of the disclosed technology.

Syntax Tables

Coding unit syntax	
<code>coding_unit(x0, y0, cbWidth, cbHeight, treeType) {</code>	Descriptor
<code>if(tile_group_type != I sps_ibc_enabled_flag) {</code>	
<code>if(treeType != DUAL_TREE_CHROMA)</code>	
<code>cu_skip_flag[x0][y0]</code>	<code>ae(v)</code>
<code>if(cu_skip_flag[x0][y0] == 0 && tile_group_type != I)</code>	
<code>pred_mode_flag</code>	<code>ae(v)</code>

Coding unit syntax	
	Descriptor
coding_unit(x0, y0, cbWidth, cbHeight, treeType) {	
if(((tile_group_type == I && cu_skip_flag[x0][y0] == 0) (tile_group_type != I && CuPredMode[x0][y0] != MODE_INTRA)) && sps_ibc_enabled_flag) pred_mode_ibc_flag	ae(v)
}	
if(CuPredMode[x0][y0] == MODE_INTRA) {	
if(sps_pcm_enabled_flag && cbWidth >= MinIpcmCbSizeY && cbWidth <= MaxIpcmCbSizeY && cbHeight >= MinIpcmCbSizeY && cbHeight <= MaxIpcmCbSizeY) pcm_flag[x0][y0]	ae(v)
if(pcm_flag[x0][y0]) {	
while(!byte_aligned()) pcm_alignment_zero_bit	f(1)
pcm_sample(cbWidth, cbHeight, treeType)	
} else {	
if(treeType == SINGLE_TREE treeType == DUAL_TREE_LUMA) {	
if(Abs(Log2(cbWidth) - Log2(cbHeight)) <= 2)	
intra_lwip_flag[x0][y0]	ae(v)
if(intra_lwip_flag[x0][y0]) {	
intra_lwip_mpm_flag[x0][y0]	ae(v)
if(intra_lwip_mpm_flag[x0][y0])	
intra_lwip_mpm_idx[x0][y0]	ae(v)
else	
intra_lwip_mpm_remainder[x0][y0]	ae(v)
} else {	
if((y0 % CtbSizeY) > 0)	
intra_luma_ref_idx[x0][y0]	ae(v)
if(intra_luma_ref_idx[x0][y0] == 0 && (cbWidth <= MaxTbSizeY cbHeight <= MaxTbSizeY) && (cbWidth * cbHeight > MinTbSizeY * MinTbSizeY))	
intra_subpartitions_mode_flag[x0][y0]	ae(v)
if(intra_subpartitions_mode_flag[x0][y0] == 1 && cbWidth <= MaxTbSizeY && cbHeight <= MaxTbSizeY)	
intra_subpartitions_split_flag[x0][y0]	ae(v)
if(intra_luma_ref_idx[x0][y0] == 0 && intra_subpartitions_mode_flag[x0][y0] == 0)	
intra_luma_mpm_flag[x0][y0]	ae(v)
if(intra_luma_mpm_flag[x0][y0])	
intra_luma_mpm_idx[x0][y0]	ae(v)
else	
intra_luma_mpm_remainder[x0][y0]	ae(v)
}	
}	
if(treeType == SINGLE_TREE treeType == DUAL_TREE_CHROMA)	
intra_chroma_pred_mode[x0][y0]	ae(v)
}	
} else if(treeType != DUAL_TREE_CHROMA) { /* MODE_INTER or MODE_IBC */	
... }	

Semantics

<begin>intra_lwip_flag[x0][y0] equal to 1 specifies that the intra prediction type for luma samples is affine linear weighted intra prediction. intra_lwip_flag[x0][y0] equal to 0 specifies that the intra prediction type for luma samples is not affine linear weighted intra prediction.

When intra_lwip_flag[x0][y0] is not present, it is inferred to be equal to 0.

The syntax elements intra_lwip_mpm_flag[x0][y0], intra_lwip_mpm_idx[x0][y0] and intra_lwip_mpm_remainder[x0][y0] specify the affine linear weighted intra prediction mode for luma samples. The array indices x0, y0 specify the location (x0, y0) of the top-left luma sample of the considered coding block relative to the top-left luma sample of the picture. When intra_lwip_mpm_flag[x0][y0] is equal to 1, the affine linear weighted intra prediction mode is inferred from a neighboring intra-predicted coding unit according to clause 8.4.X.

When intra_lwip_mpm_flag[x0][y0] is not present, it is inferred to be equal to 1.<end>

intra_subpartitions_split_flag[x0][y0] specifies whether the intra subpartitions split type is horizontal or vertical. When intra_subpartitions_split_flag[x0][y0] is not present, it is inferred as follows:

If intra_lwip_flag[x0][y0] is equal to 1, intra_subpartitions_split_flag[x0][y0] is inferred to be equal to 0.

Otherwise, the following applies:

If cbHeight is greater than MaxTbSizeY, intra_subpartitions_split_flag[x0][y0] is inferred to be equal to 0. Otherwise (cbWidth is greater than MaxTbSizeY), intra_subpartitions_split_flag[x0][y0] is inferred to be equal to 1.

Decoding Process

8.4.1 General Decoding Process for Coding Units Coded in Intra Prediction Mode

Inputs to this process are:

- a luma location (xCb, yCb) specifying the top-left sample of the current coding block relative to the top-left luma sample of the current picture,
- a variable cbWidth specifying the width of the current coding block in luma samples,

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- a variable cbHeight specifying the height of the current coding block in luma samples,
- a variable treeType specifying whether a single or a dual tree is used and if a dual tree is used, it specifies whether the current tree corresponds to the luma or chroma components.

Output of this process is a modified reconstructed picture before in-loop filtering.

The derivation process for quantization parameters as specified in clause 8.7.1 is invoked with the luma location (xCb, yCb), the width of the current coding block in luma samples cbWidth and the height of the current coding block in luma samples cbHeight, and the variable treeType as inputs.

When treeType is equal to SINGLE_TREE or treeType is equal to DUAL_TREE_LUMA, the decoding process for luma samples is specified as follows:

If pcm_flag[xCb][yCb] is equal to 1, the reconstructed picture is modified as follows:

$$S_L[xCb+i][yCb+j] = \text{pcm_sample_luma}[(cbHeight*j) + i] << (\text{BitDepth}_Y - \text{PcmBitDepth}_Y), \quad (8-6)$$

with $i=0 \dots cbWidth-1$, $j=0 \dots cbHeight-1$

Otherwise, the following applies:

1. The luma intra prediction mode is derived as follows:

If intra_lwip_flag[xCb][yCb] is equal to 1, the derivation process for the affine linear weighted intra prediction mode as specified in clause 8.4.X is invoked with the luma location (xCb, yCb), the width of the current coding block in luma samples cbWidth and the height of the current coding block in luma samples cbHeight as input.

Otherwise, the derivation process for the luma intra prediction mode as specified in clause 8.4.2 is invoked with the luma location (xCb, yCb), the width of the current coding block in luma samples cbWidth and the height of the current coding block in luma samples cbHeight as input.

2. The general decoding process for intra blocks as specified in clause 8.4.4.1 is invoked with the luma location (xCb, yCb), the tree type treeType, the variable nTbW set equal to cbWidth, the variable nTbH set equal to cbHeight, the variable predModeIntra set equal to IntraPredModeY[xCb][yCb], and the variable cIdx set equal to 0 as inputs, and the output is a modified reconstructed picture before in-loop filtering.

...
<begin>

8.4.X Derivation Process for Affine Linear Weighted Intra Prediction Mode

Input to this process are:

- a luma location (xCb, yCb) specifying the top-left sample of the current luma coding block relative to the top-left luma sample of the current picture,
- a variable cbWidth specifying the width of the current coding block in luma samples,
- a variable cbHeight specifying the height of the current coding block in luma samples.

In this process, the affine linear weighted intra prediction mode IntraPredModeY[xCb][yCb] is derived. IntraPredModeY[xCb][yCb] is derived by the following ordered steps:

1. The neighboring locations (xNbA, yNbA) and (xNbB, yNbB) are set equal to (xCb-1, yCb) and (xCb, yCb-1), respectively.
2. For X being replaced by either A or B, the variables candLwipModeX are derived as follows:

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The availability derivation process for a block as specified in clause 6.4.X [Ed. (BB): Neighboring blocks availability checking process tbd] is invoked with the location (xCurr, yCurr) set equal to (xCb, yCb) and the neighboring location (xNbY, yNbY) set equal to (xNbX, yNbX) as inputs, and the output is assigned to availableX.

The candidate affine linear weighted intra prediction mode candLwipModeX is derived as follows:

If one or more of the following conditions are true, candLwipModeX is set equal to -1.

The variable availableX is equal to FALSE.

CuPredMode[xNbX][yNbX] is not equal to MODE_INTRA and mh_intra_flag[xNbX][yNbX] is not equal to 1.

pcm_flag[xNbX][yNbX] is equal to 1.

X is equal to B and yCb-1 is less than ((yCb >> CtbLog2SizeY) << CtbLog2SizeY).

Otherwise, the following applies:

The size type derivation process for a block as specified in clause 8.4.X.1 is invoked with the width of the current coding block in luma samples cbWidth and the height of the current coding block in luma samples cbHeight as input, and the output is assigned to variable sized.

If intra_lwip_flag[xNbX][yNbX] is equal to 1, the size type derivation process for a block as specified in clause 8.4.X.1 is invoked with the width of the neighboring coding block in luma samples nbWidthX and the height of the neighboring coding block in luma samples nbHeightX as input, and the output is assigned to variable sizedX.

If sized is equal to sizedX, candLwipModeX is set equal to IntraPredModeY[xNbX][yNbX].

Otherwise, candLwipModeX is set equal to -1.

Otherwise, candLwipModeX is derived using IntraPredModeY[xNbX][yNbX] and sized as specified in Table 8-X1.

3. The candLwipModeList[x] with $x=0 \dots 2$ is derived as follows, using lwipMpmCand[sized] as specified in Table 8-X2:

If candLwipModeA and candLwipModeB are both equal to -1, the following applies:

$$\text{candLwipModeList}[0] = \text{lwipMpmCand}[\text{sized}][0] \quad (8-X1)$$

$$\text{candLwipModeList}[1] = \text{lwipMpmCand}[\text{sized}][1] \quad (8-X2)$$

$$\text{candLwipModeList}[2] = \text{lwipMpmCand}[\text{sized}][2] \quad (8-X3)$$

Otherwise, the following applies:

If candLwipModeA is equal to candLwipModeB or if either candLwipModeA or candLwipModeB is equal to -1, the following applies:

$$\text{candLwipModeList}[0] = (\text{candLwipModeA} \neq -1) ? \text{candLwipModeA} : \text{candLwipModeB} \quad (8-X4)$$

If candLwipModeList[0] is equal to lwipMpmCand[sized][0], the following applies:

$$\text{candLwipModeList}[1] = \text{lwipMpmCand}[\text{sized}][1] \quad (8-X5)$$

$$\text{candLwipModeList}[2] = \text{lwipMpmCand}[\text{sized}][2] \quad (8-X6)$$

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Otherwise, the following applies:

candLwipModeList[1]=lwipMpmCand[sizeId][0] (8-X7)

candLwipModeList[2]=(candLwipModeList[0]
!=lwipMpmCand[sizeId][1])?lwipMpmCand
[sizeId][1]:lwipMpmCand[sizeId][2] (8-X8)

Otherwise, the following applies:

candLwipModeList[0]=candLwipModeA (8-X9)

candLwipModeList[1]=candLwipModeB (8-X10)

If candLwipModeA and candLwipModeB are
both not equal to lwipMpmCand[sizeId][0], the
following applies:

candLwipModeList[2]=lwipMpmCand[sizeId][0] (8-X11)

Otherwise, the following applies:

If candLwipModeA and candLwipModeB are
both not equal to lwipMpmCand[sizeId][1], the
following applies:

candLwipModeList[2]=lwipMpmCand[sizeId][1] (8-X12)

Otherwise, the following applies:

candLwipModeList[2]=lwipMpmCand[sizeId][2] (8-X13)

4. IntraPredModeY[xCb][yCb] is derived by applying the
following procedure:

If intra_lwip_mpm_flag[xCb][yCb] is equal to 1, the
IntraPredModeY[xCb][yCb] is set equal to cand-
LwipModeList[intra_lwip_mpm_idx[xCb][yCb]].

Otherwise, IntraPredModeY[xCb][yCb] is derived by
applying the following ordered steps:

1. When candLwipModeList[i] is greater than cand-
LwipModeList[j] for i=0 . . . 1 and for each i,
j=(i+1) . . . 2, both values are swapped as follows:

(candLwipModeList[i],candLwipModeList[j])=Swap
(candLwipModeList[j],candLwipModeList[i]) (8-X14)

2. IntraPredModeY[xCb][yCb] is derived by the
following ordered steps:

i. IntraPredModeY[xCb][yCb] is set equal to
intra_lwip_mpm_remainder[xCb][yCb].

ii. For i equal to 0 to 2, inclusive, when IntraPred-
ModeY[xCb][yCb] is greater than or equal to
candLwipModeList[i], the value of IntraPred-
ModeY[xCb][yCb] is incremented by one.

The variable IntraPredModeY[x][y] with x=xCb . . . xCb+
cbWidth-1 and y=yCb . . . yCb+cbHeight-1 is set to be
equal to IntraPredModeY[xCb][yCb].

8.4.X.1 Derivation Process for Prediction Block Size Type
Input to this process are:

a variable cbWidth specifying the width of the current
coding block in luma samples,

a variable cbHeight specifying the height of the current
coding block in luma samples.

Output of this process is a variable sizeId.

The variable sizeId is derived as follows:

If both cbWidth and cbHeight are equal to 4, sizeId is set
equal to 0.

Otherwise, if both cbWidth and cbHeight are less than or
equal to 8, sizeId is set equal to 1.

Otherwise, sizeId is set equal to 2.

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TABLE 8-X1

Specification of mapping between intra prediction and affine linear weighted intra prediction modes				
IntraPredModeY[xNbX][yNbX]	block size type sizeId			
	0	1	2	
0	17	0	5	
1	17	0	1	
2, 3	17	10	3	
4, 5	9	10	3	
6, 7	9	10	3	
8, 9	9	10	3	
10, 11	9	10	0	
12, 13	17	4	0	
14, 15	17	6	0	
16, 17	17	7	4	
18, 19	17	7	4	
20, 21	17	7	4	
22, 23	17	5	5	
24, 25	17	5	1	
26, 27	5	0	1	
28, 29	5	0	1	
30, 31	5	3	1	
32, 33	5	3	1	
34, 35	34	12	6	
36, 37	22	12	6	
38, 39	22	12	6	
40, 41	22	12	6	
42, 43	22	14	6	
44, 45	34	14	10	
46, 47	34	14	10	
48, 49	34	16	9	
50, 51	34	16	9	
52, 53	34	16	9	
54, 55	34	15	9	
56, 57	34	13	9	
58, 59	26	1	8	
60, 61	26	1	8	
62, 63	26	1	8	
64, 65	26	1	8	
66	26	1	8	

TABLE 8-X2

Specification of affine linear weighted intra prediction candidate modes			
	candidate mode		
	0	1	2
lwipMpmCand[0]	17	34	5
lwipMpmCand[1]	0	7	16
lwipMpmCand[2]	1	4	6

<end>

8.4.2. Derivation Process for Luma Intra Prediction Mode

Input to this process are:

a luma location (xCb, yCb) specifying the top-left sample
of the current luma coding block relative to the top-left
luma sample of the current picture,

a variable cbWidth specifying the width of the current
coding block in luma samples,

a variable cbHeight specifying the height of the current
coding block in luma samples.

In this process, the luma intra prediction mode IntraPred-
ModeY[xCb][yCb] is derived.

Table 8-1 specifies the value for the intra prediction mode
IntraPredModeY[xCb][yCb] and the associated names.

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TABLE 8-1

Specification of intra prediction mode and associated names	
Intra prediction mode	Associated name
0	INTRA_PLANAR
1	INTRA_DC
2 . . . 66	INTRA_ANGULAR2 . . . INTRA_ANGULAR66
81 . . . 83	INTRA_LT_CCLM, INTRA_L_CCLM, INTRA_T_CCLM

NOTE—:

The intra prediction modes INTRA_LT_CCLM, INTRA_L_CCLM and INTRA_T_CCLM are only applicable to chroma components.

IntraPredModeY[xCb][yCb] is derived by the following ordered steps:

1. The neighboring locations (xNbA, yNbA) and (xNbB, yNbB) are set equal to (xCb-1, yCb+cbHeight-1) and (xCb+cbWidth-1, yCb-1), respectively.
2. For X being replaced by either A or B, the variables candIntraPredModeX are derived as follows:

The availability derivation process for a block as specified in clause <begin>6.4.X [Ed. (BB): Neighboring blocks availability checking process tbd]<end> is invoked with the location (xCurr, yCurr) set equal to (xCb, yCb) and the neighboring location (xNbY, yNbY) set equal to (xNbX, yNbX) as inputs, and the output is assigned to availableX.

The candidate intra prediction mode candIntraPredModeX is derived as follows:

If one or more of the following conditions are true, candIntraPredModeX is set equal to INTRA_PLANAR.

The variable availableX is equal to FALSE.

CuPredMode[xNbX][yNbX] is not equal to MODE_INTRA and ciip_flag[xNbX][yNbX] is not equal to 1.

pcm_flag[xNbX][yNbX] is equal to 1.

X is equal to B and yCb-1 is less than ((yCb>>CtbLog 2SizeY)<<CtbLog 2SizeY).

Otherwise, candIntraPredModeX is derived as follows:

If intra_lwip_flag[xCb][yCb] is equal to 1, candIntraPredModeX is derived by the following ordered steps:

- i. The size type derivation process for a block as specified in clause 8.4.X.1 is invoked with the width of the current coding block in luma samples cbWidth and the height of the current coding block in luma samples cbHeight as input, and the output is assigned to variable sizeId.
- ii. candIntraPredModeX is derived using IntraPredModeY[xNbX][yNbX] and sizeId as specified in Table 8-X3.

Otherwise, candIntraPredModeX is set equal to IntraPredModeY[xNbX][yNbX].

3. The variables ispDefaultMode1 and ispDefaultMode2 are defined as follows:

If IntraSubPartitionsSplitType is equal to ISP_HOR_SPLIT, ispDefaultMode1 is set equal to INTRA_ANGULAR18 and ispDefaultMode2 is set equal to INTRA_ANGULAR5.

Otherwise, ispDefaultMode1 is set equal to INTRA_ANGULAR50 and ispDefaultMode2 is set equal to INTRA_ANGULAR63.

...

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TABLE 8-X3

Specification of mapping between affine linear weighted intra prediction and intra prediction modes			
IntraPredModeY[xNbX][yNbX]	block size type sizeId 2		
	0	1	2
0	0	0	1
1	18	1	1
2	18	0	1
3	0	1	1
4	18	0	18
5	0	22	0
6	12	18	1
7	0	18	0
8	18	1	1
9	2	0	50
10	18	1	0
11	12	0	
12	18	1	
13	18	0	
14	1	44	
15	18	0	
16	18	50	
17	0	1	
18	0	0	
19	50		
20	0		
21	50		
22	0		
23	56		
24	0		
25	50		
26	66		
27	50		
28	56		
29	50		
30	50		
31	1		
32	50		
33	50		
34	50		

8.4.3 Derivation Process for Chroma Intra Prediction Mode

Input to this process are:

- a luma location (xCb, yCb) specifying the top-left sample of the current chroma coding block relative to the top-left luma sample of the current picture,
- a variable cbWidth specifying the width of the current coding block in luma samples,
- a variable cbHeight specifying the height of the current coding block in luma samples.

In this process, the chroma intra prediction mode IntraPredModeC[xCb][yCb] is derived.

The corresponding luma intra prediction mode lumaIntraPredMode is derived as follows:

If intra_lwip_flag[xCb][yCb] is equal to 1, lumaIntraPredMode is derived by the following ordered steps:

- i. The size type derivation process for a block as specified in clause 8.4.X.1 is invoked with the width of the current coding block in luma samples cbWidth and the height of the current coding block in luma samples cbHeight as input, and the output is assigned to variable sizeId.
- ii. The luma intra prediction mode is derived using IntraPredModeY[xCb+cbWidth/2][yCb+cbHeight/2] and sizeId as specified in Table 8-X3 and assigning the value of candIntraPredModeX to lumaIntraPredMode.

Otherwise, lumaIntraPredMode is set equal to IntraPredModeY[xCb+cbWidth/2][yCb+cbHeight/2].

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The chroma intra prediction mode $\text{IntraPredModeC}[\text{xCb}][\text{yCb}]$ is derived using $\text{intra_chroma_pred_mode}[\text{xCb}][\text{yCb}]$ and lumaIntraPredMode as specified in Table 8-2 and Table 8-3.

...
xxx. Intra sample prediction

<begin>

Inputs to this process are:

- a sample location $(\text{xTbCmp}, \text{yTbCmp})$ specifying the top-left sample of the current transform block relative to the top-left sample of the current picture,
- a variable predModelIntra specifying the intra prediction mode,
- a variable nTbW specifying the transform block width,
- a variable nTbH specifying the transform block height,
- a variable nCbW specifying the coding block width,
- a variable nCbH specifying the coding block height,
- a variable cldx specifying the colour component of the current block.

Outputs of this process are the predicted samples $\text{predSamples}[\text{x}][\text{y}]$, with $\text{x}=0 \dots \text{nTbW}-1$, $\text{y}=0 \dots \text{nTbH}-1$.

The predicted samples $\text{predSamples}[\text{x}][\text{y}]$ are derived as follows:

If $\text{intra_lwip_flag}[\text{xTbCmp}][\text{yTbCmp}]$ is equal to 1 and cldx is equal to 0, the affine linear weighted intra sample prediction process as specified in clause 8.4.4.2.X1 is invoked with the location $(\text{xTbCmp}, \text{yTbCmp})$, the intra prediction mode predModelIntra , the transform block width nTbW and height nTbH as inputs, and the output is predSamples .

Otherwise, the general intra sample prediction process as specified in clause 8.4.4.2.X1. is invoked with the location $(\text{xTbCmp}, \text{yTbCmp})$, the intra prediction mode predModelIntra , the transform block width nTbW and height nTbH , the coding block width nCbW and height nCbH , and the variable cldx as inputs, and the output is predSamples .

8.4.4.2.X1 Affine linear weighted intra sample prediction

Inputs to this process are:

- a sample location $(\text{xTbCmp}, \text{yTbCmp})$ specifying the top-left sample of the current transform block relative to the top-left sample of the current picture,
- a variable predModelIntra specifying the intra prediction mode,
- a variable nTbW specifying the transform block width,
- a variable nTbH specifying the transform block height.

Outputs of this process are the predicted samples $\text{predSamples}[\text{x}][\text{y}]$, with $\text{x}=0 \dots \text{nTbW}-1$, $\text{y}=0 \dots \text{nTbH}-1$.

The size type derivation process for a block as specified in clause 8.4.X.1 is invoked with the transform block width nTbW and the transform block height nTbH as input, and the output is assigned to variable sizeId .

Variables numModes , boundarySize , predW , predH and predC are derived using sizeId as specified in Table 8-X4.

TABLE 8-X4

Specification of number of modes, boundary sample size and prediction sizes depending on sizeId					
sizeId	numModes	boundarySize	predW	predH	predC
0	35	2	4	4	4
1	19	4	4	4	4
2	11	4	$\text{Min}(\text{nTbW}, 8)$	$\text{Min}(\text{nTbH}, 8)$	8

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The flag isTransposed is derived as follows:

$$\text{isTransposed} = (\text{predModelIntra} > (\text{numModes}/2)) ? 1 : 0 \quad (8\text{-X15})$$

The flags needUpsBdryHor and needUpsBdryVer are derived as follows:

$$\text{needUpsBdryHor} = (\text{nTbW} > \text{predW}) ? \text{TRUE} : \text{FALSE} \quad (8\text{-X16})$$

$$\text{needUpsBdryVer} = (\text{nTbH} > \text{predH}) ? \text{TRUE} : \text{FALSE} \quad (8\text{-X17})$$

The variables upsBdryW and upsBdryH are derived as follows:

$$\text{upsBdryW} = (\text{nTbH} > \text{nTbW}) ? \text{nTbW} : \text{predW} \quad (8\text{-X18})$$

$$\text{upsBdryH} = (\text{nTbH} > \text{nTbW}) ? \text{predH} : \text{nTbH} \quad (8\text{-X19})$$

The variables lwipW and lwipH are derived as follows:

$$\text{lwipW} = (\text{isTransposed} == 1) ? \text{predH} : \text{predW} \quad (8\text{-X20})$$

$$\text{lwipH} = (\text{isTransposed} == 1) ? \text{predW} : \text{predH} \quad (8\text{-X21})$$

For the generation of the reference samples $\text{refT}[\text{x}]$ with $\text{x}=0 \dots \text{nTbW}-1$ and $\text{refL}[\text{y}]$ with $\text{y}=0 \dots \text{nTbH}-1$, the reference sample derivation process as specified in clause 8.4.4.2.X2 is invoked with the sample location $(\text{xTbCmp}, \text{yTbCmp})$, the transform block width nTbW , the transform block height nTbH as inputs, and top and left reference samples $\text{refT}[\text{x}]$ with $\text{x}=0 \dots \text{nTbW}-1$ and $\text{refL}[\text{y}]$ with $\text{y}=0 \dots \text{nTbH}-1$, respectively, as outputs.

For the generation of the boundary samples $\text{p}[\text{x}]$ with $\text{x}=0 \dots 2 \cdot \text{boundarySize}-1$, the following applies:

The boundary reduction process as specified in clause 8.4.4.2.X3 is invoked for the top reference samples with the block size nTbW , the reference samples refT , the boundary size boundarySize , the upsampling boundary flag needUpsBdryVer , and the upsampling boundary size upsBdryW as inputs, and reduced boundary samples $\text{redT}[\text{x}]$ with $\text{x}=0 \dots \text{boundarySize}-1$ and upsampling boundary samples $\text{upsBdryT}[\text{x}]$ with $\text{x}=0 \dots \text{upsBdryW}-1$ as outputs.

The boundary reduction process as specified in clause 8.4.4.2.X3 is invoked for the left reference samples with the block size nTbH , the reference samples refL , the boundary size boundarySize , the upsampling boundary flag needUpsBdryHor , and the upsampling boundary size upsBdryH as inputs, and reduced boundary samples $\text{redL}[\text{x}]$ with $\text{x}=0 \dots \text{boundarySize}-1$ and upsampling boundary samples $\text{upsBdryL}[\text{x}]$ with $\text{x}=0 \dots \text{upsBdryH}-1$ as outputs.

The reduced top and left boundary samples redT and redL are assigned to the boundary sample array p as follows:

If isTransposed is equal to 1, $\text{p}[\text{x}]$ is set equal to $\text{redL}[\text{x}]$ with $\text{x}=0 \dots \text{boundarySize}-1$ and $\text{p}[\text{x}+\text{boundarySize}]$ is set equal to $\text{redT}[\text{x}]$ with $\text{x}=0 \dots \text{boundarySize}-1$.

Otherwise, $\text{p}[\text{x}]$ is set equal to $\text{redT}[\text{x}]$ with $\text{x}=0 \dots \text{boundarySize}-1$ and $\text{p}[\text{x}+\text{boundarySize}]$ is set equal to $\text{redL}[\text{x}]$ with $\text{x}=0 \dots \text{boundarySize}-1$.

For the intra sample prediction process according to predModelIntra , the following ordered steps apply:

1. The affine linear weighted samples $\text{predLwip}[\text{x}][\text{y}]$, with $\text{x}=0 \dots \text{lwipW}-1$, $\text{y}=0 \dots \text{lwipH}-1$ are derived as follows:

The variable modelId is derived as follows:

$$\text{modelId} = \text{predModelIntra} - (\text{isTransposed} == 1) ? (\text{numModes}/2) : 0 \quad (8\text{-X22})$$

The weight matrix $\text{mWeight}[\text{x}][\text{y}]$ with $\text{x}=0 \dots 2 \cdot \text{boundarySize}-1$, $\text{y}=0 \dots \text{predC} \cdot \text{predC}-1$ is derived using sizeId and modelId as specified in Table 8-XX [TBD: add weight matrices].

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The bias vector $vBias[y]$ with $y=0 \dots predC*predC-1$ is derived using $sizeId$ and $modelId$ as specified in Table 8-XX [TBD: add bias vectors].

The variable sW is derived using $sizeId$ and $modelId$ as specified in Table 8-X5.

The affine linear weighted samples $predLwip[x][y]$, with $x=0.3wipW-1$, $y=0 \dots lwipH-1$ are derived as follows:

$$oW=1<<(sW-1) \quad (8-X23)$$

$$sB=BitDepth_Y-1 \quad (8-X24)$$

$$incW=(predC>lwipW)?2:1 \quad (8-X25)$$

$$incH=(predC>lwipH)?2:1 \quad (8-X26)$$

$$predLwip[x][y]=((\sum_{i=0}^{2*boundarySize-1} mWeight[i] \\ [y*incH*predC+x*incW]*p[i]+(vBias \\ [y*incH*predC+x*incW]<<sB)+oW)>>sW) \quad (8-X27)$$

2. The predicted samples $predSamples[x][y]$, with $x=0 \dots nTbW-1$, $y=0 \dots nTbH-1$ are derived as follows:

When $isTransposed$ is equal to 1, $predLwip[x][y]$, with $x=0 \dots predW-1$, $y=0 \dots predH-1$ is set equal to $predLwip[y][x]$.

If $needUpsBdryVer$ is equal to TRUE or $needUpsBdryHor$ is equal to TRUE, the prediction upsampling process as specified in clause 8.4.4.2.X4 is invoked with the input block width $predW$, the input block height $predH$, affine linear weighted samples $predLwip$, the transform block width $nTbW$, the transform block height $nTbH$, the upsampling boundary width $upsBdryW$, the upsampling boundary height $upsBdryH$, the top upsampling boundary samples $upsBdryT$, and the left upsampling boundary samples $upsBdryL$ as inputs, and the output is the predicted sample array $predSamples$.

Otherwise, $predSamples[x][y]$, with $x=0 \dots nTbW-1$, $y=0 \dots nTbH-1$ is set equal to $predLwip[x][y]$.

TABLE 8-X5

Specification of weight shifts sW depending on $sizeId$ and $modelId$																		
$sizeId$	$modelId$																	
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
0	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
1	8	8	8	9	8	8	8	8	9	8								
2	8	8	8	8	8	8												

8.4.4.2.X2 Reference Sample Derivation Process

Inputs to this process are:

- a sample location $(xTbY, yTbY)$ specifying the top-left luma sample of the current transform block relative to the top-left luma sample of the current picture,
- a variable $nTbW$ specifying the transform block width,
- a variable $nTbH$ specifying the transform block height.

Outputs of this process are the top and left reference samples $refT[x]$ with $x=0 \dots nTbW-1$ and $refL[y]$ with $y=0 \dots nTbH-1$, respectively.

The neighboring samples $refT[x]$ with $x=0 \dots nTbW-1$ and $refL[y]$ with $y=0 \dots nTbH-1$ are constructed samples prior to the in-loop filter process and derived as follows:

The top and left neighboring luma locations $(xNbT, yNbT)$ and $(xNbL, yNbL)$ are specified by:

$$(xNbT, yNbT)=(xTbY+x, yTbY-1) \quad (8-X28)$$

$$(xNbL, yNbL)=(xTbY-1, yTbY+y) \quad (8-X29)$$

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The availability derivation process for a block as specified in clause 6.4.X [Ed. (BB): Neighboring blocks availability checking process tbd] is invoked with the current luma location $(xCurr, yCurr)$ set equal to $(xTbY, yTbY)$ and the top neighboring luma location $(xNbT, yNbT)$ as inputs, and the output is assigned to $availTop[x]$ with $x=0 \dots nTbW-1$.

The availability derivation process for a block as specified in clause 6.4.X [Ed. (BB): Neighboring blocks availability checking process tbd] is invoked with the current luma location $(xCurr, yCurr)$ set equal to $(xTbY, yTbY)$ and the left neighboring luma location $(xNbL, yNbL)$ as inputs, and the output is assigned to $availLeft[y]$ with $y=0 \dots nTbH-1$.

The top reference samples $refT[x]$ with $x=0 \dots nTbW-1$ are derived as follows:

If all $availTop[x]$ with $x=0 \dots nTbW-1$ are equal to TRUE, the sample at the location $(xNbT, yNbT)$ is assigned to $refT[x]$ with $x=0 \dots nTbW-1$.

Otherwise, if $availTop[0]$ is equal to FALSE, all $refT[x]$ with $x=0 \dots nTbW-1$ are set equal to $1<<(\text{BitDepth}_Y-1)$.

Otherwise, reference samples $refT[x]$ with $x=0 \dots nTbW-1$ are derived by the following ordered steps:

1. The variable $lastT$ is set equal to the position x of the first element in the sequence $availTop[x]$ with $x=1 \dots nTbW-1$ that is equal to FALSE.
2. For every $x=0 \dots lastT-1$, the sample at the location $(xNbT, yNbT)$ is assigned to $refT[x]$.
3. For every $x=lastT \dots nTbW-1$, $refT[x]$ is set equal to $refT[lastT-1]$.

The left reference samples $refL[y]$ with $x=0 \dots nTbH-1$ are derived as follows:

If all $availLeft[y]$ with $y=0 \dots nTbH-1$ are equal to TRUE, the sample at the location $(xNbL, yNbL)$ is assigned to $refL[y]$ with $y=0 \dots nTbH-1$.

Otherwise, if $availLeft[0]$ is equal to FALSE, all $refL[y]$ with $y=0 \dots nTbH-1$ are set equal to $1<<(\text{BitDepth}_Y-1)$.

Otherwise, reference samples $refL[y]$ with $y=0 \dots nTbH-1$ are derived by the following ordered steps:

1. The variable $lastL$ is set equal to the position y of the first element in the sequence $availLeft[y]$ with $y=1 \dots nTbH-1$ that is equal to FALSE.
2. For every $y=0 \dots lastL-1$, the sample at the location $(xNbL, yNbL)$ is assigned to $refL[y]$.
3. For every $y=lastL \dots nTbH-1$, $refL[y]$ is set equal to $refL[lastL-1]$.

Specification of the boundary reduction process

Inputs to this process are:

- a variable $nTbX$ specifying the transform block size,
- reference samples $refX[x]$ with $x=0 \dots nTbX-1$,
- a variable $boundarySize$ specifying the downsampled boundary size,

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a flag needUpsBdryX specifying whether intermediate boundary samples are required for upsampling,
a variable upsBdrySize specifying the boundary size for upsampling.

Outputs of this process are the reduced boundary samples redX[x] with $x=0 \dots \text{boundarySize}-1$ and upsampling boundary samples upsBdryX[x] with $x=0 \dots \text{upsBdrySize}-1$.

The upsampling boundary samples upsBdryX[x] with $x=0 \dots \text{upsBdrySize}-1$ are derived as follows:

If needUpsBdryX is equal to TRUE and upsBdrySize is less than nTbX, the following applies:

$$\text{uDwn} = \text{nTbX} / \text{upsBdrySize} \quad (8\text{-X30})$$

$$\text{upsBdryX}[x] = (\sum_{i=0}^{\text{uDwn}-1} \text{refX}[x * \text{uDwn} + i] + (1 \ll (\text{Log } 2(\text{uDwn}) - 1))) \gg \text{Log } 2(\text{uDwn}) \quad (8\text{-X31})$$

Otherwise (upsBdrySize is equal to nTbX), upsBdryX[x] is set equal to refX[x].

The reduced boundary samples redX[x] with $x=0 \dots \text{boundarySize}-1$ are derived as follows:

If boundarySize is less than upsBdrySize, the following applies:

$$\text{bDwn} = \text{upsBdrySize} / \text{boundarySize} \quad (8\text{-X32})$$

$$\text{redX}[x] = (\sum_{i=0}^{\text{bDwn}-1} \text{upsBdryX}[x * \text{bDwn} + i] + (1 \ll (\text{Log } 2(\text{bDwn}) - 1))) \gg \text{Log } 2(\text{bDwn}) \quad (8\text{-X33})$$

Otherwise (boundarySize is equal to upsBdrySize), redX[x] is set equal to upsBdryX[x].

8.4.4.2.X4 Specification of the Prediction Upsampling Process

Inputs to this process are:

a variable predW specifying the input block width,
a variable predH specifying the input block height,
affine linear weighted samples predLwip[x][y], with $x=0 \dots \text{predW}-1, y=0 \dots \text{predH}-1$,
a variable nTbW specifying the transform block width,
a variable nTbH specifying the transform block height,
a variable upsBdryW specifying the upsampling boundary width,
a variable upsBdryH specifying the upsampling boundary height,
top upsampling boundary samples upsBdryT[x] with $x=0 \dots \text{upsBdryW}-1$,
left upsampling boundary samples upsBdryL[x] with $x=0 \dots \text{upsBdryH}-1$.

Outputs of this process are the predicted samples predSamples[x][y], with $x=0 \dots \text{nTbW}-1, y=0 \dots \text{nTbH}-1$.

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The sparse predicted samples predSamples[m][n] are derived from predLwip[x][y], with $x=0 \dots \text{predW}-1, y=0 \dots \text{predH}-1$ as follows:

$$\text{upHor} = \text{nTbW} / \text{predW} \quad (8\text{-X34})$$

$$\text{upVer} = \text{nTbH} / \text{predH} \quad (8\text{-X35})$$

$$\text{predSamples}[(x+1) * \text{upHor} - 1][(y+1) * \text{upVer} - 1] = \text{predLwip}[x][y] \quad (8\text{-X36})$$

The top boundary samples upsBdryT[x] with $x=0 \dots \text{upsBdryW}-1$ are assigned to predSamples[m][-1] as follows:

$$\text{predSamples}[(x+1) * (\text{nTbW} / \text{upsBdryW}) - 1][-1] = \text{upsBdryT}[x] \quad (8\text{-X37})$$

The left boundary samples upsBdryL[y] with $y=0 \dots \text{upsBdryH}-1$ are assigned to predSamples[-1 n] as follows:

$$\text{predSamples}[-1][(y+1) * (\text{nTbH} / \text{upsBdryH}) - 1] = \text{upsBdryL}[y] \quad (8\text{-X38})$$

The predicted samples predSamples[x][y], with $x=0 \dots \text{nTbW}-1, y=0 \dots \text{nTbH}-1$ are derived as follows:

If nTbH is greater than nTbW, the following ordered steps apply:

1. When upHor is greater than 1, horizontal upsampling for all sparse positions (xHor, yHor)=(m*upHor-1, n*upVer-1) with $m=0 \dots \text{predW}-1, n=1 \dots \text{predH}$ is applied with dX=1 ... upHor-1 as follows:

$$\text{predSamples}[x\text{Hor} + dX][y\text{Hor}] = ((\text{upHor} - dX) * \text{predSamples}[x\text{Hor}][y\text{Hor}] + dX * \text{predSamples}[x\text{Hor} + \text{upHor}][y\text{Hor}]) / \text{upHor} \quad (8\text{-X39})$$

2. Vertical upsampling for all sparse positions (xVer, yVer)=(m, n*upVer-1) with $m=0 \dots \text{nTbW}-1, n=0 \dots \text{predH}-1$ is applied with dY=1 ... upVer-1 as follows:

$$\text{predSamples}[x\text{Ver}][y\text{Ver} + dY] = ((\text{upVer} - dY) * \text{predSamples}[x\text{Ver}][y\text{Ver}] + dY * \text{predSamples}[x\text{Ver}][y\text{Ver} + \text{upVer}]) / \text{upVer} \quad (8\text{-X40})$$

Otherwise, the following ordered steps apply:

1. When upVer is greater than 1, vertical upsampling for all sparse positions (xVer, yVer)=(m*upHor-1, n*upVer-1) with $m=1 \dots \text{predW}, n=0 \dots \text{predH}-1$ is applied with dY=1 ... upVer-1 as specified in (8-X40).

2. Horizontal upsampling for all sparse positions (xHor, yHor)=(m*upHor-1, n) with $m=0 \dots \text{predW}-1, n=0 \dots \text{nTbH}-1$ is applied with dX=1 ... upHor-1 as specified in (8-X39).

<end>

TABLE 9-9

Syntax elements and associated binarizations			
Syntax structure	Syntax element	Binarization	
		Process	Input parameters
coding_unit()	cu_skip_flag[][]	FL	cMax = 1
	pred_mode_ibc_flag	FL	cMax = 1
	pred_mode_flag	FL	cMax = 1
	<begin>intra_lwip_flag[][]	FL	cMax = 1
	intra_lwip_mpm_flag[][]	FL	cMax = 1
	intra_lwip_mpm_idx[][]	TR	cMax = 2, cRiceParam = 0
	intra_lwip_mpm_remainder[][]	FL	cMax = (cbWidth == 4 && cbHeight == 4) ? 31 : ((cbWidth <= 8 && cbHeight <= 8) ? 15 : 7)
	...		

TABLE 9-15

Assignment of ctxInc to syntax elements with context coded bins						
Syntax element	binIdx					
	0	1	2	3	4	≥ 5
...	terminate	na	na	na	na	na
intra_lwip_flag[][]	(Abs(Log2(cbWidth) – Log2(cbHeight)) > 1) ? 3 : (0,1,2 (clause 9.5.4.2.2))	na	na	na	na	na
intra_lwip_mpm_flag[][]	0	na	na	na	na	na
intra_lwip_mpm_idx[][]	bypass	bypass	na	na	na	na
intra_lwip_mpm_remainder[][]	bypass	bypass	bypass	bypass	bypass	na

TABLE 9-16

Specification of ctxInc using left and above syntax elements			
Syntax element	condL	condA	ctxSetIdx
...			
intra_lwip_flag[x0][y0] intra_lwip_flag[xNbL][yNbL] intra_lwip_flag[xNbA][yNbA]			0
...			

<end>

Summary of ALWIP

For predicting the samples of a rectangular block of width W and height H, affine linear weighted intra prediction (ALWIP) takes one line of H reconstructed neighboring boundary samples left of the block and one line of W reconstructed neighboring boundary samples above the block as input. If the reconstructed samples are unavailable, they are generated as it is done in the conventional intra prediction. ALWIP is only applied to luma intra block. For chroma intra block, the conventional intra coding modes are applied.

The generation of the prediction signal is based on the following three steps:

1. Out of the boundary samples, four samples in the case of W=H=4 and eight samples in all other cases are extracted by averaging.
2. A matrix vector multiplication, followed by addition of an offset, is carried out with the averaged samples as an input. The result is a reduced prediction signal on a subsampled set of samples in the original block.
3. The prediction signal at the remaining positions is generated from the prediction signal on the subsampled set by linear interpolation which is a single step linear interpolation in each direction.

If an ALWIP mode is to be applied, the index predmode of the ALWIP mode is signaled using a MPM-list with 3 MPMS. Here, the derivation of the MPMS is performed using the intra-modes of the above and the left PU as follows. There are three fixed tables map_angular_to_alwip_idx, idx $\in\{0,1,2\}$ that assign to each conventional intra prediction mode predmode_{Angular} an ALWIP mode

$$\text{predmode}_{ALWIP} = \text{map_angular_to_alwip_idx}[\text{predmode}_{Angular}].$$

For each PU of width W and height H one defines an index

$$\text{idx}(PU) = \text{idx}(W, H) \in \{0, 1, 2\}$$

that indicates from which of the three sets the ALWIP-parameters are to be taken.

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If the above Prediction Unit PU_{above} is available, belongs to the same CTU as the current PU and is in intra mode, if idx(PU)=idx(PU above) and if ALWIP is applied on PU_{above} with ALWIP-mode predmode_{ALWIP}^{above}, one puts

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$$\text{mode}_{ALWIP}^{\text{above}} = \text{predmode}_{ALWIP}^{\text{above}}.$$

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If the above PU is available, belongs to the same CTU as the current PU and is in intra mode and if a conventional intra prediction mode predmode_{Angular}^{above} is applied on the above PU, one puts

$$\text{mode}_{ALWIP}^{\text{above}} = \text{map_angular_to_alwip_idx}(PU_{\text{above}})[\text{predmode}_{Angular}^{\text{above}}].$$

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In all other cases, one puts

$$\text{mode}_{ALWIP}^{\text{above}} = -1$$

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which means that this mode is unavailable. In the same way but without the restriction that the left PU needs to belong to the same CTU as the current PU, one derives a mode mode_{ALWIP}^{left}.

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Finally, three fixed default lists list_{idx}, idx $\in\{0,1,2\}$ are provided, each of which contains three distinct ALWIP modes. Out of the default list list_{idx(PU)} and the modes mode_{ALWIP}^{above} and mode_{ALWIP}^{left}, one constructs three distinct MPMS by substituting -1 by default values as well as eliminating repetitions.

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For the luma MPM-list derivation, whenever a neighboring luma block is encountered which uses an ALWIP-mode predmode_{ALWIP}, this block is treated as if it was using the conventional intra-prediction mode predmode_{Angular}.

$$\text{predmode}_{Angular} = \text{map_alwip_to_angular_idx}(PU)[\text{predmode}_{ALWIP}].$$

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3 Transform in VVC

3.1 Multiple Transform Selection (MTS)

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In addition to DCT-II which has been employed in HEVC, a Multiple Transform Selection (MTS) scheme is used for residual coding both inter and intra coded blocks. It uses multiple selected transforms from the DCT8/DST7. The newly introduced transform matrices are DST-VII and DCT-VIII.

3.2 Reduced Secondary Transform (RST) Proposed in JVET-N0193

Reduced secondary transform (RST) applies 16×16 and 16×64 non-separable transform for 4×4 and 8×8 blocks, respectively. Primary forward and inverse transforms are still performed the same way as two 1-D horizontal/vertical transform passes. Secondary forward and inverse transforms are a separate process step from that of primary transforms. For encoder, primary forward transform is performed first, then followed by secondary forward transform and quantization, and CABAC bit encoding. For decoder, CABAC bit decoding and inverse quantization, then Secondary inverse transform is performed first, then followed by primary inverse transform. RST applies only to intra coded TUs in both intra slice and inter slices.

3.3 A Unified MPM List for Intra Mode Coding in JVET-N0185

A unified 6-MPM list is proposed for intra blocks irrespective of whether Multiple Reference Line (MRL) and Intra sub-partition (ISP) coding tools are applied or not. The MPM list is constructed based on intra modes of the left and above neighboring block as in VTM4.0. Suppose the mode of the left is denoted as Left and the mode of the above block is denoted as Above, the unified MPM list is constructed as follows:

When a neighboring block is not available, its intra mode is set to Planar by default.

If both modes Left and Above are non-angular modes:

a. MPM list → {Planar, DC, V, H, V-4, V+4}

If one of modes Left and Above is angular mode, and the other is non-angular:

a. Set a mode Max as the larger mode in Left and Above
b. MPM list → {Planar, Max, DC, Max-1, Max+1, Max-2}

If Left and Above are both angular and they are different:

a. Set a mode Max as the larger mode in Left and Above
b. if the difference of mode Left and Above is in the range of 2 to 62, inclusive

i. MPM list → {Planar, Left, Above, DC, Max-1, Max+1}

c. Otherwise

i. MPM list → {Planar, Left, Above, DC, Max-2, Max+2}

If Left and Above are both angular and they are the same:

a. MPM list → {Planar, Left, Left-1, Left+1, DC, Left-2}

Besides, the first bin of the MPM index codeword is CABAC context coded. In total three contexts are used, corresponding to whether the current intra block is MRL enabled, ISP enabled, or a normal intra block.

The left neighboring block and above neighboring block used in the unified MPM list construction is A2 and B2 as shown in FIG. 10.

One MPM flag is firstly coded. If the block is coded with one of mode in the MPM list, an MPM index is further coded. Otherwise, an index to the remaining modes (excluding MPMs) is coded.

4 Examples of Drawbacks in Existing Implementations

The design of ALWIP in JVET-N0217 has the following problems:

- 1) At the March 2019 JVET meeting, a unified 6-MPM list generation was adopted for MRL mode, ISP mode, and normal intra mode. But the affine linear weighted prediction mode uses a different 3-MPM list construction which makes the MPM list construction complicated. A complex MPM list construction might compromise the throughput of the decoder, in particular for small blocks such as 4×4 samples.

2) ALWIP is only applied to luma component of the block. For the chroma component of an ALWP coded block, a chroma mode index is coded and sent to decoder, which could result in unnecessary signaling.

3) The interactions of ALWIP with other coding tools should be considered.

4) When calculating upsBdryX in upsBdryX[x] = $(\sum_{i=0}^{uDwn-1} \text{refX}[x * uDwn + i] + (1 << (\text{Log}_2(uDwn) - 1))) >> \text{Log}_2(uDwn)$ (8-X31), it is possible that $\text{Log}_2(uDwn) - 1$ is equal to -1, while left shifted with -1 is undefined.

5) When upsampling the prediction samples, no rounding is applied.

6) In the deblocking process, ALWIP coded blocks are treated as normal intra-blocks.

5 Exemplary Methods for Matrix-Based Intra Coding

Embodiments of the presently disclosed technology overcome drawbacks of existing implementations, thereby providing video coding with higher coding efficiencies but lower computational complexity. Matrix-based intra prediction methods for video coding, and as described in the present document, may enhance both existing and future video coding standards, is elucidated in the following examples described for various implementations. The examples of the disclosed technology provided below explain general concepts and are not meant to be interpreted as limiting. In an example, unless explicitly indicated to the contrary, the various features described in these examples may be combined.

In the following discussion, an intra-prediction mode refers to an angular intra prediction mode (including DC, planar, CCLM and other possible intra prediction modes); while an intra mode refers to normal intra mode, or MRL, or ISP or ALWIP.

In the following discussion, “Other intra modes” may refer to one or multiple intra modes except ALWIP, such as normal intra mode, or MRL, or ISP.

In the following discussion, $\text{SatShift}(x, n)$ is defined as

$$\text{SatShift}(x, n) = \begin{cases} (x + \text{offset0}) >> n & \text{if } x \geq 0 \\ -((-x + \text{offset1}) >> n) & \text{if } x < 0 \end{cases}$$

$\text{Shift}(x, n)$ is defined as $\text{Shift}(x, n) = (x + \text{offset0}) >> n$.

In one example, offset0 and/or offset1 are set to $(1 << n) >> 1$ or $(1 << (n-1))$. In another example, offset0 and/or offset1 are set to 0.

In another example, $\text{offset0} = \text{offset1} = ((1 << n) >> 1) - 1$ or $((1 << (n-1))) - 1$.

$\text{Clip3}(\text{min}, \text{max}, x)$ is defined as

$$\text{Clip3}(\text{Min}, \text{Max}, x) = \begin{cases} \text{Min} & \text{if } x < \text{Min} \\ \text{Max} & \text{if } x > \text{Max} \\ x & \text{Otherwise} \end{cases}$$

MPM List Construction for ALWIP

1. It is proposed that the whole or partial of the MPM list for ALWIP may be constructed according to the whole or partial procedure to construct the MPM list for non-ALWIP intra mode (such as normal intra mode, MRL, or ISP).

a. In one example, the size of the MPM list for ALWIP may be the same as that of the MPM list for non-ALWIP intra mode.

i. For example, the size of MPM list is 6 for both ALWIP and non-ALWIP intra modes.

- b. In one example, the MPM list for ALWIP may be derived from the MPM list for non-ALWIP intra mode.
- i. In one example, the MPM list for non-ALWIP intra mode may be firstly constructed. Afterwards, partial or all of them may be converted to the MPMs which may be further added to the MPM list for ALWIP coded blocks.
 - 1) Alternatively, furthermore, when adding a converted MPM to the MPM list for ALWIP coded blocks, pruning may be applied.
 - 2) Default modes may be added to the MPM list for ALWIP coded blocks.
 - a. In one example, default modes may be added before those converted from the MPM list of non-ALWIP intra mode.
 - b. Alternatively, default modes may be added after those converted from the MPM list of non-ALWIP intra mode.
 - c. Alternatively, default modes may be added in an interleaved way with those converted from the MPM list of non-ALWIP intra mode.
 - d. In one example, the default modes may be fixed to be the same for all kinds of blocks.
 - e. Alternatively, the default modes may be determined according to coded information, such as availability of neighboring blocks, mode information of neighboring blocks, block dimension.
- ii. In one example, one intra-prediction mode in the MPM list for non-ALWIP intra mode may be converted to its corresponding ALWIP intra-prediction mode, when it is put into the MPM list for ALWIP.
 - 1) Alternatively, all the intra-prediction modes in the MPM list for non-ALWIP intra modes may be converted to corresponding ALWIP intra-prediction modes before being used to construct the MPM list for ALWIP.
 - 2) Alternatively, all the candidate intra-prediction modes (may include the intra-prediction modes from neighboring blocks and default intra-prediction modes such as Planar and DC) may be converted to corresponding ALWIP intra-prediction modes before being used to construct the MPM list for non-ALWIP intra modes, if the MPM list for non-ALWIP intra modes may be further used to derive the MPM list for ALWIP.
 - 3) In one example, two converted ALWIP intra-prediction modes may be compared.
 - a. In one example, if they are the same, only one of them may be put into the MPM list for ALWIP.
 - b. In one example, if they are the same, only one of them may be put into the MPM list for non-ALWIP intra modes.
- iii. In one example, K out of S intra-prediction modes in the MPM list for non-ALWIP intra modes may be picked as the MPM list for ALWIP mode. E.g., K is equal to 3 and S is equal to 6.
 - 1) In one example, the first K intra-prediction modes in the MPM list for non-ALWIP intra modes may be picked as the MPM list for ALWIP mode.
- 2. It is proposed that the one or multiple neighboring blocks used to derive the MPM list for ALWIP may also

- be used to used derive the MPM list for non-ALWIP intra modes (such as normal intra mode, MRL, or ISP).
- a. In one example, the neighboring block left to the current block used to derive the MPM list for ALWIP should be the same as that used to derive the MPM list for non-ALWIP intra modes.
 - i. Suppose the top-left corner of the current block is (xCb, yCb), the width and height of the current block are W and H, then in one example, the left neighboring block used to derive the MPM list for both ALWIP and non-ALWIP intra modes may cover the position (xCb-1, yCb). In an alternative example, the left neighboring block used to derive the MPM list for both ALWIP and non-ALWIP intra modes may cover the position (xCb-1, yCb+H-1).
 - ii. For example, the left neighboring block and above neighboring block used in the unified MPM list construction is A2 and B2 as shown in FIG. 10.
- b. In one example, the neighboring block above to the current block used to derive the MPM list for ALWIP should be the same as that used to derive the MPM list for non-ALWIP intra modes.
 - i. Suppose the top-left corner of the current block is (xCb, yCb), the width and height of the current block are W and H, then in one example, the above neighboring block used to derive the MPM list for both ALWIP and non-ALWIP intra modes may cover the position (xCb, yCb-1). In an alternative example, the above neighboring block used to derive the MPM list for both ALWIP and non-ALWIP intra modes may cover the position (xCb+W-1, yCb-1).
 - ii. For example, the left neighboring block and above neighboring block used in the unified MPM list construction is A1 and B1 as shown in FIG. 10.
- 3. It is proposed that the MPM list for ALWIP may be constructed in different ways according to the width and/or height of the current block.
 - a. In one example, different neighboring blocks may be accessed for different block dimensions.
- 4. It is proposed that the MPM list for ALWIP and the MPM list for non-ALWIP intra modes may be constructed with the same procedure but with different parameters.
 - a. In one example, K out of S intra-prediction modes in the MPM list construction procedure of non-ALWIP intra modes may be derived for the MPM list used in ALWIP mode. E.g., K is equal to 3 and S is equal to 6.
 - i. In one example, the first K intra-prediction modes in the MPM list construction procedure may be derived for the MPM list used in ALWIP mode.
 - b. In one example, the first mode in the MPM list may be different.
 - i. For example, the first mode in the MPM list for non-ALWIP intra modes may be Planar, but it may be a Mode X0 in the MPM list for ALWIP.
 - 1) In one example, X0 may be the ALWIP intra-prediction mode converted from Planar.
 - c. In one example, stuffing modes in the MPM list may be different.
 - i. For example, the first three stuffing modes in the MPM list for non-ALWIP intra modes may be DC, Vertical and Horizontal, but they may be Mode X1, X2, X3 in the MPM list for ALWIP.

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- 1) In one example, X1, X2, X3 may be different for different sized.
- ii. In one example, the number of stuffing mode may be different.
- d. In one example, neighboring modes in the MPM list may be different.
 - i. For example, the normal intra-prediction modes of neighboring blocks are used to construct the MPM list for non-ALWIP intra modes. And they are converted to ALWIP intra-prediction modes to construct the MPM list for ALWIP mode.
- e. In one example, the shifted modes in the MPM list may be different.
 - i. For example, X+K0 where X is a normal intra-prediction mode and K0 is an integer may be put into the MPM list for non-ALWIP intra modes. And Y+K1 where Y is an ALWIP intra-prediction mode and K1 is an integer may be put into the MPM list for ALWIP, where K0 may be different from K1.
 - 1) In one example, K1 may depend on the width and height.
5. It is proposed that a neighboring block is treated as unavailable if it is coded with ALWIP when constructing the MPM list for the current block with non-ALWIP intra modes.
 - a. Alternatively, a neighboring block is treated as being coded with a predefined intra-prediction mode (such as Planar) if it is coded with ALWIP when constructing the MPM list for the current block with non-ALWIP intra modes.
6. It is proposed that a neighboring block is treated as unavailable if it is coded with non-ALWIP intra modes when constructing the MPM list for the current block with ALWIP mode.
 - a. Alternatively, a neighboring block is treated as being coded with a predefined ALWIP intra-prediction mode X if it is coded with non-ALWIP intra modes when constructing the MPM list for the current block with ALWIP mode.
 - i. In one example, X may depend on the block dimensions, such as width and/or height.
7. It is proposed to remove the storage of ALWIP flag from line buffer.
 - a. In one example, when the 2nd block to be accessed is located in a different LCU/CTU row/region compared to the current block, the conditional check of whether the 2nd block is coded with ALWIP is skipped.
 - b. In one example, when the 2nd block to be accessed is located in a different LCU/CTU row/region compared to the current block, the 2nd block is treated in the same way as non-ALWIP mode, such as treated as normal intra coded block.
8. When encoding the ALWIP flag, no more than K (K>=0) contexts may be used.
 - a. In one example, K=1.
9. It is proposed to store the converted intra prediction mode of ALWIP coded blocks instead of directly storing the mode index associated with the ALWIP mode.
 - a. In one example, the decoded mode index associated with one ALWIP coded block is mapped to the normal intra mode, such as according to map_alwip_to_angular as described in Section 2.5.7.
 - b. Alternatively, furthermore, the storage of ALWIP flag is totally removed.

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- c. Alternatively, furthermore, the storage of ALWIP mode is totally removed.
 - d. Alternatively, furthermore, condition check of whether one neighboring/current block is coded with ALWIP flag may be skipped.
 - e. Alternatively, furthermore, the conversion of modes assigned for ALWIP coded blocks and normal intra predictions associated with one accessed block may be skipped.
10. ALWIP on Different Color Components
10. It is proposed that an inferred chroma intra mode (e.g., DM mode) might be always applied if the corresponding luma block is coded with ALWIP mode.
 - a. In one example, chroma intra mode is inferred to be DM mode without signaling if the corresponding luma block is coded with ALWIP mode.
 - b. In one example, the corresponding luma block may be the one covering the corresponding sample of a chroma sample located at a given position (e.g., top-left of current chroma block, center of current chroma block).
 - c. In one example, the DM mode may be derived according to the intra prediction mode of the corresponding luma block, such as via mapping the (ALWIP) mode to one of the normal intra mode.
 11. When the corresponding luma block of the chroma blocks is coded with ALWIP mode, several DM modes may be derived.
 12. It is proposed that a special mode is assigned to the chroma blocks if one corresponding luma block is coded with ALWIP mode.
 - a. In one example, the special mode is defined to be a given normal intra prediction mode regardless the intra prediction mode associated with the ALWIP coded blocks.
 - b. In one example, different ways of intra prediction may be assigned to this special mode.
 13. It is proposed that ALWIP may also be applied to chroma components.
 - a. In one example, the matrix and/or bias vector may be different for different color components.
 - b. In one example, the matrix and/or bias vector may be predefined jointly for Cb and Cr.
 - i. In one example, Cb and Cr component may be concatenated.
 - ii. In one example, Cb and Cr component may be interleaved.
 - c. In one example, the chroma component may share the same ALWIP intra-prediction mode as the corresponding luma block.
 - i. In one example, the same ALWIP intra-prediction mode is applied on the chroma component if the corresponding luma block applies the ALWIP mode and the chroma block is coded with DM mode.
 - ii. In one example, the same ALWIP intra-prediction mode is applied on the chroma component and the linear interpolation thereafter can be skipped.
 - iii. In one example, the same ALWIP intra-prediction mode is applied on the chroma component with a subsampled matrix and/or bias vector.
 - d. In one example, the number of ALWIP intra-prediction modes for different component may be different.
 - i. For example, the number of ALWIP intra-prediction modes for chroma components may be less than that for luma component for the same block width and height.

Applicability of ALWIP

14. It is proposed that whether ALWIP can be applied may be signaled.
- For example, it may be signaled at sequence level (e.g. in SPS), at picture level (e.g. in PPS or picture header), at slice level (e.g. in slice header), at tile group level (e.g. in tile group header), at tile level, at CTU row level, or at CTU level.
 - For example, `intra_lwip_flag` may not be signaled and inferred to be 0 if ALWIP cannot be applied.
15. It is proposed that whether ALWIP can be applied may depend on the block width (W) and/or height (H).
- For example, ALWIP may not be applied if $W \geq T1$ (or $W > T1$) and $H \geq T2$ (or $H > T2$). E.g. $T1=T2=32$;
 - For example, ALWIP may not be applied if $W \leq T1$ (or $W < T1$) and $H \leq T2$ (or $H < T2$). E.g. $T1=T2=32$;
 - For example, ALWIP may not be applied if $W \geq T1$ (or $W > T1$) or $H \geq T2$ (or $H > T2$). E.g. $T1=T2=32$;
 - For example, ALWIP may not be applied if $W \leq T1$ (or $W < T1$) or $H \leq T2$ (or $H < T2$). E.g. $T1=T2=32$, or $T1=T2=8$;
 - For example, ALWIP may not be applied if $W+H \geq T$ (or $W*H > T$). E.g. $T=256$;
 - For example, ALWIP may not be applied if $W+H \leq T$ (or $W*H < T$). E.g. $T=256$;
 - For example, ALWIP may not be applied if $W*H \geq T$ (or $W*H > T$). E.g. $T=256$;
 - For example, ALWIP may not be applied if $W*H \leq T$ (or $W*H < T$). E.g. $T=256$;
 - For example, `intra_lwip_flag` may not be signaled and inferred to be 0 if ALWIP cannot be applied.

Calculation Problems in ALWIP

16. It is proposed that any shift operation involved in ALWIP can only left shift or right shift a number by S, where S must be larger or equal to 0.
- In one example, the right shift operation may be different when S is equal to 0 or larger than 0.
 - In one example, $\text{upsBdryX}[x]$ should be calculated as $\text{upsBdryX}[x] = (\sum_{i=0}^{uDwn-1} \text{refX}[x*uDwn+i] + (1 \ll (\text{Log } 2(uDwn)-1))) \gg \text{Log } 2(uDwn)$ when $uDwn > 1$, and $\text{upsBdryX}[x] = \sum_{i=0}^{uDwn-1} \text{refX}[x*uDwn+i]$ when $uDwn$ is equal to 1.
 - In one example, $\text{upsBdryX}[x]$ should be calculated as $\text{upsBdryX}[x] = (\sum_{i=0}^{uDwn-1} \text{refX}[x*uDwn+i] + (1 \ll (\text{Log } 2(uDwn)-1))) \gg \text{Log } 2(uDwn)$
17. It is proposed that the results should be rounded toward-zero or away-from-zero in the up-sampling process of ALWIP.
- In one example,

$$\text{predSamples}[xHor+dX][yHor] = ((\text{upHor}-dX)*\text{predSamples}[xHor][yHor]+dX*\text{predSamples}[xHor+\text{upHor}][yHor]+\text{offsetHor})/\text{upHor} \quad (8-X39)$$

and

$$\text{predSamples}[xVer][yVer+dY] = ((\text{upVer}-dY)*\text{predSamples}[xVer][yVer]+dY*\text{predSamples}[xVer][yVer+\text{upVer}]+\text{offsetVer})/\text{upVer} \quad (8-X40)$$

where `offsetHor` and `offsetVer` are integers. For example, `offsetHor=upHor/2` and `offsetVer=upVer/2`. Interaction with other coding tools

18. It is proposed that ALWIP may be used for a CIIP-coded block.
- In one example, in a CIIP-coded block, it may be explicitly signaled whether an ALWIP intra-predic-

- tion mode or a normal intra prediction mode such as Planar is used to generate the intra prediction signal.
- In one example, it may be implicitly inferred whether an ALWIP intra-prediction mode or a normal intra prediction mode such as Planar may be used to generate the intra prediction signal.
 - In one example, ALWIP intra-prediction mode may never be used in a CIIP coded block.
 - Alternatively, normal intra prediction may never be used in a CIIP coded block.
 - In one example, it may be inferred from information of neighboring blocks whether an ALWIP intra-prediction mode or a normal intra prediction mode such as Planar is used to generate the intra prediction signal.
19. It is proposed that the whole or partial of the procedure used to down-sample the neighboring luma samples in the CCLM mode may be used to down-sample the neighboring samples in the ALWIP mode.
- Alternatively, the whole or partial of the procedure used to down-sample the neighboring luma samples in the ALWIP mode may be used to down-sample the neighboring samples in the CCLM mode.
 - The down-sampling procedure may be invoked with different parameters/arguments when it is used in the CCLM process and ALWIP process.
 - In one example, the down-sampling method (such as selection of neighboring luma locations, down-sampling filters) in the CCLM process may be utilized in the ALWIP process.
 - The procedure used to down-sample the neighboring luma samples at least include the selection of down-sampled positions, the down-sampling filters, the rounding and clipping operations.
20. It is proposed that a block coded with ALWIP mode cannot apply RST or/and secondary transform or/and rotation transform or/and Non-Separable Secondary Transform (NSST).
- In one example, whether such constraint may be applied or not may depend on the dimension information of the block, e.g., same as conditions described in (15).
 - Alternatively, ALWIP mode may be disallowed when RST or/and secondary transform or/and rotation transform or/and NSST is applied.
 - Alternatively, a block coded with ALWIP mode may apply RST or/and secondary transform or/and rotation transform or/and Non-Separable Secondary Transform (NSST).
 - In one example, the selection of transform matrix may depend the ALWIP intra-prediction mode.
 - In one example, the selection of transform matrix may depend the normal intra-prediction mode which is converted from the ALWIP intra-prediction mode.
 - In one example, the selection of transform matrix may depend the classification on the normal intra-prediction mode which is converted from the ALWIP intra-prediction mode.
21. It is proposed that a block coded with ALWIP mode cannot apply Block-based DPCM (BDPCM) or Residue RDPCM.
- Alternatively, ALWIP mode may be disallowed when BDPCM or RDPCM is applied.
22. It is proposed that a block coded with ALWIP mode may only use DCT-II as the transform.

- a. In one example, the signalling of transform matrix indices is always skipped.
- b. Alternatively, it is proposed that the transform used for a block coded with ALWIP mode may be implicitly derived instead of explicitly signaled. For example, the transform may be selected following the way proposed in JVET-M0303.
- c. Alternatively, it is proposed that a block coded with ALWIP mode may only use transform skip.
 - i. Alternatively, furthermore, when ALWIP is used, the signalling of indication of usage of transform skip is skipped.
- d. In one example, ALWIP mode information (such as enabled/disabled, prediction mode index) may be conditionally signalled after indications of transform matrix.
 - i. In one example, for a given transform matrix (such as transform skip or DCT-II), the indications of ALWIP mode information may be signalled.
 - ii. Alternatively, furthermore, the indications of ALWIP mode information may be skipped for some pre-defined transform matrices.
- 23. It is proposed that a block coded with ALWIP mode is regarded to be coded with a normal intra-prediction converted from the ALWIP intra-prediction mode when the selected transform is mode-dependent.
- 24. ALWIP mode may not use transform skip.
 - a. For example, there is no need to further signal the indication of usage of transform skip in this case.
 - b. Alternatively, ALWIP mode may be disallowed when transform skip is applied.
 - i. For example, there is no need to signal ALWIP mode information when transform skip is applied in this case.
- 25. In the filtering process, such as deblocking filter, sample adaptive offset (SAO), adaptive loop filter (ALF), how to select the filters and/or whether to filter samples may be determined by the usage of ALWIP.
- 26. Unfiltered neighboring samples may be used in ALWIP mode.
 - a. Alternatively, filtered neighboring samples may be used in ALWIP mode.
 - b. In one example, filtered neighboring samples may be used for down sampling and unfiltered neighboring samples may be used for up sampling.
 - c. In one example, unfiltered neighboring samples may be used for down sampling and filtered neighboring samples may be used for up sampling.
 - d. In one example, filtered left neighboring samples may be used in up sampling and unfiltered above neighboring samples may be used in up sampling.
 - e. In one example, unfiltered left neighboring samples may be used in up sampling and filtered above neighboring samples may be used in up sampling.
 - f. In one example, whether filter or unfiltered neighboring samples is used may depend on the ALWIP mode.
 - i. In one example, ALWIP mode may be converted to traditional intra prediction mode, and whether filtered or unfiltered neighboring samples is used may depend on the converted traditional intra prediction mode. For example, such decision is same as traditional intra prediction modes.
 - ii. Alternatively, whether filter or unfiltered neighboring samples is used for ALWIP mode may be signalled.

- g. In one example, the filtered samples may be generated same as traditional intra prediction modes.
- 27. Which matrices or/and offset vectors are used may depend on reshaping (a.k.a. LMCS, luma mapping with chroma scaling) information.
 - a. In one example, different matrices or/and offset vectors may be used when reshaping is on and off.
 - b. In one example, different matrices or/and offset vectors may be used for different reshaping parameters.
 - c. In one example, ALWIP may be always performed in original domain.
 - i. For example, neighboring sample are mapped to the original domain (if reshaping is applied) before used in ALWIP.
- 28. ALWIP may be disabled when reshaping is applied.
 - a. Alternatively, reshaping may be disabled when ALWIP is enabled.
 - b. In one example, ALWIP may be disabled for HDR (high dynamic range) content when reshaping is applied.
- 29. The matrices used in ALWIP may depend on sample bit-depth.
- 30. It is proposed that the prediction generated by ALWIP may be treated as an intermediate signal which will be processed to obtain the prediction signal to be further used.
 - a. In one example, Position Dependent Intra Prediction Combination (PDPC) may be applied on the prediction generated by ALWIP to generate the prediction signal to be further used.
 - b. In one example, the boundary samples prediction generated by ALWIP may be filtered with neighbouring samples to generate the prediction signal to be further used.
- 31. It is proposed that interpolation filters other than bilinear interpolation filter may be used in the up-sampling process of ALWIP.
 - a. In one example, 4-tap interpolation filters may be used in the up-sampling process of ALWIP.
 - i. For example, the 4-tap interpolation filters in VVC used to do the motion compensation for chroma components may be used in the up-sampling process of ALWIP.
 - ii. For example, the 4-tap interpolation filters in VVC used to do angular intra-prediction may be used in the up-sampling process of ALWIP.
 - iii. For example, the 8-tap interpolation filters in VVC used to do the motion compensation for luma component may be used in the up-sampling process of ALWIP.

The examples described above may be incorporated in the context of the methods described below, e.g., methods **1100** to **1400**, and **2000** to **3800**, which may be implemented at a video encoder and/or decoder.

FIG. 11 shows a flowchart of an exemplary method for video processing. The method **1100** includes, at step **1102**, determining that a current video block is coded using an affine linear weighted intra prediction (ALWIP) mode.

The method **1100** includes, at step **1104**, constructing, based on the determining, at least a portion of a most probable mode (MPM) list for the ALWIP mode based on at least a portion of an MPM list for a non-ALWIP intra mode.

The method **1100** includes, at step **1106**, performing, based on the MPM list for the ALWIP mode, a conversion

between the current video block and a bitstream representation of the current video block.

In some embodiments, a size of the MPM list of the ALWIP mode is identical to a size of the MPM list for the non-ALWIP intra mode. In an example, the size of the MPM list of the ALWIP mode is 6.

In some embodiments, the method **1100** further comprises the step of inserting default modes to the MPM list for the ALWIP mode. In an example, the default modes are inserted prior to the portion of a MPM list for the ALWIP mode that is based on the MPM list for the non-ALWIP intra mode. In another example, the default modes are inserted subsequent to the portion of a MPM list for the ALWIP mode that is based on the MPM list for the non-ALWIP intra mode. In yet another example, the default modes are inserted in an interleaved manner with the portion of a MPM list for the ALWIP mode that is based on the MPM list for the non-ALWIP intra mode.

In some embodiments, constructing the MPM list for the ALWIP mode and the MPM list for the non-ALWIP intra mode is based on one or more neighboring blocks.

In some embodiments, constructing the MPM list for the ALWIP mode and the MPM list for the non-ALWIP intra mode is based a height or a width of the current video block.

In some embodiments, constructing the MPM list for the ALWIP mode is based on a first set of parameters that is different from a second set of parameters used to construct the MPM list for the non-ALWIP intra mode.

In some embodiments, the method **1100** further includes the step of determining that a neighboring block of the current video block has been coded with the ALWIP mode, and designating, in constructing the MPM list for the non-ALWIP intra mode, the neighboring block as unavailable.

In some embodiments, the method **1100** further includes the step of determining that a neighboring block of the current video block has been coded with the non-ALWIP intra mode, and designating, in constructing the MPM list for the ALWIP mode, the neighboring block as unavailable.

In some embodiments, the non-ALWIP intra mode is based on a normal intra mode, a multiple reference line (MRL) intra prediction mode or an intra sub-partition (ISP) tool.

FIG. **12** shows a flowchart of an exemplary method for video processing. The method **1200** includes, at step **1210**, determining that a luma component of a current video block is coded using an affine linear weighted intra prediction (ALWIP) mode.

The method **1200** includes, at step **1220**, inferring, based on the determining, a chroma intra mode.

The method **1200** includes, at step **1230**, performing, based on the chroma intra mode, a conversion between the current video block and a bitstream representation of the current video block.

In some embodiments, the luma component covers a predetermined chroma sample of the chroma component. In an example, the predetermined chroma sample is a top-left sample or a center sample of the chroma component.

In some embodiments, the inferred chroma intra mode is a DM mode.

In some embodiments, the inferred chroma intra mode is the ALWIP mode.

In some embodiments, the ALWIP mode is applied to one or more chroma components of the current video block.

In some embodiments, different matrix or bias vectors of the ALWIP mode are applied to different color components of the current video block. In an example, the different matrix or bias vectors are predefined jointly for Cb and Cr

components. In another example, the Cb and Cr components are concatenated. In yet another example, the Cb and Cr components are interleaved.

FIG. **13** shows a flowchart of an exemplary method for video processing. The method **1300** includes, at step **1302**, determining that a current video block is coded using an affine linear weighted intra prediction (ALWIP) mode.

The method **1300** includes, at step **1304**, performing, based on the determining, a conversion between the current video block and a bitstream representation of the current video block.

In some embodiments, the determining is based on signaling in a sequence parameter set (SPS), a picture parameter set (PPS), a slice header, a tile group header, a tile header, a coding tree unit (CTU) row or a CTU region.

In some embodiments, the determining is based on a height (H) or a width (W) of the current video block. In an example, $W > T1$ or $H > T2$. In another example, $W \geq T1$ or $H \geq T2$. In yet another example, $W < T1$ or $H < T2$. In yet another example, $W \leq T1$ or $H \leq T2$. In yet another example, $T1 = 32$ and $T2 = 32$.

In some embodiments, the determining is based on a height (H) or a width (W) of the current video block. In an example, $W + H \leq T$. In another example, $W + H \geq T$. In yet another example, $W \times H \leq T$. In yet another example, $W \times H \geq T$. In yet another example, $T = 256$.

FIG. **14** shows a flowchart of an exemplary method for video processing. The method **1400** includes, at step **1402**, determining that a current video block is coded using a coding mode different from an affine linear weighted intra prediction (ALWIP) mode.

The method **1400** includes, at step **1404**, performing, based on the determining, a conversion between the current video block and a bitstream representation of the current video block.

In some embodiments, the coding mode is a combined intra and inter prediction (CIIP) mode, and method **1400** further includes the step of performing a selection between the ALWIP mode and a normal intra prediction mode. In an example, performing the selection is based on an explicit signaling in the bitstream representation of the current video block. In another example, performing the selection is based on predetermined rule. In yet another example, the predetermined rule always selects the ALWIP mode when the current video block is coded using the CIIP mode. In yet another example, the predetermined rule always selects the normal intra prediction mode when the current video block is coded using the CIIP mode.

In some embodiments, the coding mode is a cross-component linear model (CCLM) prediction mode. In an example, a downsampling procedure for the ALWIP mode is based on a downsampling procedure for the CCLM prediction mode. In another example, the downsampling procedure for the ALWIP mode is based on a first set of parameters, and where the downsampling procedure for the CCLM prediction mode is based on a second set of parameters different from the first set of parameters. In yet another example, the downsampling procedure for the ALWIP mode or the CCLM prediction mode comprises at least one of a selection of downsampled positions, a selection of downsampling filters, a rounding operation or a clipping operation.

In some embodiments, the method **1400** further includes the step of applying one or more of a Reduced Secondary Transform (RST), a secondary transform, a rotation transform or a Non-Separable Secondary Transform (NSST).

In some embodiments, the method **1400** further includes the step of applying block-based differential pulse coded modulation (DPCM) or residual DPCM.

6 Example Implementations of the Disclosed Technology

FIG. **15** is a block diagram of a video processing apparatus **1500**. The apparatus **1500** may be used to implement one or more of the methods described herein. The apparatus **1500** may be embodied in a smartphone, tablet, computer, Internet of Things (IoT) receiver, and so on. The apparatus **1500** may include one or more processors **1502**, one or more memories **1504** and video processing hardware **1506**. The processor(s) **1502** may be configured to implement one or more methods (including, but not limited to, methods **1100** to **1400**, and **2000** to **3800**) described in the present document. The memory (memories) **1504** may be used for storing data and code used for implementing the methods and techniques described herein. The video processing hardware **1506** may be used to implement, in hardware circuitry, some techniques described in the present document.

In some embodiments, the video coding methods may be implemented using an apparatus that is implemented on a hardware platform as described with respect to FIG. **15**.

FIG. **16** is a block diagram showing an example video processing system **1600** in which various techniques disclosed herein may be implemented. Various implementations may include some or all of the components of the system **1600**. The system **1600** may include input **1602** for receiving video content. The video content may be received in a raw or uncompressed format, e.g., 8 or 10 bit multi-component pixel values, or may be in a compressed or encoded format. The input **1602** may represent a network interface, a peripheral bus interface, or a storage interface. Examples of network interface include wired interfaces such as Ethernet, passive optical network (PON), etc. and wireless interfaces such as Wi-Fi or cellular interfaces.

The system **1600** may include a coding component **1604** that may implement the various coding or encoding methods described in the present document. The coding component **1604** may reduce the average bitrate of video from the input **1602** to the output of the coding component **1604** to produce a coded representation of the video. The coding techniques are therefore sometimes called video compression or video transcoding techniques. The output of the coding component **1604** may be either stored, or transmitted via a communication connected, as represented by the component **1606**. The stored or communicated bitstream (or coded) representation of the video received at the input **1602** may be used by the component **1608** for generating pixel values or displayable video that is sent to a display interface **1610**. The process of generating user-viewable video from the bitstream representation is sometimes called video decompression. Furthermore, while certain video processing operations are referred to as “coding” operations or tools, it will be appreciated that the coding tools or operations are used at an encoder and corresponding decoding tools or operations that reverse the results of the coding will be performed by a decoder.

Examples of a peripheral bus interface or a display interface may include universal serial bus (USB) or high definition multimedia interface (HDMI) or Displayport, and so on. Examples of storage interfaces include SATA (serial advanced technology attachment), PCI, IDE interface, and the like. The techniques described in the present document may be embodied in various electronic devices such as

mobile phones, laptops, smartphones or other devices that are capable of performing digital data processing and/or video display.

Some embodiments of the disclosed technology include making a decision or determination to enable a video processing tool or mode. In an example, when the video processing tool or mode is enabled, the encoder will use or implement the tool or mode in the processing of a block of video but may not necessarily modify the resulting bitstream based on the usage of the tool or mode. That is, a conversion from the block of video to the bitstream representation of the video will use the video processing tool or mode when it is enabled based on the decision or determination. In another example, when the video processing tool or mode is enabled, the decoder will process the bitstream with the knowledge that the bitstream has been modified based on the video processing tool or mode. That is, a conversion from the bitstream representation of the video to the block of video will be performed using the video processing tool or mode that was enabled based on the decision or determination.

Some embodiments of the disclosed technology include making a decision or determination to disable a video processing tool or mode. In an example, when the video processing tool or mode is disabled, the encoder will not use the tool or mode in the conversion of the block of video to the bitstream representation of the video. In another example, when the video processing tool or mode is disabled, the decoder will process the bitstream with the knowledge that the bitstream has not been modified using the video processing tool or mode that was disabled based on the decision or determination.

FIG. **17** is a block diagram that illustrates an example video coding system **100** that may utilize the techniques of this disclosure. As shown in FIG. **17**, video coding system **100** may include a source device **110** and a destination device **120**. Source device **110** generates encoded video data which may be referred to as a video encoding device. Destination device **120** may decode the encoded video data generated by source device **110** which may be referred to as a video decoding device. Source device **110** may include a video source **112**, a video encoder **114**, and an input/output (I/O) interface **116**.

Video source **112** may include a source such as a video capture device, an interface to receive video data from a video content provider, and/or a computer graphics system for generating video data, or a combination of such sources. The video data may comprise one or more pictures. Video encoder **114** encodes the video data from video source **112** to generate a bitstream. The bitstream may include a sequence of bits that form a coded representation of the video data. The bitstream may include coded pictures and associated data. The coded picture is a coded representation of a picture. The associated data may include sequence parameter sets, picture parameter sets, and other syntax structures. I/O interface **116** may include a modulator/demodulator (modem) and/or a transmitter. The encoded video data may be transmitted directly to destination device **120** via I/O interface **116** through network **130a**. The encoded video data may also be stored onto a storage medium/server **130b** for access by destination device **120**.

Destination device **120** may include an I/O interface **126**, a video decoder **124**, and a display device **122**.

I/O interface **126** may include a receiver and/or a modem. I/O interface **126** may acquire encoded video data from the source device **110** or the storage medium/server **130b**. Video decoder **124** may decode the encoded video data. Display device **122** may display the decoded video data to a user.

Display device **122** may be integrated with the destination device **120** or may be external to destination device **120** which be configured to interface with an external display device.

Video encoder **114** and video decoder **124** may operate according to a video compression standard, such as the High Efficiency Video Coding (HEVC) standard, Versatile Video Coding (VVC) standard and other current and/or further standards.

FIG. **18** is a block diagram illustrating an example of video encoder **200**, which may be video encoder **114** in the system **100** illustrated in FIG. **17**.

Video encoder **200** may be configured to perform any or all of the techniques of this disclosure. In the example of FIG. **18**, video encoder **200** includes a plurality of functional components. The techniques described in this disclosure may be shared among the various components of video encoder **200**. In some examples, a processor may be configured to perform any or all of the techniques described in this disclosure.

The functional components of video encoder **200** may include a partition unit **201**, a predication unit **202** which may include a mode select unit **203**, a motion estimation unit **204**, a motion compensation unit **205** and an intra prediction unit **206**, a residual generation unit **207**, a transform unit **208**, a quantization unit **209**, an inverse quantization unit **210**, an inverse transform unit **211**, a reconstruction unit **212**, a buffer **213**, and an entropy encoding unit **214**.

In other examples, video encoder **200** may include more, fewer, or different functional components. In an example, predication unit **202** may include an intra block copy (IBC) unit. The IBC unit may perform predication in an IBC mode in which at least one reference picture is a picture where the current video block is located.

Furthermore, some components, such as motion estimation unit **204** and motion compensation unit **205** may be highly integrated but are represented in the example of FIG. **18** separately for purposes of explanation.

Partition unit **201** may partition a picture into one or more video blocks. Video encoder **200** and video decoder **300** may support various video block sizes.

Mode select unit **203** may select one of the coding modes, intra or inter, e.g., based on error results, and provide the resulting intra- or inter-coded block to a residual generation unit **207** to generate residual block data and to a reconstruction unit **212** to reconstruct the encoded block for use as a reference picture. In some example, Mode select unit **203** may select a combination of intra and inter predication (CIIP) mode in which the predication is based on an inter predication signal and an intra predication signal. Mode select unit **203** may also select a resolution for a motion vector (e.g., a sub-pixel or integer pixel precision) for the block in the case of inter-predication.

To perform inter prediction on a current video block, motion estimation unit **204** may generate motion information for the current video block by comparing one or more reference frames from buffer **213** to the current video block. Motion compensation unit **205** may determine a predicted video block for the current video block based on the motion information and decoded samples of pictures from buffer **213** other than the picture associated with the current video block.

Motion estimation unit **204** and motion compensation unit **205** may perform different operations for a current video block, for example, depending on whether the current video block is in an I slice, a P slice, or a B slice.

In some examples, motion estimation unit **204** may perform uni-directional prediction for the current video block, and motion estimation unit **204** may search reference pictures of list 0 or list 1 for a reference video block for the current video block. Motion estimation unit **204** may then generate a reference index that indicates the reference picture in list 0 or list 1 that contains the reference video block and a motion vector that indicates a spatial displacement between the current video block and the reference video block. Motion estimation unit **204** may output the reference index, a prediction direction indicator, and the motion vector as the motion information of the current video block. Motion compensation unit **205** may generate the predicted video block of the current block based on the reference video block indicated by the motion information of the current video block.

In other examples, motion estimation unit **204** may perform bi-directional prediction for the current video block, motion estimation unit **204** may search the reference pictures in list 0 for a reference video block for the current video block and may also search the reference pictures in list 1 for another reference video block for the current video block. Motion estimation unit **204** may then generate reference indexes that indicate the reference pictures in list 0 and list 1 containing the reference video blocks and motion vectors that indicate spatial displacements between the reference video blocks and the current video block. Motion estimation unit **204** may output the reference indexes and the motion vectors of the current video block as the motion information of the current video block. Motion compensation unit **205** may generate the predicted video block of the current video block based on the reference video blocks indicated by the motion information of the current video block.

In some examples, motion estimation unit **204** may output a full set of motion information for decoding processing of a decoder.

In some examples, motion estimation unit **204** may do not output a full set of motion information for the current video. Rather, motion estimation unit **204** may signal the motion information of the current video block with reference to the motion information of another video block. For example, motion estimation unit **204** may determine that the motion information of the current video block is sufficiently similar to the motion information of a neighboring video block.

In one example, motion estimation unit **204** may indicate, in a syntax structure associated with the current video block, a value that indicates to the video decoder **300** that the current video block has the same motion information as the another video block.

In another example, motion estimation unit **204** may identify, in a syntax structure associated with the current video block, another video block and a motion vector difference (MVD). The motion vector difference indicates a difference between the motion vector of the current video block and the motion vector of the indicated video block. The video decoder **300** may use the motion vector of the indicated video block and the motion vector difference to determine the motion vector of the current video block.

As discussed above, video encoder **200** may predictively signal the motion vector. Two examples of predictive signaling techniques that may be implemented by video encoder **200** include advanced motion vector predication (AMVP) and merge mode signaling.

Intra prediction unit **206** may perform intra prediction on the current video block. When intra prediction unit **206** performs intra prediction on the current video block, intra

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prediction unit **206** may generate prediction data for the current video block based on decoded samples of other video blocks in the same picture. The prediction data for the current video block may include a predicted video block and various syntax elements.

Residual generation unit **207** may generate residual data for the current video block by subtracting (e.g., indicated by the minus sign) the predicted video block(s) of the current video block from the current video block. The residual data of the current video block may include residual video blocks that correspond to different sample components of the samples in the current video block.

In other examples, there may be no residual data for the current video block for the current video block, for example in a skip mode, and residual generation unit **207** may not perform the subtracting operation.

Transform processing unit **208** may generate one or more transform coefficient video blocks for the current video block by applying one or more transforms to a residual video block associated with the current video block.

After transform processing unit **208** generates a transform coefficient video block associated with the current video block, quantization unit **209** may quantize the transform coefficient video block associated with the current video block based on one or more quantization parameter (QP) values associated with the current video block.

Inverse quantization unit **210** and inverse transform unit **211** may apply inverse quantization and inverse transforms to the transform coefficient video block, respectively, to reconstruct a residual video block from the transform coefficient video block. Reconstruction unit **212** may add the reconstructed residual video block to corresponding samples from one or more predicted video blocks generated by the prediction unit **202** to produce a reconstructed video block associated with the current block for storage in the buffer **213**.

After reconstruction unit **212** reconstructs the video block, loop filtering operation may be performed reduce video blocking artifacts in the video block.

Entropy encoding unit **214** may receive data from other functional components of the video encoder **200**. When entropy encoding unit **214** receives the data, entropy encoding unit **214** may perform one or more entropy encoding operations to generate entropy encoded data and output a bitstream that includes the entropy encoded data.

FIG. **19** is a block diagram illustrating an example of video decoder **300** which may be video decoder **114** in the system **100** illustrated in FIG. **17**.

The video decoder **300** may be configured to perform any or all of the techniques of this disclosure. In the example of FIG. **19**, the video decoder **300** includes a plurality of functional components. The techniques described in this disclosure may be shared among the various components of the video decoder **300**. In some examples, a processor may be configured to perform any or all of the techniques described in this disclosure.

In the example of FIG. **19**, video decoder **300** includes an entropy decoding unit **301**, a motion compensation unit **302**, an intra prediction unit **303**, an inverse quantization unit **304**, an inverse transformation unit **305**, and a reconstruction unit **306** and a buffer **307**. Video decoder **300** may, in some examples, perform a decoding pass generally reciprocal to the encoding pass described with respect to video encoder **200** (FIG. **18**).

Entropy decoding unit **301** may retrieve an encoded bitstream. The encoded bitstream may include entropy coded video data (e.g., encoded blocks of video data).

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Entropy decoding unit **301** may decode the entropy coded video data, and from the entropy decoded video data, motion compensation unit **302** may determine motion information including motion vectors, motion vector precision, reference picture list indexes, and other motion information. Motion compensation unit **302** may, for example, determine such information by performing the AMVP and merge mode.

Motion compensation unit **302** may produce motion compensated blocks, possibly performing interpolation based on interpolation filters. Identifiers for interpolation filters to be used with sub-pixel precision may be included in the syntax elements.

Motion compensation unit **302** may use interpolation filters as used by video encoder **20** during encoding of the video block to calculate interpolated values for sub-integer pixels of a reference block. Motion compensation unit **302** may determine the interpolation filters used by video encoder **200** according to received syntax information and use the interpolation filters to produce predictive blocks.

Motion compensation unit **302** may use some of the syntax information to determine sizes of blocks used to encode frame(s) and/or slice(s) of the encoded video sequence, partition information that describes how each macroblock of a picture of the encoded video sequence is partitioned, modes indicating how each partition is encoded, one or more reference frames (and reference frame lists) for each inter-encoded block, and other information to decode the encoded video sequence.

Intra prediction unit **303** may use intra prediction modes for example received in the bitstream to form a prediction block from spatially adjacent blocks. Inverse quantization unit **303** inverse quantizes, i.e., de-quantizes, the quantized video block coefficients provided in the bitstream and decoded by entropy decoding unit **301**. Inverse transform unit **303** applies an inverse transform.

Reconstruction unit **306** may sum the residual blocks with the corresponding prediction blocks generated by motion compensation unit **202** or intra-prediction unit **303** to form decoded blocks. If desired, a deblocking filter may also be applied to filter the decoded blocks in order to remove blockiness artifacts. The decoded video blocks are then stored in buffer **307**, which provides reference blocks for subsequent motion compensation/intra prediction and also produces decoded video for presentation on a display device.

In some embodiments, the ALWIP mode or the MIP mode is used to compute a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation (or an averaging operation), followed by a matrix vector multiplication operation, and selectively (or optionally) followed by an upsampling operation (or a linear interpolation operation). In some embodiments, the ALWIP mode or the MIP mode is used to compute a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation (or an averaging operation) and followed by a matrix vector multiplication operation. In some embodiments, the ALWIP mode or the MIP mode can also perform an upsampling operation (or a linear interpolation operation) after performing the matrix vector multiplication operation.

FIG. **20** shows an example flowchart of an example method **2000** for matrix-based intra prediction. Operation **2002** includes generating, for a conversion between a current video block of a video and a coded representation of the current video block, a first most probable mode (MPM) list using a first procedure based on a rule. Operation **2004**

includes performing the conversion between the current video block and the coded representation of the current video block using the first MPM list, where the conversion of the current video block uses a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where the rule specifies that the first procedure used for generation of the first MPM list is same as a second procedure used for generation of a second MPM list for conversion of other video blocks of the video that are coded using a non-MIP intra mode that is different from the MIP mode, and where at least a portion of the first MPM list is generated based on at least a portion of the second MPM list.

In some embodiments for method 2000, a size of the first MPM list for the MIP mode is same as that of the second MPM list for the non-MIP intra mode. In some embodiments for method 2000, the size of the first MPM list for the MIP mode and of the second MPM list for the non-MIP intra mode is 6. In some embodiments for method 2000, the second MPM list for the non-MIP intra mode is constructed before the first MPM list for the MIP mode is constructed. In some embodiments for method 2000, the portion or all of the second MPM list for the non-MIP intra mode is converted to MPMs that are added to the portion of the first MPM list for the MIP mode.

In some embodiments for method 2000, a subset of the MPMs are added to the portion of the first MPM list for the MIP mode by pruning the MPMs. In some embodiments, the method 2000 further includes adding default intra-prediction modes to the first MPM list for the MIP mode. In some embodiments for method 2000, the default intra-prediction modes are added to the first MPM list for the MIP mode prior to the portion of the first MPM list for the MIP mode that is based on the portion of the second MPM list for the non-MIP intra mode. In some embodiments for method 2000, the default intra-prediction modes are added to the first MPM list for the MIP mode after the portion of the first MPM list for the MIP mode that is based on the portion of the second MPM list for the non-MIP intra mode. In some embodiments for method 2000, the default intra-prediction modes are added to the first MPM list for the MIP mode in an interleaved manner with the portion of the first MPM list for the MIP mode that is based on the portion of the second MPM list for the non-MIP intra mode. In some embodiments for method 2000, the default intra-prediction modes are same for a plurality of types of video blocks. In some embodiments for method 2000, the default intra-prediction modes are determined based on coded information of the current video block. In some embodiments for method 2000, the coded information includes availability of neighboring video blocks, mode information of the neighboring video blocks, or block dimension of the current video block.

In some embodiments for method 2000, one intra-prediction mode in the second MPM list for the non-MIP intra mode is converted to its corresponding MIP mode to obtain a converted MIP mode that is added to the first MPM list for the MIP mode. In some embodiments for method 2000, all intra-prediction modes in the second MPM list for the non-MIP intra mode are converted to its corresponding MIP mode to obtain a plurality of converted MIP modes that are added to the first MPM list for the MIP mode. In some embodiments for method 2000, all intra-prediction modes are converted to its corresponding MIP mode to obtain a plurality of converted MIP modes that are used to construct

the second MPM list for the non-MIP intra mode. In some embodiments for method 2000, the all intra-prediction modes include intra-prediction modes from neighboring video blocks of the current video block and default intra-prediction modes. In some embodiments for method 2000, the default intra-prediction modes include a planar mode and a direct current (DC) mode.

In some embodiments for method 2000, two intra-prediction modes in the second MPM list for the non-MIP intra mode are converted to its corresponding MIP modes to obtain two converted MIP modes, and one of two converted MIP modes is added to the first MPM list for the MIP mode in response to the two converted MIP modes being same. In some embodiments for method 2000, two intra-prediction modes in the second MPM list for the non-MIP intra mode are converted to its corresponding MIP modes to obtain two converted MIP modes, and one of two converted MIP modes is added to the second MPM list for the non-MIP intra mode in response to the two converted MIP modes being same. In some embodiments for method 2000, the second MPM list for non-MIP intra mode includes S intra-prediction modes, and where K out of S intra-prediction modes is or are selected to be included in the first MPM list for MIP mode. In some embodiments for method 2000, K is 3 and S is 6. In some embodiments for method 2000, the first K intra-prediction modes in the second MPM list for non-MIP intra mode is selected to be included in the first MPM list for MIP mode. In some embodiments for method 2000, the first MPM list for the MIP mode and the second MPM list for the non-MIP intra mode is derived or constructed based on one or more neighboring video blocks of the current video block. In some embodiments for method 2000, the first MPM list for the MIP mode and the second MPM list for the non-MIP intra mode is derived or constructed based on a same neighboring video block located to a left of the current video block.

In some embodiments for method 2000, the same neighboring video block located to the left is directly to left of the current video block and is aligned with the current video block. In some embodiments for method 2000, the same neighboring video block located to the left and below the current video block. In some embodiments for method 2000, the first MPM list for the MIP mode and the second MPM list for the non-MIP intra mode is derived or constructed based on a same neighboring video block, where the same neighboring video block is located to left of the current video block at a bottom-most position or the same neighboring video block is located above the current video block at a right-most position. In some embodiments for method 2000, the first MPM list for the MIP mode and the second MPM list for the non-MIP intra mode is derived or constructed based on a same neighboring video block located above the current video block.

In some embodiments for method 2000, the same neighboring video block located to above is directly above the current video block and is aligned with the current video block. In some embodiments for method 2000, the same neighboring video block located to left and above the current video block. In some embodiments for method 2000, the first MPM list for the MIP mode and the second MPM list for the non-MIP intra mode is derived or constructed based on a same neighboring video block, where the same neighboring video block is located above the current video block at a left-most position or the same neighboring video block is located to left of the current video block at a top-most position. In some embodiments for method 2000, the non-MIP intra mode is based on an intra prediction mode, a

multiple reference line (MRL) intra prediction mode, or an intra sub-partition (ISP) tool. In some embodiments for method **2000**, the first MPM list for the MIP mode is further based on a height or a width of the current video block.

In some embodiments for method **2000**, the first MPM list for the MIP mode is further based on a height or a width of a neighboring video block of the current video block. In some embodiments for method **2000**, the first MPM list for the MIP mode is constructed based on a first set of parameters that is different from a second set of parameters used to construct the second MPM list for the non-MIP intra mode. In some embodiments for method **2000**, the second MPM list for non-MIP intra mode includes S intra-prediction modes, and where K out of S intra-prediction modes are derived to be included in the first MPM list for MIP mode. In some embodiments for method **2000**, K is 3 and S is 6.

In some embodiments for method **2000**, the first K intra-prediction modes in the second MPM list for non-MIP intra mode is derived to be included in the first MPM list for MIP mode. In some embodiments for method **2000**, a first mode listed in the first MPM list for MIP mode is different from a first mode listed in the second MPM list for non-MIP intra mode. In some embodiments for method **2000**, the first mode listed in the first MPM list for MIP mode is a first mode, and where the first mode listed in the second MPM list for non-MIP intra mode is a planar mode. In some embodiments for method **2000**, the first mode in the first MPM list for MIP mode is converted from the planar mode in the second MPM list for non-MIP intra mode.

In some embodiments for method **2000**, the first MPM list for MIP mode includes a first set of stuffing modes that are different from a second set of stuffing modes included in the second MPM list for non-MIP intra mode. In some embodiments for method **2000**, the first set of stuffing modes include a first mode, a second mode, and a third mode, and where the second set of stuffing modes include a direct current (DC) mode, a vertical mode, and a horizontal mode. In some embodiments for method **2000**, the first mode, the second mode, and the third mode are included in the first set of stuffing modes based on a size of the current video block. In some embodiments for method **2000**, the first MPM list for MIP mode includes a first set of intra-prediction modes of neighboring video blocks of the current video block, the second MPM list for non-MIP intra mode includes a second set of intra-prediction modes of the neighboring video blocks of the current video block, and the first set of intra-prediction modes are different from the second set of intra-prediction modes.

In some embodiments for method **2000**, the second set of intra-prediction modes include intra-prediction modes that are converted to MIP modes that are included in the first set of intra-prediction modes. In some embodiments for method **2000**, the first MPM list for MIP mode includes a first set of shifted intra-prediction modes, the second MPM list for non-MIP intra mode includes a second set of shifted intra-prediction modes, and the first set of shifted intra-prediction modes are different from the second set of shifted intra-prediction modes. In some embodiments for method **2000**, the first set of shifted intra-prediction modes includes an MIP mode (Y) shifted by K1 according to a first formula $Y+K1$, the second set of shifted intra-prediction modes includes a non-MIP intra mode (X) shifted by K0 according to a second formula $X+K0$, and K1 is different from K0. In some embodiments for method **2000**, K1 depends on a width and a height of the current video block.

In some embodiments, the method **2000** further includes performing a first determination that a neighboring video

block of the current video block is coded with a non-MIP intra mode; and performing a second determination, in response to the first determination, that the neighboring video block is unavailable for the constructing of the first MPM list for the MIP mode.

In some embodiments, the method **2000** further includes performing a first determination that a neighboring video block of the current video block is coded with a non-MIP intra mode; and performing, in response to the first determination, a second determination that the neighboring video block is coded with a pre-defined MIP intra-prediction mode, where the first MPM list for the MIP mode is constructed using the pre-defined MIP intra-prediction mode. In some embodiments for method **2000**, the pre-defined MIP intra-prediction mode depends on a width and/or a height of the current video block.

FIG. **21** shows an example flowchart of an example method **2100** for matrix-based intra prediction. Operation **2100** includes generating, for a conversion between a current video block of a video and a coded representation of the current video block, a most probable mode (MPM) list based on a rule, where the rule is based on whether a neighboring video block of the current video block is coded with a matrix based intra prediction (MIP) mode in which a prediction block of the neighboring video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation **2104** includes performing the conversion between the current video block and the coded representation of the current video block using the MPM list, where the conversion applies a non-MIP mode to the current video block, and where the non-MIP mode is different from the MIP mode.

In some embodiments for method **2100**, the rule specifies that the neighboring video block coded with the MIP mode is treated as unavailable for generating the MPM list for the non-MIP coded current video block. In some embodiments for method **2100**, the rule specifies that the neighboring video block coded with the MIP mode is determined to be coded with a pre-defined intra-prediction mode. In some embodiments for method **2100**, the rule specifies that the pre-defined intra-prediction mode includes a planar mode.

FIG. **22** shows an example flowchart of an example method **2200** for matrix-based intra prediction. Operation **2202** includes decoding a current video block of a video that is coded in a coded representation of the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation **2204** includes updating a line buffer associated with the decoding without storing in the line buffer information indicating whether the current video block is coded using the MIP mode.

In some embodiments, the method **2200** further includes accessing a second video block of the video, where the second video block is decoded before the decoding the current video block, where the second video block is located at a different Largest Coding Unit (LCU) or Coding Tree Unit (CTU) row or CTU region compared to that of the current video block, and where the current video block is decoded without determining whether the second video block is coded using the MIP mode. In some embodiments, the method **2200** further includes accessing a second video block by determining that the second video block is coded

using a non-MIP intra mode without determining whether the second video block is coded using the MIP mode, where the second video located at a different Largest Coding Unit (LCU) or Coding Tree Unit (CTU) row or CTU region compared to that of the current video block; and decoding, based on the accessing the second video block, the current video block.

FIG. 23 shows an example flowchart of an example method 2300 for matrix-based intra prediction. Operation 2302 includes performing a conversion between a current video block and a bitstream representation of the current video block, where the current video block is coded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where a flag is coded in the bitstream presentation using at most K contexts in an arithmetic coding or a decoding process, where the flag indicates whether the current video block is coded using the MIP mode, and where K is greater than or equal to zero. In some embodiments for method 2300, K is 1. In some embodiments for method 2300, K is 4.

FIG. 24 shows an example flowchart of an example method 2400 for matrix-based intra prediction. Operation 2402 includes generating, for a conversion between a current video block of a video and a bitstream representation of the current video block, an intra prediction mode for the current video block coded with a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation 2404 includes determining a rule for storing information indicative of the intra prediction mode based on whether the current video block is coded with the MIP mode. Operation 2406 includes performing the conversion according to the rule, where the rule defines that a syntax element for the intra prediction mode is stored in the bitstream representation for the current video block, and where the rule defines that a mode index for the MIP mode for the current video block is not stored in the bitstream representation.

In some embodiments for method 2400, the mode index for the MIP mode is associated with the intra prediction mode. In some embodiments for method 2400, the rule defines that the bitstream representation excludes a flag that indicates that the current video block is coded with the MIP mode. In some embodiments for method 2400, the rule defines that the bitstream representation excludes a storage of information indicative of the MIP mode associated with the current video block. In some embodiments, the method 2400 further includes performing, after the conversion, a second conversion between a second video block of the video and the bitstream representation of the second video block, where the second video block is a neighboring video block of the current video block, and where the second conversion is performed without determining whether the second video block is coded using the MIP mode.

FIG. 25 shows an example flowchart of an example method 2500 for matrix-based intra prediction. Operation 2502 includes performing a first determination that a luma video block of a video is coded using a matrix based intra prediction (MIP) mode in which a prediction block of the luma video block is determined by performing, on previously coded samples of the video, a boundary downsam-

pling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation 2504 includes performing, based on the first determination, a second determination about a chroma intra mode to be used for a chroma video block associated with the luma video block. Operation 2506 includes performing, based on the second determination, a conversion between the chroma video block and a bitstream representation of the chroma video block.

In some embodiments for method 2500, the performing the second determination includes determining that the chroma intra mode is a Derived Mode (DM) without depending on a signaling. In some embodiments for method 2500, the luma video block covers a predetermined corresponding chroma sample of the chroma video block. In some embodiments for method 2500, the predetermined corresponding chroma sample is a top-left sample or a center sample of the chroma video block. In some embodiments for method 2500, the chroma intra mode includes a Derived Mode (DM). In some embodiments for method 2500, the DM is used to derive a first intra prediction mode for the chroma video block from a second intra prediction mode of the luma video block. In some embodiments for method 2500, the MIP mode associated with the luma video block is mapped to a pre-defined intra mode.

In some embodiments for method 2500, a plurality of Derived Modes (DMs) are derived in response to the first determination that the luma video block is coded with the MIP mode. In some embodiments for method 2500, the chroma intra mode includes a specified intra prediction mode. In some embodiments for method 2500, the pre-defined intra mode or the specified intra mode is a Planar mode. In some embodiments for method 2500, the chroma video block is coded using the MIP mode. In some embodiments for method 2500, the chroma video block is coded using a first matrix or a first bias vector that are different from a second matrix or a second bias vector of a second chroma video block. In some embodiments for method 2500, the chroma video block is a blue color component, where the first matrix or the first bias vector is predefined for the blue color component, where the second chroma video block is a red color component, where the second matrix or the second bias vector is predefined for the red color component.

In some embodiments for method 2500, the blue color component and the red color component are concatenated. In some embodiments for method 2500, the blue color component and the red color component are interleaved. In some embodiments for method 2500, the chroma video block is coded using the MIP mode that is same as the MIP mode used for the luma video block. In some embodiments for method 2500, the chroma video block is coded using a Derived Mode (DM). In some embodiments for method 2500, the upsampling operation (or a linear interpolation technique) is skipped after the chroma video block is coded using the MIP mode. In some embodiments for method 2500, the chroma video block is coded using the MIP mode with a subsampled matrix and/or a bias vector. In some embodiments for method 2500, a number of MIP modes for the luma video block and the chroma video block are different. In some embodiments for method 2500, the chroma video block is associated with a first number of MIP modes, the luma video block is associated with a second number of MIP modes, and the first number of MIP modes is less than the second number of MIP modes. In some embodiments for method 2500, the chroma video block and the luma video block have a same size.

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FIG. 26 shows an example flowchart of an example method 2600 for matrix-based intra prediction. Operation 2602 includes performing a conversion between a current video block of a video and a coded representation of the current video block, where the conversion is based on a determination (or based on determining) whether to code the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation.

In some embodiments of method 2600, a syntax element is included in the coded representation in response to the determination that the current video block is to be coded with the MIP mode, and the syntax element indicates the use of the MIP mode for coding the current video block. In some embodiments of method 2600, the coded representation includes a syntax element that indicates that the MIP mode is enabled or disabled for the current video block, and where the syntax element indicates that the MIP mode is allowed or disallowed for coding the current video block. In some embodiments of method 2600, a syntax element is not included in the coded representation in response to the determination that the current video block is not to be coded with the MIP mode, and the syntax element indicates whether the MIP mode is used for the current video block.

In some embodiments of method 2600, a second syntax element indicating enabling of the MIP mode is further included in a sequence parameter set (SPS), a picture parameter set (PPS), a slice header, a tile group header, a tile header, a coding tree unit (CTU) row or a CTU region. In some embodiments of method 2600, the determination whether to use the MIP mode for the conversion is based on a height (H) and/or a width (W) of the current video block. In some embodiments of method 2600, the current video block is determined not to be coded using the MIP mode in response to $W \geq T1$ and $H \geq T2$. In some embodiments of method 2600, the current video block is determined not to be coded using the MIP mode in response to $W \leq T1$ and $H \leq T2$. In some embodiments of method 2600, the current video block is determined not to be coded using the MIP mode in response to $W \geq T1$ or $H \geq T2$. In some embodiments of method 2600, the current video block is determined not to be coded using the MIP mode in response to $W \leq T1$ or $H \leq T2$. In some embodiments of method 2600, $T1=32$ and $T2=32$. In some embodiments of method 2600, the current video block is determined not to be coded using the MIP mode in response to $W+H \geq T$. In some embodiments of method 2600, the current video block is determined not to be coded using the MIP mode in response to $W+H \leq T$. In some embodiments of method 2600, the current video block is determined not to be coded using the MIP mode in response to $W \times H \geq T$. In some embodiments of method 2600, $T=256$.

FIG. 27A shows an example flowchart of an example video encoding method 2700A for matrix-based intra prediction. Operation 2702A includes determining, for encoding a current video block of a video into a bitstream representation of the current video block, according to a rule whether to use a matrix based intra prediction (MIP) mode and a coding mode that is different from the MIP mode for encoding the current video block, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the

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video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation 2704A includes adding, based on the determining, an encoded representation of the current video block to the bitstream representation.

FIG. 27B shows an example flowchart of an example video decoding method 2700B for matrix-based intra prediction. Operation 2702B includes determining that a current video block of a video is encoded in a bitstream representation using a matrix based intra prediction (MIP) mode and a coding mode that is different from the MIP mode, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation 2704B includes generating a decoded representation of the current video block by parsing and decoding the bitstream representation.

In some embodiments for methods 2700A and/or 2700B, the coding mode is a combined intra and inter prediction (CIIP) mode, and where the method further comprises generating an intra prediction signal for the current video block by performing a selection between the MIP mode and an intra prediction mode. In some embodiments for methods 2700A and/or 2700B, the performing the selection is based on a signaling in the bitstream representation of the current video block. In some embodiments for methods 2700A and/or 2700B, the performing the selection is based on a predetermined rule. In some embodiments for methods 2700A and/or 2700B, the predetermined rule selects the MIP mode in response to the current video block being coded using the CIIP mode. In some embodiments for methods 2700A and/or 2700B, the predetermined rule selects the intra prediction mode in response to the current video block being coded using the CIIP mode. In some embodiments for method 2700A and/or 2700B, the predetermined rule selects a Planar mode in response to the current video block being coded using the CIIP mode. In some embodiments for methods 2700A and/or 2700B, the performing the selection is based on information associated with neighboring video blocks of the current video block.

In some embodiments for methods 2700A and/or 2700B, the coding mode is a cross-component linear model (CCLM) prediction mode. In some embodiments for methods 2700A and/or 2700B, a first downsampling procedure for downsampling a neighboring sample of the current video block in the MIP mode uses at least some of a second downsampling procedure for the CCLM prediction mode in which neighboring luma samples of the current video block are downsampled. In some embodiments for methods 2700A and/or 2700B, a first downsampling procedure for downsampling a neighboring sample of the current video block in the CCLM prediction mode uses at least some of a second downsampling procedure for the MIP prediction mode in which neighboring luma samples of the current video block are downsampled. In some embodiments for methods 2700A and/or 2700B, the first downsampling procedure for the MIP mode is based on a first set of parameters, and the first downsampling procedure for the CCLM prediction mode is based on a second set of parameters different from the first set of parameters.

In some embodiments for methods 2700A and/or 2700B, the first downsampling procedure for the MIP mode and the second downsampling procedure for the CCLM prediction mode include selection of neighboring luma locations or

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selection of downsampling filter. In some embodiments for methods **2700A** and/or **2700B**, the neighboring luma samples are downsampled using at least one of a selection of downsampled positions, a selection of downsampling filters, a rounding operation or a clipping operation. In some embodiments for methods **2700A** and/or **2700B**, the rule defines that a block-based differential pulse coded modulation (BDPCM) or a residual DPCM is not applied to the current video block coded with the MIP mode. In some embodiments for methods **2700A** and/or **2700B**, the rule defines that the MIP mode is disallowed for applying to the current video block in response to applying a block-based differential pulse-coded modulation (BDPCM) or a residual DPCM.

FIG. **28** shows an example flowchart of an example method **2800** for matrix-based intra prediction. Operation **2802** includes making a determination about an applicability of a loop filter to a reconstructed block of a current video block of a video in a conversion between the video and a coded representation of the current video block, where the current video block is coded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation **2804** includes processing the current video block according to the determination.

In some embodiments for method **2800**, the loop filter includes a deblocking filter. In some embodiments for method **2800**, the loop filter includes a sample adaptive offset (SAO). In some embodiments for method **2800**, the loop filter includes an adaptive loop filter (ALF).

FIG. **29A** shows an example flowchart of an example video encoding method **2900A** for matrix-based intra prediction. Operation **2902A** includes determining, for encoding a current video block of a video into a bitstream representation of the current video block, a type of neighboring samples of the current video block to be used according to a rule. Operation **2904A** includes adding, based on the determining, an encoded representation of the current video block to the bitstream representation, where the current video block is encoded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation.

FIG. **29B** shows an example flowchart of an example video decoding method **2900B** for matrix-based intra prediction. Operation **2902B** includes determining that a current video block of a video is encoded in a bitstream representation using a matrix based intra prediction (MIP) mode and using a type of neighboring samples of the current video block according to a rule, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation **2904B** includes generating a decoded representation of the current video block by parsing and decoding the bitstream representation.

In some embodiments for methods **2900A** and/or **2900B**, the rule defines that the type of neighboring samples are unfiltered neighboring samples. In some embodiments for

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methods **2900A** and/or **2900B**, the rule defines that the type of neighboring samples that include the unfiltered neighboring samples are used for an upsampling process, the rule defines that the type of neighboring samples that include the unfiltered neighboring samples are used for a downsampling process, and the rule defines that the type of neighboring samples that include the unfiltered neighboring samples are used for the matrix vector multiplication operation. In some embodiments for methods **2900A** and/or **2900B**, the rule defines that the type of neighboring samples are filtered neighboring samples. In some embodiments for methods **2900A** and/or **2900B**, the rule defines that the type of neighboring samples that include unfiltered neighboring samples are used for an upsampling technique, and the rule defines that the type of neighboring samples that include filtered neighboring samples are used for a downsampling technique.

In some embodiments for methods **2900A** and/or **2900B**, the rule defines that the type of neighboring samples that include unfiltered neighboring samples are used for a downsampling technique, and the rule defines that the type of neighboring samples that include filtered neighboring samples are used for an upsampling technique. In some embodiments for methods **2900A** and/or **2900B**, the rule defines that the type of neighboring samples that include unfiltered above neighboring samples are used for an upsampling technique, and the rule defines that the type of neighboring samples that include filtered left neighboring samples are used for an upsampling technique. In some embodiments for methods **2900A** and/or **2900B**, the rule defines that the type of neighboring samples that include unfiltered left neighboring samples are used for an upsampling technique, and the rule defines that the type of neighboring samples that include filtered above neighboring samples are used for an upsampling technique.

In some embodiments for methods **2900A** and/or **2900B**, the type of neighboring samples include unfiltered neighboring samples or filtered neighboring samples, and the rule defines that whether the unfiltered neighboring samples or the filtered neighboring samples are used is based on the MIP mode of the current video block.

In some embodiments, methods **2900A** and/or **2900B** further comprise converting the MIP mode to an intra prediction mode, where the type of neighboring samples include unfiltered neighboring samples or filtered neighboring samples, and where the rule defines that whether the unfiltered neighboring samples or the filtered neighboring samples are used is based on the intra prediction mode. In some embodiments for methods **2900A** and/or **2900B**, the type of neighboring samples includes unfiltered neighboring samples or filtered neighboring samples, and the rule defines that whether the unfiltered neighboring samples or the filtered neighboring samples are used is based on a syntax element or signaling. In some embodiments for methods **2900A** and/or **2900B**, the filtered neighboring samples or the filtered left neighboring samples or the filtered above neighboring samples are generated using an intra prediction mode.

FIG. **30** shows an example flowchart of an example method **3000** for matrix-based intra prediction. Operation **3002** includes performing a conversion between a current video block of a video and a bitstream representation of the current video block, where the conversion includes generating, by using a matrix based intra prediction (MIP) mode, a prediction block for the current video block by selecting and applying matrix multiplication using matrices of samples and/or by selecting and adding an offset using offset

vectors for the current video block, where the samples are obtained from a row and column wise averaging of previously coded samples of the video, and where the selecting is based on reshaping information associated with applying a luma mapping with chroma scaling (LMCS) technique on a reference picture of the current video block. In some embodiments for method **3000**, when a reshaping technique is enabled for the reference picture, the reshaping information is associated with the matrices and/or the offset vectors that are different from those associated with the reshaping information when the reshaping technique is disabled for the reference picture. In some embodiments for method **3000**, different matrices and/or the offset vectors are used for different reshaping parameters of the reshaping information. In some embodiments for method **3000**, the performing the conversion includes performing intra prediction on the current video block using the MIP mode in an original domain. In some embodiments for method **3000**, neighboring samples of the current video block are mapped to the original domain in response to applying the reshaping information, and where the neighboring samples are mapped to the original domain before being used in MIP mode.

FIG. **31** shows an example flowchart of an example method **3100** for matrix-based intra prediction. Operation **3102** includes determining that a current video block is to be coded using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation **3104** includes performing, based on the determining, a conversion between the current video block and a bitstream representation of the current video block, where the performing the conversion is based on a rule of co-application of the MIP mode and another coding technique.

In some embodiments for method **3100**, the rule specifies that the MIP mode and the another coding technique are used mutually exclusively. In some embodiments for method **3100**, the rule specifies that the another coding technique is high dynamic range (HDR) technique or a reshaping technique.

FIG. **32** shows an example flowchart of an example method **3200** for matrix-based intra prediction. Operation **3202** includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode, where the performing the conversion using the MIP mode includes generating a prediction block by applying matrix multiplication using matrices of samples obtained from a row and column wise averaging of previously coded samples of the video, and where the matrices depend on a bit-depth of a sample.

FIG. **33** shows an example flowchart of an example method **3300** for matrix-based intra prediction. Operation **3302** includes generating, for a current video block of a video, an intermediate prediction signal using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation **3304** includes generating, based on the intermediate prediction signal, a final prediction signal. Operation **3306** includes performing, based on the final

prediction signal, a conversion between the current video block and a bitstream representation of the current video block.

In some embodiments for method **3300**, the generating the final prediction signal is performed by applying a Position Dependent Intra Prediction Combination (PDPC) to the intermediate prediction signal. In some embodiments for method **3300**, the generating the final prediction signal is performed by generating the previously coded samples of the video with the MIP mode and by filtering the previously coded samples with neighboring samples of the current video block.

FIG. **34** shows an example flowchart of an example method **3400** for matrix-based intra prediction. Operation **3402** includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode, where the performing the conversion includes using an interpolation filter in an upsampling process for the MIP mode, where, in the MIP mode, a matrix multiplication is applied to a first set of samples obtained from a row and column wise averaging of previously coded samples of the video, and the interpolation filter is applied to a second set of samples obtained from the matrix multiplication, and where interpolation filter excludes a bilinear interpolation filter.

In some embodiments for method **3400**, the interpolation filter includes a 4-tap interpolation filter. In some embodiments for method **3400**, a motion compensation for chroma components of the current video block is performed using the 4-tap interpolation filter. In some embodiments for method **3400**, an angular intra-prediction of the current video block is performed using the 4-tap interpolation filter. In some embodiments for method **3400**, the interpolation filter includes an 8-tap interpolation filter, and a motion compensation for luma component of the current video block is performed using the 8-tap interpolation filter.

FIG. **35** shows an example flowchart of an example method **3500** for matrix-based intra prediction. Operation **3502** includes performing a conversion between a current video block of a video and a bitstream representation of the current video block according to a rule, where the rule specifies a relationship between applicability of a matrix based intra prediction (MIP) mode or a transform mode during the conversion, where the MIP mode includes determining a prediction block of the current video block by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, and where the transform mode specifies use of a transform operation for the determining the prediction block for the current video block.

In some embodiments for method **3500**, the transform mode includes a Reduced Secondary Transform (RST), a secondary transform, a rotation transform or a Non-Separable Secondary Transform (NSST). In some embodiments for method **3500**, the rule indicates that the transform operation using the transform mode is not to be applied in response to the MIP mode being applied to the current video block. In some embodiments for method **3500**, the rule indicates whether to apply the transform operation using the transform mode based on a height (H) or a width (W) of the current video block. In some embodiments for method **3500**, the rule indicates that the transform operation using the transform mode is not applied to the current video block in response to $W \geq T1$ and $H \geq T2$. In some embodiments for method **3500**, the rule indicates that the transform operation

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using the transform mode is not applied to the current video block in response to $W \leq T1$ and $H \leq T2$. In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is not applied to the current video block in response to $W \geq T1$ or $H \geq T2$. In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is not applied to the current video block in response to $W \leq T1$ or $H \leq T2$. In some embodiments for method 3500, where $T1=32$ and $T2=32$. In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is not applied to the current video block in response to $W+H \geq T$. In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is not applied to the current video block in response to $W+H \leq T$. In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is not applied to the current video block in response to $W \times H \geq T$. In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is not applied to the current video block in response to $W \times H \leq T$. In some embodiments for method 3500, $T=256$.

In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is to be applied in response to the current video block being coded with MIP mode. In some embodiments for method 3500, a selection of a transform matrix or kernel for the transform operation is based on the current video block being coded with the MIP mode. In some embodiments, the method 3500 further includes converting the MIP mode to an intra prediction mode; and selecting a transform matrix or kernel based on the converted intra prediction mode. In some embodiments, the method 3500 further includes converting the MIP mode to an intra prediction mode; and selecting a transform matrix or kernel based on a classification of the converted intra prediction mode. In some embodiments for method 3500, the converted intra prediction mode includes a planar mode.

In some embodiments for method 3500, the rule indicates that the transform operation using the transform mode is applied to the current video block when the MIP mode is disallowed from being applied to the current video block. In some embodiments for method 3500, the rule indicates that a Discrete Cosine Transform Type II (DCT-II) transform coding technique is applied to the current video block coded using the MIP mode. In some embodiments for method 3500, the bitstream representation of the current video block excludes signaling of transform matrix indexes for the DCT-II transform coding technique. In some embodiments for method 3500, the performing the conversion includes deriving a transform matrix to be used by the DCT-II transform coding technique. In some embodiments for method 3500, the bitstream representation includes information related to the MIP mode that is signaled after an indication of a transform matrix.

In some embodiments for method 3500, the bitstream representation includes an indication of the MIP mode for the transform matrix. In some embodiments for method 3500, the rule indicates that the bitstream representation excludes an indication of the MIP mode for pre-defined transform matrices. In some embodiments for method 3500, the rule indicates that the transform operation using a transform skip technique is applied to the current video block in response to the current video block being coded with the MIP mode. In some embodiments for method 3500, the bitstream representation of the current video block excludes signaling of the transform skip technique when the

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current video block is coded with the MIP mode. In some embodiments for method 3500, the rule indicates that, for the current video block coded with the MIP mode, the MIP mode is converted to a pre-defined intra prediction mode when the transform operation is performed by selecting a mode dependent transform matrix or kernel. In some embodiments for method 3500, the rule indicates that the transform operation using a transform skip technique is disallowed for the current video block that is coded with the MIP mode. In some embodiments for method 3500, the bitstream representation excludes a signaling that indicates a usage of the transform skip technique. In some embodiments for method 3500, the rule indicates that the transform operation using a transform skip technique is allowed for the current video block that is not coded with the MIP mode. In some embodiments for method 3500, the bitstream representation excludes a signaling that indicates using the MIP mode.

FIG. 36 shows an example flowchart of an example method 3600 for matrix-based intra prediction. Operation 3602 includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where the performing the conversion includes deriving, according to a rule, boundary samples by applying a left bit shift operation or a right bit shift operation on a sum of at least one reference boundary sample, and where the rule determines whether to apply the left bit shift operation or the right bit shift operation.

In some embodiments for method 3600, the rule defines that the right bit shift operation is applied in response to a number of shifted bits being greater than or equal to zero. In some embodiments for method 3600, the rule defines that the right bit shift operation is applied using a first technique in response to the number of shifted bits being greater than zero, and where the rule defines that the right bit shift operation is applied using a second technique in response to the number of shifted bits being equal to zero, where the first technique is different from the second technique.

In some embodiments for method 3600, the boundary samples $\text{upsBdryX}[x]$ are calculated using one of following equations: $\text{upsBdryX}[x] = (\sum_{i=0}^{uDwn-1} \text{refX}[x * uDwn + i] + (1 \ll (\text{Log}_2(uDwn) - 1))) \gg \text{Log}_2(uDwn)$, if $uDwn > 1$, or $\text{upsBdryX}[x] = \text{refX}[x]$ if $uDwn = 1$, where $uDwn$ is equal to a function of the current video block size and a boundary size, where $\text{refX}[x]$ indicates a number of reference samples x , where $\gg$ indicates the right bit shift operation, and where $\ll$ indicates the left bit shift operation. In some embodiments for method 3600, the boundary samples $\text{upsBdryX}[x]$ are calculated using following equation: $\text{upsBdryX}[x] = (\sum_{i=0}^{uDwn-1} \text{refX}[x * uDwn + i] + (1 \ll (\text{Log}_2(uDwn) \gg 1))) \gg \text{Log}_2(uDwn)$, where $uDwn$ is equal to a function of the current video block and a boundary size, where $\text{refX}[x]$ indicates a number of reference samples x , where $\gg$ indicates the right bit shift operation, and where $\ll$ indicates the left bit shift operation.

In some embodiments for method 3600, the boundary size is predefined based on the current video block size. In some embodiments for method 3600, $uDwn$ is calculated as $uDwn = nTbs / \text{boundarySize}$, wherein $nTbs$ and boundarySize represent the current video block size and the boundary size, respectively. In some embodiments for method 3600, the

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boundary samples $\text{upsBdryX}[x]$ are calculated using one of following equations: $\text{upsBdryX}[x] = (\sum_{i=0}^{u\text{Dwn}-1} \text{refX}[x * u\text{Dwn} + i] + (1 << (\text{Log } 2(u\text{Dwn}) - 1))) >> \text{Log } 2(u\text{Dwn})$, if $n\text{Tbs} > \text{boundarySize}$, or $\text{upsBdryX}[x] = \text{refX}[x]$ if $n\text{Tbs} = \text{boundarySize}$.

FIG. 37 shows an example flowchart of an example method 3700 for matrix-based intra prediction. Operation 3702 includes performing a conversion between a current video block of a video and a bitstream representation of the current video block using a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation, where the prediction samples $\text{predSamples}[x\text{Hor} + dX][y\text{Hor}]$ are determined according to following formula in the upsampling operation: $\text{predSamples}[x\text{Hor} + dX][y\text{Hor}] = ((\text{upHor} - dX) * \text{predSamples}[x\text{Hor}][y\text{Hor}] + dX * \text{predSamples}[x\text{Hor} + \text{upHor}][y\text{Hor}] + \text{offsetHor}) / \text{upHor}$, and $\text{predSamples}[x\text{Ver}][y\text{Ver} + dY] = ((\text{upVer} - dY) * \text{predSamples}[x\text{Ver}][y\text{Ver}] + dY * \text{predSamples}[x\text{Ver}][y\text{Ver} + \text{upVer}] + \text{offsetVer}) / \text{upVer}$, where offsetHor and offsetVer are integers, where upHor is a function of the current video block width and a pre-defined value based on the current video block size, where upVer is a function of the current video block height and the pre-defined value based on the current video block size; where dX is $1 \dots \text{upHor} - 1$, where dY is $1 \dots \text{upVer} - 1$, and where $x\text{Hor}$ are positions based on the upHor and $y\text{Hor}$ are positions based on the upVer . In some embodiments for method 3700, $\text{offsetHor} = \text{upHor} / 2$, and $\text{offsetVer} = \text{upVer} / 2$.

FIG. 38 shows an example flowchart of an example method 3800 for matrix-based intra prediction. Operation 3802 includes generating, for a conversion between a current video block of a video and a bitstream representation of the current video block, an intra prediction mode for the current video block coded with a matrix based intra prediction (MIP) mode in which a prediction block of the current video block is determined by performing, on previously coded samples of the video, a boundary downsampling operation, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation. Operation 3804 includes determining a rule for storing information indicative of the intra prediction mode based on whether the current video block is coded with the MIP mode. Operation 3806 includes performing the conversion according to the rule, where the rule defines that the bitstream representation excludes a storage of information indicative of the MIP mode associated with the current video block.

This patent document may refer to bitstream representation to mean coded representation and vice versa. From the foregoing, it will be appreciated that specific embodiments of the presently disclosed technology have been described herein for purposes of illustration, but that various modifications may be made without deviating from the scope of the invention. Accordingly, the presently disclosed technology is not limited except as by the appended claims.

Implementations of the subject matter and the functional operations described in this patent document can be implemented in various systems, digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Implementations of the subject matter described in this specification can be implemented as one or more computer program products, i.e., one or more modules of computer program instructions encoded on a tangible and non-transi-

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tory computer readable medium for execution by, or to control the operation of, data processing apparatus. The computer readable medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them. The term “data processing unit” or “data processing apparatus” encompasses all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them.

A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit).

Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of nonvolatile memory, media, and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

It is intended that the specification, together with the drawings, be considered exemplary only, where exemplary means an example. As used herein, the use of “or” is intended to include “and/or”, unless the context clearly indicates otherwise.

While this patent document contains many specifics, these should not be construed as limitations on the scope of any invention or of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments of particular inventions. Certain features that are described in this patent document in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Moreover, the separation of various system components in the embodiments described in this patent document should not be understood as requiring such separation in all embodiments.

Only a few implementations and examples are described, and other implementations, enhancements and variations can be made based on what is described and illustrated in this patent document.

What is claimed is:

1. A video processing method, comprising:

determining, for a conversion between a video block of a video and a bitstream of the video, that a first intra mode is applied on the video block of the video,

performing, in the first intra mode, a boundary downsampling operation on reference samples of the video block based on a size of the video block, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation to generate prediction samples for the video block, and

performing the conversion based on the prediction samples and residual samples of the video block,

wherein a secondary transform tool is applied on the video block, and a secondary transform matrix is selected to generate the residual samples for the video block, and

wherein the secondary transform tool includes applying, during encoding, a forward secondary transform to an output of a forward primary transform before applying a quantization process, or applying, during decoding, an inverse secondary transform to an output of a dequantization process before applying an inverse primary transform,

wherein in response to a width-height ratio of the video block being greater than 2, a context with an increment value of 3 is used for a first bin of a second syntax element,

wherein in response to the width-height ratio of the video block being smaller than or equal to 2, a single context selected from contexts with increment values of 0, 1 or 2 is used for the first bin of the second syntax element,

wherein the boundary downsampling operation includes deriving, according to a rule, boundary samples by applying a left bit shift operation or a right bit shift operation on a sum of at least one reference boundary sample, wherein the rule determines whether to apply the left bit shift operation or the right bit shift operation, wherein the rule defines that a right bit shift operation is applied in response to a number of shifted bits being

greater than zero, and wherein the boundary samples $\text{redBdryS}[x]$ are calculated using one of following equations:

$\text{redBdryS}[x] =$

$$\left(\sum_{i=0}^{bDwn-1} \text{refS}[x * bDwn + 1] + (1 \ll (\text{Log2}(bDwn) - 1)) \right) \gg \text{Log2}(bDwn),$$

if $bDwn > 1$, or $\text{redBdryS}[x] = \text{refS}[x]$ if $bDwn = 1$, wherein $bDwn$ is equal to a function of the size of the video block and a boundary size, wherein $\text{refS}[x]$ indicates a number of reference samples x ,

wherein $\gg$ indicates the right bit shift operation, and wherein $\ll$ indicates the left bit shift operation, wherein the boundary size is predefined based on the size of the video block, and

wherein $bDwn$ is calculated as $bDwn = nTbs / \text{boundarySize}$, wherein $nTbs$ and boundarySize represent the size of the video block and the boundary size, respectively.

2. The method of claim 1, wherein the secondary transform tool includes a non-separable secondary transform.

3. The method of claim 1, wherein the secondary transform tool includes a Reduced Secondary Transform (RST) or a rotation transform.

4. The method of claim 1, wherein based on the first intra mode being applied on the video block, a second intra mode is converted from the first intra mode, and wherein a secondary transform classification is derived based on the second intra mode, wherein the secondary transform matrix is determined based on the secondary transform classification.

5. The method of claim 1, wherein a down-sampling factor derived in a boundary down-sampling operation is larger than or equal to 1, and wherein a one-dimensional vector array is further derived based on concatenating down-sampled reference samples derived from the boundary down-sampling operation and the one-dimensional vector array is used as input of the matrix vector multiplication operation.

6. The method of claim 1, wherein based on the first intra mode being applied on the video block, a second intra mode is converted from the first intra mode, and wherein a secondary transform classification is derived based on the second intra mode, wherein in the second intra mode, distance-based weighted calculations are applied on reference values in vertical direction and horizontal direction to derive prediction values.

7. The method of claim 1, wherein based on the first intra mode being applied on the video block, a second intra mode is converted from the first intra mode, and wherein a secondary transform classification is derived based on the second intra mode, wherein the second intra mode includes a planar mode.

8. The method of claim 1, wherein whether to apply the first intra mode is specified by a first syntax element included in a sequence parameter set and the second syntax element included in a coding unit level set.

9. The method of claim 1, wherein the first intra mode includes multiple types, and a type index for the video block is derived excluding referring to type indices of previous video blocks.

10. The method of claim 9, wherein the type index for the video block is explicitly included in the bitstream.

11. The method of claim 1, wherein whether to apply the secondary transform tool is based on a height (H) or a width (W) of the video block.

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12. The method of claim 1, wherein the conversion includes encoding the video block into the bitstream.

13. The method of claim 1, wherein the conversion includes decoding the video block from the bitstream.

14. The method of claim 1, wherein the boundary size is predefined based on the current video block size.

15. The method of claim 1, wherein bDwn is calculated as $bDwn = nTbs / boundarySize$, wherein nTbs and boundarySize represent the current video block size and the boundary size, respectively.

16. An apparatus for processing video data comprising a processor and a non-transitory memory with instructions thereon, wherein the instructions upon execution by the processor, cause the processor to:

determine, for a conversion between a video block of a video and a bitstream of the video, that a first intra mode is applied on the video block of the video,

perform, in the first intra mode, a boundary downsampling operation on reference samples of the video block based on a size of the video block, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation to generate prediction samples for the video block, and

perform the conversion based on the prediction samples and residual samples of the video block,

wherein a secondary transform tool is applied on the video block, and a secondary transform matrix is selected to generate the residual samples for the video block, and

wherein the secondary transform tool includes applying, during encoding, a forward secondary transform to an output of a forward primary transform before applying a quantization process, or applying, during decoding, an inverse secondary transform to an output of a dequantization process before applying an inverse primary transform,

wherein in response to a width-height ratio of the video block being greater than 2, a context with an increment value of 3 is used for a first bin of a second syntax element, wherein in response to the width-height ratio of the video block being smaller than or equal to 2, a single context selected from contexts with increment values of 0, 1 or 2 is used for the first bin of the second syntax element,

wherein the boundary downsampling operation includes deriving, according to a rule, boundary samples by applying a left bit shift operation or a right bit shift operation on a sum of at least one reference boundary sample, wherein the rule determines whether to apply the left bit shift operation or the right bit shift operation, wherein the rule defines that a right bit shift operation is applied in response to a number of shifted bits being greater than zero, and wherein the boundary samples $redBdryS[x]$ are calculated using one of following equations:

$$redBdryS[x] =$$

$$\left(\sum_{i=0}^{bDwn-1} refS[x * bDwn + 1] + (1 << (\log_2(bDwn) - 1)) \right) >> \log_2(bDwn),$$

if $bDwn > 1$, or $redBdryS[x] = refS[x]$ if $bDwn = 1$,

wherein bDwn is equal to a function of the size of the video block and a boundary size,

wherein refS [x] indicates a number of reference samples x,

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wherein >> indicates the right bit shift operation, and wherein << indicates the left bit shift operation,

wherein the boundary size is predefined based on the size of the video block, and

wherein bDwn is calculated as $bDwn = nTbs / boundarySize$, wherein nTbs and boundarySize represent the size of the video block and the boundary size, respectively.

17. The apparatus of claim 16, wherein the secondary transform tool includes a non-separable secondary transform.

18. A non-transitory computer-readable storage medium storing instructions that cause a processor to:

determine, for a conversion between a video block of a video and a bitstream of the video, that a first intra mode is applied on the video block of the video,

perform, in the first intra mode, a boundary downsampling operation on reference samples of the video block based on a size of the video block, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation to generate prediction samples for the video block, and

perform the conversion based on the prediction samples and residual samples of the video block,

wherein a secondary transform tool is applied on the video block, and a secondary transform matrix is selected to generate the residual samples for the video block, and

wherein the secondary transform tool includes applying, during encoding, a forward secondary transform to an output of a forward primary transform before applying a quantization process, or applying, during decoding, an inverse secondary transform to an output of a dequantization process before applying an inverse primary transform,

wherein in response to a width-height ratio of the video block being greater than 2, a context with an increment value of 3 is used for a first bin of a second syntax element, wherein in response to the width-height ratio of the video block being smaller than or equal to 2, a single context selected from contexts with increment values of 0, 1 or 2 is used for the first bin of the second syntax element,

wherein the boundary downsampling operation includes deriving, according to a rule, boundary samples by applying a left bit shift operation or a right bit shift operation on a sum of at least one reference boundary sample, wherein the rule determines whether to apply the left bit shift operation or the right bit shift operation, wherein the rule defines that a right bit shift operation is applied in response to a number of shifted bits being greater than zero, and wherein the boundary samples $redBdryS[x]$ are calculated using one of following equations:

$$redBdryS[x] =$$

$$\left(\sum_{i=0}^{bDwn-1} refS[x * bDwn + 1] + (1 << (\log_2(bDwn) - 1)) \right) >> \log_2(bDwn),$$

if $bDwn > 1$, or $redBdryS[x] = refS[x]$ if $bDwn = 1$,

wherein bDwn is equal to a function of the size of the video block and a boundary size,

wherein refS [x] indicates a number of reference samples x,

wherein >> indicates the right bit shift operation, and wherein << indicates the left bit shift operation,

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wherein the boundary size is predefined based on the size of the video block, and

wherein $bDwn$ is calculated as $bDwn = nTbs / \text{boundarySize}$, wherein $nTbs$ and boundarySize represent the size of the video block and the boundary size, respectively.

19. The non-transitory computer-readable storage medium of claim 18, wherein the secondary transform tool includes a non-separable secondary transform.

20. A method for storing a bitstream of a video, comprising:

determining that a first intra mode is applied on a video block of the video,

performing, in the first intra mode, a boundary downsampling operation on reference samples of the video block based on a size of the video block, followed by a matrix vector multiplication operation, and selectively followed by an upsampling operation to generate prediction samples for the video block, and

generating the bitstream based on the prediction samples and residual samples of the video block; and

storing the bitstream in a non-transitory computer-readable recoding medium,

wherein a secondary transform tool is applied on the video block, and a secondary transform matrix is selected to generate the residual samples for the video block, and

wherein the secondary transform tool includes applying, during encoding, a forward secondary transform to an output of a forward primary transform before applying a quantization process, or applying, during decoding, an inverse secondary transform to an output of a dequantization process before applying an inverse primary transform,

wherein in response to a width-height ratio of the video block being greater than 2, a context with an increment value of 3 is used for a first bin of a second syntax element,

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wherein in response to the width-height ratio of the video block being smaller than or equal to 2, a single context selected from contexts with increment values of 0, 1 or 2 is used for the first bin of the second syntax element,

wherein the boundary downsampling operation includes deriving, according to a rule, boundary samples by applying a left bit shift operation or a right bit shift operation on a sum of at least one reference boundary sample, wherein the rule determines whether to apply the left bit shift operation or the right bit shift operation, wherein the rule defines that a right bit shift operation is applied in response to a number of shifted bits being greater than zero, and wherein the boundary samples $\text{redBdryS}[x]$ are calculated using one of following equations:

$$\text{redBdryS}[x] = \left(\sum_{i=0}^{bDwn-1} \text{refS}[x * bDwn + i] + (1 \ll (\text{Log2}(bDwn) - 1)) \right) \gg \text{Log2}(bDwn),$$

$(1 \ll (\text{Log2}(bDwn) - 1)) \gg \text{Log2}(bDwn)$, if $bDwn > 1$, or $\text{redBdryS}[x] = \text{refS}[x]$ if $bDwn = 1$,

wherein $bDwn$ is equal to a function of the size of the video block and a boundary size,

wherein $\text{refS}[x]$ indicates a number of reference samples x ,

wherein $\gg$ indicates the right bit shift operation, and wherein $\ll$ indicates the left bit shift operation,

wherein the boundary size is predefined based on the size of the video block, and

wherein $bDwn$ is calculated as $bDwn = nTbs / \text{boundarySize}$, wherein $nTbs$ and boundarySize represent the size of the video block and the boundary size, respectively.

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